ASD-STE100 SIMPLIFIED TECHNICAL ENGLISH EDITION

Linear Algebra: The Math Book

Complete Equation Derivations from +2 Base Knowledge, SVG Diagrams, Verbatim Book Examples, and Solved Problem Sets

SYLLABUS COVERAGE (GILBERT STRANG, 4TH EDITION)

- **Chapter 1 (Sections 1.2 & 1.3):** Lengths, Dot Products, Cosine Formula, Schwarz & Triangle Inequalities, Difference Matrices (pp. 11–21, 29–30).
- **Chapter 2 (Section 4.1):** Orthogonality of the Four Fundamental Subspaces, Orthogonal Complements, Fundamental Theorem Part 2 (pp. 195–205).
- **Chapter 3 (Section 4.2):** Projections onto Lines and Subspaces, Projection Matrices $P = A(A^TA)^{-1}A^T$, Normal Equations (pp. 206–217).
- **Chapter 4 (Section 4.3):** Least Squares Approximations, Line and Parabola Fitting, Orthogonalized Time Shift (pp. 218–229).
- **Chapter 5 (Section 4.4):** Orthogonal Bases, Orthogonal Matrices Q, Gram-Schmidt Process, $A = QR$ Factorization (pp. 230–243).
- **Chapter 6 (Sections 6.5 & 6.7):** Positive Definite Matrices, Energy $\boldsymbol{x}^TA\boldsymbol{x} > 0$, Cholesky $A = R^TR$, Ellipses, Hessian, SVD & Eigenvalue Table (pp. 342–354, 373–374).
- **Chapter 7 (Sections 8.6 & 8.7):** Weighted Least Squares, Covariance Matrix Σ, PCA, Homogeneous Coordinates in Computer Graphics (pp. 453–458, 463–464).

Written in compliance with ASD-STE100 Simplified Technical English rules: active voice, direct instructions, zero ambiguity.

Contents

1 +2 Base Toolkit **3**

Module 1 Vectors, Lengths, and Dot Products **5**

2 Lengths, Dot Products, and Matrices **6**

2.1 Dot Product and Vector Length 6

2.2 Angles and Inequalities 8

2.3 Solved Problem Set 1.2 11

2.4 Solved Problem Set 1.3 14

Module 2 Orthogonality, Projections, and Least Squares **16**

3 Orthogonality of the Four Subspaces **17**

3.1 The Four Subspaces and Complements 17

3.2 Solved Problem Set 4.1 20

4 Projections **24**

4.1 Projection Onto a Line 24

4.2 Projection Onto a Subspace 25

4.3 Solved Problem Set 4.2 28

5 Least Squares Approximations **31**

5.1 Minimizing Error by Geometry, Algebra, and Calculus 31

5.2 Solved Problem Set 4.3 34

Module 3 Orthonormal Bases, Gram-Schmidt, and Positive Definite Matrices **36**

6 Orthogonal Bases and Gram-Schmidt **37**

6.1 Orthonormal Vectors and Orthogonal Matrices 37

6.2 Gram-Schmidt and A = QR 38

6.3 Solved Problem Set 4.4 40

7 Positive Definite Matrices and SVD **43**

7.1 Tests and Five Equivalent Definitions 43

7.2 Ellipses, Minima, and the SVD Table 44

7.3 Solved Problem Sets 6.5 and 6.7 46

Module 4 Applications: Statistics, Probability, and Computer Graphics **49**

8 Statistics, Probability, and Graphics **50**

8.1 Weighted Least Squares, Covariance, and PCA 50

8.2 Computer Graphics and Homogeneous Coordinates 51

8.3 Solved Problem Sets 8.6 and 8.7 52

Master Syllabus Formula Sheet (ASD-STE100) **54**

CHAPTER 1

The +2 Mathematics Toolkit

Every equation in this book comes from eight +2 (high-school grades 11–12) rules. Refer to tools T1–T8 whenever a derivation cites them.

T1 Pythagoras Theorem in 2D and 3D

For a right triangle with legs a, b and hypotenuse c, the side lengths obey:

$$a^2 + b^2 = c^2 \tag{1.1}$$

In 2D Cartesian coordinates, the distance from $(0,0)$ to (v_1, v_2) is $\sqrt{v_1^2 + v_2^2}$. Applying Pythagoras twice in a 3D box gives the diagonal length $\sqrt{v_1^2 + v_2^2 + v_3^2}$.

T2 Trigonometric Cosine Subtraction Identity

For any two angles α and β on the unit circle $\cos^2\theta + \sin^2\theta = 1$:

$$\cos(\beta - \alpha) = \cos\beta\cos\alpha + \sin\beta\sin\alpha \tag{1.2}$$

Also, $|\cos\theta| \le 1$ for every real angle θ, and $\cos 90^\circ = 0$, $\cos 0^\circ = 1$, $\cos 180^\circ = -1$.

T3 Algebraic Square Expansion and Non-Negativity

For any real numbers a and b:

$$(a-b)^2 = a^2 - 2ab + b^2 \ge 0 \implies 2ab \le a^2 + b^2 \tag{1.3}$$

Equality holds if and only if $a = b$.

T4 Slopes of Perpendicular Lines in the Plane

A line from $(0,0)$ to (v_1, v_2) has slope $m_1 = v_2/v_1$. A second line to (w_1, w_2) has slope $m_2 = w_2/w_1$. The two lines are perpendicular if and only if:

$$m_1 m_2 = \frac{v_2 w_2}{v_1 w_1} = -1 \iff v_1 w_1 + v_2 w_2 = 0 \tag{1.4}$$

T5 Calculus Minimum Test (First and Second Derivatives)

A smooth function $E(x)$ reaches its minimum where its first derivative is zero and its second derivative is positive:

$$\frac{\mathrm{d}E}{\mathrm{d}x} = 0, \qquad \frac{\mathrm{d}^2E}{\mathrm{d}x^2} > 0 \tag{1.5}$$

For a function of two variables $E(C, D)$, both partial derivatives must equal zero at the minimum: $\partial E/\partial C = 0$ and $\partial E/\partial D = 0$. By the chain rule, $\frac{\partial}{\partial D}(C + Dt - b)^2 = 2(C + Dt - b)t$.

T6 Matrix Multiplication and Transpose Rules

If A has size $m \times n$ and B has size $n \times p$, then entry (i, j) of AB is the dot product of row i of A with column j of B. The transpose $(AB)^T$ reverses the order:

$$(AB)^T = B^T A^T, \qquad (A^{-1})^T = (A^T)^{-1} \tag{1.6}$$

T7 2×2 Determinant, Trace, and Inverse

For a 2×2 matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ with eigenvalues λ_1, λ_2:

$$\det(A) = ad - bc = \lambda_1 \lambda_2, \quad \operatorname{trace}(A) = a + d = \lambda_1 + \lambda_2, \quad A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \tag{1.7}$$

T8 Basic Probability, Mean, and Variance

If outcome e_j occurs with probability p_j (where $\sum p_j = 1$), the mean m and variance σ^2 are:

$$m = \mathrm{E}[e] = \sum_j e_j p_j, \qquad \sigma^2 = \mathrm{E}[(e - m)^2] = \sum_j (e_j - m)^2 p_j \tag{1.8}$$

MODULE 1

Vectors, Lengths, and Dot Products

CHAPTER 2

Lengths, Dot Products, and Matrices [Book 1.2, 1.3]

Syllabus Equations: Dot product $\boldsymbol{v} \cdot \boldsymbol{w} = \sum v_i w_i$ • Length $\|\boldsymbol{v}\| = \sqrt{\boldsymbol{v} \cdot \boldsymbol{v}}$ • Unit vector $\boldsymbol{u} = \boldsymbol{v}/\|\boldsymbol{v}\|$ • Right angles $\boldsymbol{v} \cdot \boldsymbol{w} = 0$ • Cosine Formula $\cos\theta = \frac{\boldsymbol{v} \cdot \boldsymbol{w}}{\|\boldsymbol{v}\|\|\boldsymbol{w}\|}$ • Schwarz Inequality $|\boldsymbol{v} \cdot \boldsymbol{w}| \leq \|\boldsymbol{v}\|\|\boldsymbol{w}\|$ • Triangle Inequality $\|\boldsymbol{v} + \boldsymbol{w}\| \leq \|\boldsymbol{v}\| + \|\boldsymbol{w}\|$ • Difference Matrices $A\boldsymbol{x} = \boldsymbol{b}$.

2.1 The Dot Product, Length, and Unit Vectors [Book 1.2]

GOAL

Define the dot product $\boldsymbol{v} \cdot \boldsymbol{w}$ of two n-dimensional vectors, derive the Euclidean length $\|\boldsymbol{v}\|$ from the +2 Pythagoras theorem, and construct a unit vector $\boldsymbol{u}$ of length 1.

FROM YOUR +2 BASE

Use the +2 Pythagoras Theorem (**T1**): a right triangle in the plane with horizontal side v_1 and vertical side v_2 has hypotenuse squared equal to $v_1^2 + v_2^2$.

Step 1: Define the dot product (inner product). Take two vectors $\boldsymbol{v} = (v_1, v_2, \ldots, v_n)$ and $\boldsymbol{w} = (w_1, w_2, \ldots, w_n)$. Multiply matching components $v_i w_i$ and add the products:

$$\boldsymbol{v} \cdot \boldsymbol{w} = v_1 w_1 + v_2 w_2 \quad (\text{in } \mathbb{R}^2) \qquad \textit{Book Eq. (1), p. 11} \tag{2.1}$$

$$\boldsymbol{v} \cdot \boldsymbol{w} = v_1 w_1 + v_2 w_2 + \cdots + v_n w_n = \sum_{i=1}^{n} v_i w_i \quad (\text{in } \mathbb{R}^n) \qquad \textit{general n-D definition} \tag{2.2}$$

Because $v_i w_i = w_i v_i$ for real numbers, the order does not matter: $\boldsymbol{w} \cdot \boldsymbol{v} = \boldsymbol{v} \cdot \boldsymbol{w}$. In matrix notation, write column vectors and transpose the first vector: $\boldsymbol{v} \cdot \boldsymbol{w} = \boldsymbol{v}^T \boldsymbol{w}$.

Step 2: Derive length squared from $\boldsymbol{v} \cdot \boldsymbol{v}$. Set $\boldsymbol{w} = \boldsymbol{v}$ in Eq. (2.2). For a 2D vector $\boldsymbol{v} = (v_1, v_2)$, **T1** gives the arrow length squared as $\|\boldsymbol{v}\|^2 = v_1^2 + v_2^2 = \boldsymbol{v} \cdot \boldsymbol{v}$. For a 3D vector $\boldsymbol{v} = (v_1, v_2, v_3)$, the base vector $(v_1, v_2, 0)$ has length $\sqrt{v_1^2 + v_2^2}$. This base vector is perpendicular to the vertical vector $(0, 0, v_3)$. Apply **T1** a second time to get the box diagonal:

$$\|\boldsymbol{v}\|^2 = \left(\sqrt{v_1^2 + v_2^2}\right)^2 + v_3^2 = v_1^2 + v_2^2 + v_3^2 = \boldsymbol{v} \cdot \boldsymbol{v} \qquad \textit{Pythagoras applied twice} \tag{2.3}$$

$$\text{length} = \|\boldsymbol{v}\| = \sqrt{\boldsymbol{v} \cdot \boldsymbol{v}} = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2} \qquad \textit{square root of } \boldsymbol{v} \cdot \boldsymbol{v} \tag{2.4}$$

Step 3: Construct a unit vector $\boldsymbol{u}$. A **unit vector** $\boldsymbol{u}$ has length $\|\boldsymbol{u}\| = 1$, which means $\boldsymbol{u} \cdot \boldsymbol{u} = 1$. Divide any nonzero vector $\boldsymbol{v}$ by its scalar length $\|\boldsymbol{v}\|$:

$$\boldsymbol{u} = \frac{\boldsymbol{v}}{\|\boldsymbol{v}\|} \implies \boldsymbol{u} \cdot \boldsymbol{u} = \frac{\boldsymbol{v}}{\|\boldsymbol{v}\|} \cdot \frac{\boldsymbol{v}}{\|\boldsymbol{v}\|} = \frac{\boldsymbol{v} \cdot \boldsymbol{v}}{\|\boldsymbol{v}\|^2} = \frac{\|\boldsymbol{v}\|^2}{\|\boldsymbol{v}\|^2} = 1 \tag{2.5}$$

In the xy plane, the unit vector at angle θ from the x axis is $\boldsymbol{u} = (\cos\theta, \sin\theta)$ because $\cos^2\theta + \sin^2\theta = 1$.

RESULT (Dot Product, Length, and Unit Vector)

$$\boxed{v \cdot w = \sum_{i=1}^{n} v_i w_i} \qquad \boxed{\|v\| = \sqrt{v \cdot v}} \qquad \boxed{u = \frac{v}{\|v\|} \quad (u \cdot u = 1)} \tag{2.6}$$

BOOK EXAMPLE 1.2.1 Zero Dot Product and Perpendicular Vectors

Question (as in Book): The vectors $v = (4, 2)$ and $w = (-1, 2)$ have a zero dot product. Compute $v \cdot w$. Also check the dot product of $v = (1, 2)$ and $w = (3, 1)$, and verify that $w \cdot v$ is also 5.

BOOK EXAMPLE 1.2.2 Engineering Moments on a See-Saw

Question (as in Book): Put a weight of 4 at the point $x = -1$ (left of zero) and a weight of 2 at the point $x = 2$ (right of zero). Show why the x axis will balance on the center point (like a see-saw) using the dot product of the weight vector $(w_1, w_2) = (4, 2)$ and distance vector $(v_1, v_2) = (-1, 2)$.

BOOK EXAMPLE 1.2.3 Dot Products in Economics and Business

Question (as in Book): We have three goods to buy and sell. Their prices are (p_1, p_2, p_3) for each unit—this is the "price vector" p. The quantities we buy or sell are (q_1, q_2, q_3)—positive when we sell, negative when we buy. Write the total income as a dot product in three dimensions and explain what $q \cdot p = 0$ means.

BOOK EXAMPLE 1.2.4 Standard Unit Vectors and Unit Circle Vectors

Question (as in Book): The standard unit vectors along the x and y axes are written i and j. In the xy plane, the unit vector that makes an angle "theta" with the x axis is $(\cos\theta, \sin\theta)$. Verify that $u \cdot u = 1$, and check that dividing $(2, 2, 1)$ by its length 3 gives a unit vector $u = (\frac{2}{3}, \frac{2}{3}, \frac{1}{3})$ with $u \cdot u = 1$.

END-OF-PAGE SOLUTION FOR EXAMPLE 1.2.1 – 1.2.4 (Page 11–13 Examples)

- **Ex. 1.2.1:** $\begin{bmatrix} 4 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} -1 \\ 2 \end{bmatrix} = (4)(-1) + (2)(2) = -4 + 4 = 0$. The vectors are perpendicular (90°). For $v = (1, 2)$ and $w = (3, 1)$, $v \cdot w = (1)(3) + (2)(1) = 5$, and $w \cdot v = (3)(1) + (1)(2) = 5$.
- **Ex. 1.2.2:** The moments are $w_1 v_1$ and $w_2 v_2$. The balance equation is $w_1 v_1 + w_2 v_2 = (4)(-1) + (2)(2) = -4 + 4 = 0$. Because the dot product is zero, the see-saw balances.
- **Ex. 1.2.3:** Income $= (q_1, q_2, q_3) \cdot (p_1, p_2, p_3) = q_1 p_1 + q_2 p_2 + q_3 p_3 = q \cdot p$. When $q \cdot p = 0$, "the books balance" (total sales equal total purchases), and p is perpendicular to q in $\mathbb{R}^3$.
- **Ex. 1.2.4:** For $u = (\cos\theta, \sin\theta)$, $u \cdot u = \cos^2\theta + \sin^2\theta = 1$. For $v = (2, 2, 1)$, $\|v\| = \sqrt{4 + 4 + 1} = 3$. Dividing by 3 gives $u = (\frac{2}{3}, \frac{2}{3}, \frac{1}{3})$ and $u \cdot u = \frac{4}{9} + \frac{4}{9} + \frac{1}{9} = 1$.

2.2 The Angle Between Vectors and the Great Inequalities [Book 1.2]

GOAL

Prove that perpendicular vectors satisfy $v \cdot w = 0$, derive the Cosine Formula $\cos\theta = \frac{v \cdot w}{\|v\|\|w\|}$, and prove the Cauchy-Schwarz, AM-GM, and Triangle inequalities.

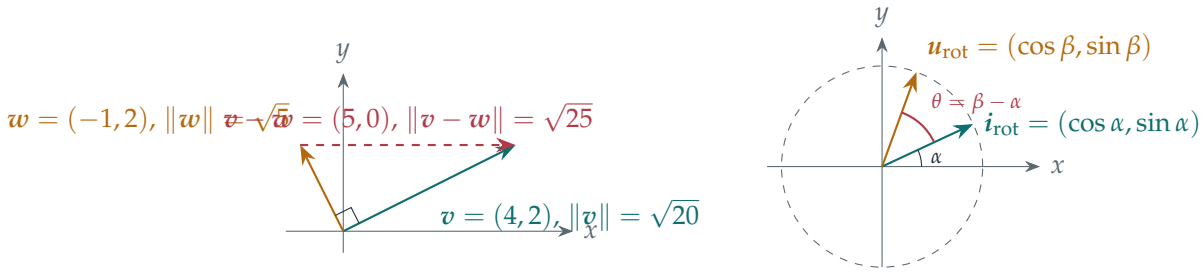


Fig. 2.1 Left: Right triangle with perpendicular sides v, w and hypotenuse $v - w$ (SVG source: `svg_dot_angle.svg`). Right: Unit vectors at angle $\theta = \beta - \alpha$.

Step 1: Prove the Right-Angle Law ($v \cdot w = 0 \iff v \perp w$). When v and w meet at 90°, they form the two legs of a right triangle. The third side connecting their tips is $v - w$. Apply the +2 Pythagoras Law (**T1**):

$$\|v\|^2 + \|w\|^2 = \|v - w\|^2 \qquad \textit{Book Eq. (2), p. 14} \tag{2.7}$$

$$\begin{aligned}
(v_1^2 + v_2^2) + (w_1^2 + w_2^2) &= (v_1 - w_1)^2 + (v_2 - w_2)^2 && \textit{Book Eq. (3), expand in 2D} \\
v_1^2 + v_2^2 + w_1^2 + w_2^2 &= (v_1^2 - 2v_1w_1 + w_1^2) + (v_2^2 - 2v_2w_2 + w_2^2) && \textit{use T3 on right side} \\
0 &= -2v_1w_1 - 2v_2w_2 && \textit{cancel } v_1^2, v_2^2, w_1^2, w_2^2 \\
v_1w_1 + v_2w_2 = 0 &\implies v \cdot w = 0 && \textit{Book Eq. (4), divide by } -2
\end{aligned} \tag{2.8}$$

In n dimensions, the same cancellation leaves $-2\sum_{i=1}^{n} v_i w_i = 0$, so $v \cdot w = 0$.

Step 2: Derive the Cosine Formula from unit vectors. Rotate the unit vector $i = (1, 0)$ through angle α to $(\cos\alpha, \sin\alpha)$. Rotate the unit vector $u = (\cos\theta, \sin\theta)$ through the same angle α to $(\cos\beta, \sin\beta)$, where $\beta = \alpha + \theta$. Take their dot product and apply **T2**:

$$u \cdot U = \cos\alpha\cos\beta + \sin\alpha\sin\beta = \cos(\beta - \alpha) = \cos\theta \tag{2.9}$$

For any nonzero vectors v and w, divide by their lengths to form unit vectors $u = v/\|v\|$ and $U = w/\|w\|$. Since $u \cdot U = \cos\theta$, we obtain the **Cosine Formula**:

$$\frac{v}{\|v\|} \cdot \frac{w}{\|w\|} = \cos\theta \implies \frac{v \cdot w}{\|v\|\|w\|} = \cos\theta \tag{2.10}$$

Step 3: Derive the Cauchy-Schwarz and Triangle Inequalities. Because $|\cos\theta| \le 1$ (**T2**), multiplying by $\|v\|\|w\|$ immediately yields the **Schwarz Inequality**:

$$\left|\frac{v \cdot w}{\|v\|\|w\|}\right| = |\cos\theta| \le 1 \implies |v \cdot w| \le \|v\|\|w\| \tag{2.11}$$

Algebraic proof in 2D (Problem 1.2.22): Subtract the left side from the right side of $(v_1w_1 + v_2w_2)^2 \leq (v_1^2 + v_2^2)(w_1^2 + w_2^2)$:

$$(v_1^2 + v_2^2)(w_1^2 + w_2^2) - (v_1w_1 + v_2w_2)^2 = v_1^2w_2^2 - 2v_1v_2w_1w_2 + v_2^2w_1^2 = (v_1w_2 - v_2w_1)^2 \geq 0 \tag{2.12}$$

To prove the **Triangle Inequality** (Problem 1.2.19–21), expand $\|v + w\|^2 = (v + w) \cdot (v + w)$ and apply $v \cdot w \leq \|v\|\|w\|$:

$$\begin{aligned} \|v + w\|^2 &= v \cdot v + 2v \cdot w + w \cdot w = \|v\|^2 + 2v \cdot w + \|w\|^2 \\ &\leq \|v\|^2 + 2\|v\|\|w\| + \|w\|^2 = (\|v\| + \|w\|)^2 \\ \implies \|v + w\| &\leq \|v\| + \|w\| \qquad \textit{take positive square root} \end{aligned} \tag{2.13}$$

Replacing w by $-w$ and using $v \cdot w = \|v\|\|w\| \cos\theta$ gives the **Law of Cosines** (Problem 1.2.20):

$$\|v - w\|^2 = (v - w) \cdot (v - w) = \|v\|^2 - 2\|v\|\|w\| \cos\theta + \|w\|^2 \tag{2.14}$$

BOOK EXAMPLE 1.2.5 Testing Cosine, Schwarz, and Triangle Inequalities

Question (as in Book): Find $\cos\theta$ for $v = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and $w = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and check both inequalities.

BOOK EXAMPLE 1.2.6 Geometric Mean vs. Arithmetic Mean from Schwarz

Question (as in Book): The dot product of $v = (a, b)$ and $w = (b, a)$ is $2ab$. Both lengths are $\sqrt{a^2 + b^2}$. Apply the Schwarz inequality to show that the geometric mean $\sqrt{xy}$ is not larger than the arithmetic mean $\frac{1}{2}(x + y)$ when $x = a^2$ and $y = b^2$. Check with $a = 2, b = 1$.

BOOK EXAMPLE 1.2 A Worked Example: Conditions for Equality

Question (as in Book): For the vectors $v = (3, 4)$ and $w = (4, 3)$ test the Schwarz inequality on $v \cdot w$ and the triangle inequality on $\|v + w\|$. Find $\cos\theta$ for the angle between v and w. When will we have *equality* $|v \cdot w| = \|v\|\|w\|$ and $\|v + w\| = \|v\| + \|w\|$?

BOOK EXAMPLE 1.2 B Worked Example: Parallel and Perpendicular Unit Vectors

Question (as in Book): Find a unit vector u in the direction of $v = (3, 4)$. Find a unit vector U that is perpendicular to u. How many possibilities for U?

BOOK EXAMPLE 1.2 C Worked Example: Finding a Vector from its Dot Products

Question (as in Book): Find a vector $x = (c, d)$ that has dot products $x \cdot r = 1$ and $x \cdot s = 0$ with the given vectors $r = (2, -1)$ and $s = (-1, 2)$.

END-OF-PAGE SOLUTION FOR EXAMPLE 1.2.5, 1.2.6, and Worked Examples 1.2 A – 1.2 C (Pages 16–18)

- **Ex. 1.2.5:** $v \cdot w = (2)(1) + (1)(2) = 4$. $\|v\| = \sqrt{5}$, $\|w\| = \sqrt{5}$. $\cos\theta = \frac{4}{\sqrt{5}\sqrt{5}} = \frac{4}{5}$. Schwarz: $4 \leq 5$. Triangle: $v + w = (3, 3)$ has length $\sqrt{18} < \sqrt{5} + \sqrt{5} = \sqrt{20}$.
- **Ex. 1.2.6:** Schwarz gives $2ab \leq \sqrt{a^2 + b^2}\sqrt{b^2 + a^2} = a^2 + b^2$, so $ab \leq \frac{a^2+b^2}{2}$. Set $x = a^2, y = b^2$ to get $\sqrt{xy} \leq \frac{x+y}{2}$. For $a = 2, b = 1$ ($x = 4, y = 1$): $\sqrt{4} = 2 \leq \frac{1+4}{2} = 2.5$.

- **Ex. 1.2 A:** $\boldsymbol{v} \cdot \boldsymbol{w} = 12 + 12 = 24$. $\|\boldsymbol{v}\| = \sqrt{9+16} = 5$, $\|\boldsymbol{w}\| = 5$. Schwarz: $24 < 25$. Triangle: $\|\boldsymbol{v} + \boldsymbol{w}\| = \|(7,7)\| = 7\sqrt{2} \approx 9.899 < 5 + 5 = 10$. $\cos\theta = \frac{24}{25}$. Equality $|\boldsymbol{v} \cdot \boldsymbol{w}| = \|\boldsymbol{v}\|\|\boldsymbol{w}\|$ holds when $\boldsymbol{w} = c\boldsymbol{v}$ ($\theta = 0^\circ$ or 180°). Equality $\|\boldsymbol{v} + \boldsymbol{w}\| = \|\boldsymbol{v}\| + \|\boldsymbol{w}\|$ holds when $\theta = 0^\circ$ ($c \geq 0$).
- **Ex. 1.2 B:** $\boldsymbol{u} = \frac{\boldsymbol{v}}{\|\boldsymbol{v}\|} = (\frac{3}{5}, \frac{4}{5})$. A perpendicular vector is $\boldsymbol{V} = (-4,3)$ because $(3)(-4) + (4)(3) = 0$. Dividing by $\|\boldsymbol{V}\| = 5$ gives $\boldsymbol{U} = (-\frac{4}{5}, \frac{3}{5})$. There are **two** possibilities in $\mathbb{R}^2$: $\boldsymbol{U}$ and $-\boldsymbol{U} = (\frac{4}{5}, -\frac{3}{5})$.
- **Ex. 1.2 C:** The dot products give $2c - d = 1$ and $-c + 2d = 0$. From the second equation, $c = 2d$. Substitute into the first: $4d - d = 1 \implies d = \frac{1}{3}, c = \frac{2}{3}$. Thus $\boldsymbol{x} = (\frac{2}{3}, \frac{1}{3}) = \frac{1}{3}(2,1)$.

2.3 All Unique Solved Problems: Problem Set 1.2 [Book pp. 19–21]

PROBLEM 1.2.1 – 1.2.3 Dot Products, Lengths, Unit Vectors, and Angles

Question 1: Calculate the dot products $u \cdot v$ and $u \cdot w$ and $u \cdot (v + w)$ and $w \cdot v$: $u = \begin{bmatrix} -.6 \\ .8 \end{bmatrix}$, $v = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$, $w = \begin{bmatrix} 8 \\ 6 \end{bmatrix}$.

Question 2: Compute the lengths $\|u\|$ and $\|v\|$ and $\|w\|$ of those vectors. Check the Schwarz inequalities $|u \cdot v| \leq \|u\|\|v\|$ and $|v \cdot w| \leq \|v\|\|w\|$.

Question 3: Find unit vectors in the directions of v and w in Problem 1, and the cosine of the angle θ. Choose vectors a, b, c that make $0°, 90°$, and $180°$ angles with w.

Solution:

1. $u \cdot v = (-.6)(3) + (.8)(4) = -1.8 + 3.2 = 1.4$. $u \cdot w = (-.6)(8) + (.8)(6) = -4.8 + 4.8 = 0$. $u \cdot (v + w) = u \cdot v + u \cdot w = 1.4 + 0 = 1.4$. $w \cdot v = (8)(3) + (6)(4) = 24 + 24 = 48$.
2. $\|u\| = \sqrt{(-.6)^2 + (.8)^2} = \sqrt{.36 + .64} = 1$. $\|v\| = \sqrt{3^2 + 4^2} = 5$. $\|w\| = \sqrt{8^2 + 6^2} = 10$. Schwarz checks: $|u \cdot v| = 1.4 \leq (1)(5) = 5$, and $|v \cdot w| = 48 \leq (5)(10) = 50$.
3. Unit vectors: $u_v = \frac{v}{5} = (.6, .8)$ and $u_w = \frac{w}{10} = (.8, .6)$. Cosine: $\cos\theta = \frac{v \cdot w}{\|v\|\|w\|} = \frac{48}{50} = 0.96$. With $w = (8, 6)$: $a = w = (8, 6)$ $(0°)$; $b = (-6, 8)$ or $u = (-.6, .8)$ $(90°)$; $c = -w = (-8, -6)$ $(180°)$.

PROBLEM 1.2.4 – 1.2.6 Unit Vector Dot Products and Perpendicular Subspaces

Question 4: For any *unit* vectors v and w, find the dot products (actual numbers) of (a) v and $-v$ (b) $v + w$ and $v - w$ (c) $v - 2w$ and $v + 2w$.

Question 5: Find unit vectors u_1 and u_2 in the directions of $v = (3, 1)$ and $w = (2, 1, 2)$. Find unit vectors U_1 and U_2 that are perpendicular to u_1 and u_2.

Question 6: (a) Describe every vector $w = (w_1, w_2)$ that is perpendicular to $v = (2, -1)$. (b) The vectors that are perpendicular to $V = (1, 1, 1)$ lie on a _____. (c) The vectors that are perpendicular to $(1, 1, 1)$ and $(1, 2, 3)$ lie on a _____.

Solution:

4. Since $\|v\| = \|w\| = 1$, $v \cdot v = 1$ and $w \cdot w = 1$: (a) $v \cdot (-v) = -1$. (b) $(v + w) \cdot (v - w) = v \cdot v - w \cdot w = 1 - 1 = 0$. (c) $(v - 2w) \cdot (v + 2w) = \|v\|^2 - 4\|w\|^2 = 1 - 4 = -3$.
5. $\|v\| = \sqrt{10} \implies u_1 = (3/\sqrt{10}, 1/\sqrt{10})$; perpendicular unit vector $U_1 = (-1/\sqrt{10}, 3/\sqrt{10})$. $\|w\| = \sqrt{4 + 1 + 4} = 3 \implies u_2 = (\frac{2}{3}, \frac{1}{3}, \frac{2}{3})$; one perpendicular unit vector is $U_2 = (1/\sqrt{2}, 0, -1/\sqrt{2})$.
6. (a) $2w_1 - w_2 = 0 \implies w = c(1, 2)$ for any scalar c. (b) They lie on a **plane** $(x + y + z = 0)$ in $\mathbb{R}^3$. (c) The intersection of two planes through the origin is a **line** in the direction $(1, -2, 1)$.

PROBLEM 1.2.7 – 1.2.11 Angles from Cosines, True/False, Slopes, and Half-Spaces

Question 7: Find the angle θ (from its cosine) between these pairs of vectors: (a) $v = \begin{bmatrix} 1 \\ \sqrt{3} \end{bmatrix}$

and $\boldsymbol{w} = \begin{bmatrix}1\\0\end{bmatrix}$ (b) $\boldsymbol{v} = \begin{bmatrix}2\\2\\-1\end{bmatrix}$ and $\boldsymbol{w} = \begin{bmatrix}2\\-1\\2\end{bmatrix}$ (c) $\boldsymbol{v} = \begin{bmatrix}1\\\sqrt{3}\end{bmatrix}$ and $\boldsymbol{w} = \begin{bmatrix}-1\\\sqrt{3}\end{bmatrix}$ (d) $\boldsymbol{v} = \begin{bmatrix}3\\1\end{bmatrix}$ and $\boldsymbol{w} = \begin{bmatrix}-1\\-2\end{bmatrix}$.

Question 8: True or false (give a reason if true or a counterexample if false): (a) If $\boldsymbol{u}$ is perpendicular (in three dimensions) to $\boldsymbol{v}$ and $\boldsymbol{w}$, those vectors $\boldsymbol{v}$ and $\boldsymbol{w}$ are parallel. (b) If $\boldsymbol{u}$ is perpendicular to $\boldsymbol{v}$ and $\boldsymbol{w}$, then $\boldsymbol{u}$ is perpendicular to $\boldsymbol{v} + 2\boldsymbol{w}$. (c) If $\boldsymbol{u}$ and $\boldsymbol{v}$ are perpendicular unit vectors then $\|\boldsymbol{u} - \boldsymbol{v}\| = \sqrt{2}$.

Question 9: The slopes of the arrows from $(0,0)$ to (v_1, v_2) and (w_1, w_2) are v_2/v_1 and w_2/w_1. Suppose the product v_2w_2/v_1w_1 of those slopes is -1. Show that $\boldsymbol{v} \cdot \boldsymbol{w} = 0$ and the vectors are perpendicular.

Question 10: Draw arrows from $(0,0)$ to the points $\boldsymbol{v} = (1,2)$ and $\boldsymbol{w} = (-2,1)$. Multiply their slopes. That answer is a signal that $\boldsymbol{v} \cdot \boldsymbol{w} = 0$ and the arrows are _____.

Question 11: If $\boldsymbol{v} \cdot \boldsymbol{w}$ is negative, what does this say about the angle between $\boldsymbol{v}$ and $\boldsymbol{w}$? Draw a 3-dimensional vector $\boldsymbol{v}$ (an arrow), and show where to find all $\boldsymbol{w}$'s with $\boldsymbol{v} \cdot \boldsymbol{w} < 0$.

Solution:

- **7:** (a) $\cos\theta = \frac{1}{(2)(1)} = \frac{1}{2} \implies \theta = 60^\circ$ $(\pi/3)$. (b) $\cos\theta = \frac{4-2-2}{(3)(3)} = 0 \implies \theta = 90^\circ$ $(\pi/2)$. (c) $\cos\theta = \frac{-1+3}{(2)(2)} = \frac{1}{2} \implies \theta = 60^\circ$ $(\pi/3)$. (d) $\cos\theta = \frac{-3-2}{\sqrt{10}\sqrt{5}} = \frac{-5}{5\sqrt{2}} = -\frac{1}{\sqrt{2}} \implies \theta = 135^\circ$ $(3\pi/4)$.
- **8:** (a) **False**: $\boldsymbol{u} = (0,0,1)$ is perpendicular to $\boldsymbol{v} = (1,0,0)$ and $\boldsymbol{w} = (0,1,0)$, which are not parallel. (b) **True**: $\boldsymbol{u} \cdot (\boldsymbol{v} + 2\boldsymbol{w}) = \boldsymbol{u} \cdot \boldsymbol{v} + 2(\boldsymbol{u} \cdot \boldsymbol{w}) = 0 + 0 = 0$. (c) **True**: $\|\boldsymbol{u} - \boldsymbol{v}\|^2 = \|\boldsymbol{u}\|^2 - 2\boldsymbol{u} \cdot \boldsymbol{v} + \|\boldsymbol{v}\|^2 = 1 - 0 + 1 = 2 \implies \|\boldsymbol{u} - \boldsymbol{v}\| = \sqrt{2}$.
- **9:** $\frac{v_2w_2}{v_1w_1} = -1 \implies v_2w_2 = -v_1w_1 \implies v_1w_1 + v_2w_2 = \boldsymbol{v} \cdot \boldsymbol{w} = 0$.
- **10:** Slopes are $2/1 = 2$ and $1/(-2) = -1/2$. Product is $(2)(-1/2) = -1$. The arrows are **perpendicular**.
- **11:** The angle is obtuse ($\theta > 90^\circ$). All $\boldsymbol{w}$'s with $\boldsymbol{v} \cdot \boldsymbol{w} < 0$ lie in the open half-space on the opposite side of the plane perpendicular to $\boldsymbol{v}$ through the origin.

PROBLEM 1.2.12 – 1.2.17 Projection Scalar, Mutual Orthogonality, and Direction Cosines

Question 12: With $\boldsymbol{v} = (1,1)$ and $\boldsymbol{w} = (1,5)$ choose a number c so that $\boldsymbol{w} - c\boldsymbol{v}$ is perpendicular to $\boldsymbol{v}$. Then find the formula that gives this number c for any nonzero $\boldsymbol{v}$ and $\boldsymbol{w}$.

Question 13: Find two vectors $\boldsymbol{v}$ and $\boldsymbol{w}$ that are perpendicular to $(1,0,1)$ and to each other.

Question 14: Find nonzero vectors $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}$ that are perpendicular to $(1,1,1,1)$ and to each other.

Question 15: The geometric mean of $x = 2$ and $y = 8$ is $\sqrt{xy} = 4$. The arithmetic mean is larger: $\frac{1}{2}(x+y) =$ _____. This would come in Example 6 from the Schwarz inequality for $\boldsymbol{v} = (\sqrt{2}, \sqrt{8})$ and $\boldsymbol{w} = (\sqrt{8}, \sqrt{2})$. Find $\cos\theta$ for this $\boldsymbol{v}$ and $\boldsymbol{w}$.

Question 16: How long is the vector $\boldsymbol{v} = (1,1,\ldots,1)$ in 9 dimensions? Find a unit vector $\boldsymbol{u}$ in the same direction as $\boldsymbol{v}$ and a unit vector $\boldsymbol{w}$ that is perpendicular to $\boldsymbol{v}$.

Question 17: What are the cosines of the angles α, β, θ between the vector $(1,0,-1)$ and the unit vectors $\boldsymbol{i}, \boldsymbol{j}, \boldsymbol{k}$ along the axes? Check the formula $\cos^2\alpha + \cos^2\beta + \cos^2\theta = 1$.

Solution:

- **12:** $v \cdot (w - cv) = 0 \implies v \cdot w - c(v \cdot v) = 0 \implies c = \frac{v \cdot w}{v \cdot v}$. For $v = (1,1)$ and $w = (1,5)$, $c = \frac{1+5}{1+1} = 3$.
- **13:** Choose $v = (0,1,0)$ and $w = (1,0,-1)$. Both have zero dot product with $(1,0,1)$ and $v \cdot w = 0$.
- **14:** Choose $u = (1,-1,0,0)$, $v = (0,0,1,-1)$, and $w = (1,1,-1,-1)$. All three are mutually orthogonal and perpendicular to $(1,1,1,1)$.
- **15:** $\frac{1}{2}(2+8) = 5$. For $v = (\sqrt{2}, \sqrt{8})$ and $w = (\sqrt{8}, \sqrt{2})$, $v \cdot w = 4+4 = 8$ and $\|v\| = \|w\| = \sqrt{10}$, so $\cos\theta = \frac{8}{10} = \frac{4}{5}$.
- **16:** $\|v\| = \sqrt{1^2 + \cdots + 1^2} = \sqrt{9} = 3$. Unit vector $u = \frac{1}{3}(1,1,1,1,1,1,1,1,1)$. Perpendicular unit vector: $w = \frac{1}{\sqrt{2}}(1,-1,0,0,0,0,0,0,0)$.
- **17:** $\|v\| = \sqrt{2}$. $\cos\alpha = \frac{1}{\sqrt{2}}$, $\cos\beta = 0$, $\cos\theta = -\frac{1}{\sqrt{2}}$. Sum of squares: $\frac{1}{2} + 0 + \frac{1}{2} = 1$.

PROBLEM 1.2.18 – 1.2.27 Triangle Identities, Proofs, and Bounds

Question 18: The parallelogram with sides $v = (4,2)$ and $w = (-1,2)$ is a rectangle. Check the Pythagoras formula $a^2 + b^2 = c^2$: (length of v)^2 + (length of w)^2 = (length of $v + w$)^2.
Question 19–24: (Solved in Section 1.2 derivations above; for Problem 23: $\cos\beta = w_1/\|w\|$ and $\sin\beta = w_2/\|w\|$; for Problem 24 with $u = (.6,.8)$ and $U = (.8,.6)$, $|u \cdot U| = .48 + .48 = .96 = \cos\theta \leq 1$.)
Question 25: Why is $|\cos\theta|$ never greater than 1 in the first place?
Question 26: If $v = (1,2)$ draw all vectors $w = (x,y)$ in the xy plane with $v \cdot w = x + 2y = 5$. Which is the shortest w?
Question 27: If $\|v\| = 5$ and $\|w\| = 3$, what are the smallest and largest values of $\|v - w\|$? What are the smallest and largest values of $v \cdot w$?

Solution:

- **18:** $\|v\|^2 = 16 + 4 = 20$; $\|w\|^2 = 1 + 4 = 5$. $v + w = (3,4) \implies \|v + w\|^2 = 9 + 16 = 25 = 20 + 5$.
- **25:** In a right triangle, $\cos\theta$ = adjacent/hypotenuse. By Pythagoras ($a^2 + b^2 = c^2$), the adjacent side cannot exceed the hypotenuse c.
- **26:** The vectors $w = (x,y)$ end on the line $x + 2y = 5$. The shortest w is perpendicular to the line (therefore parallel to $v = (1,2)$): $w = v = (1,2)$ with $\|w\| = \sqrt{5}$.
- **27:** Smallest $\|v - w\| = 5 - 3 = 2$ (when $\theta = 0°$); largest $\|v - w\| = 5 + 3 = 8$ (when $\theta = 180°$). Smallest $v \cdot w = -(5)(3) = -15$; largest $v \cdot w = +(5)(3) = 15$.

PROBLEM 1.2.28 – 1.2.32 Challenge Problems: Negative Dot Products, Hadamard, Random Vectors

Question 28: Can three vectors in the xy plane have $u \cdot v < 0$ and $v \cdot w < 0$ and $u \cdot w < 0$? How many vectors in xyz space can have all negative dot products?
Question 29: Pick any numbers that add to $x + y + z = 0$. Find the angle between your vector $v = (x,y,z)$ and the vector $w = (z,x,y)$. Explain why $v \cdot w/\|v\|\|w\|$ is always $-\frac{1}{2}$.
Question 30: How could you prove $\sqrt[3]{xyz} \leq \frac{1}{3}(x + y + z)$ (geometric mean $\leq$ arithmetic mean)?
Question 31: Find four perpendicular unit vectors with all components equal to $\frac{1}{2}$ or $-\frac{1}{2}$.
Question 32: What is the average size of the dot products $|u \cdot U_j|$ of random unit vectors in $\mathbb{R}^3$?

Solution:

- **28: Yes** in 2D: three vectors 120° apart, e.g., $(1,0)$, $(-1/2,\sqrt{3}/2)$, $(-1/2,-\sqrt{3}/2)$. In 3D (xyz space), up to **four** vectors can have all negative dot products (pointing to the vertices of a regular tetrahedron, e.g., $(1,1,1),(1,-1,-1),(-1,1,-1),(-1,-1,1)$).
- **29:** Since $x+y+z=0$, $0=(x+y+z)^2=(x^2+y^2+z^2)+2(xz+xy+yz)=\|\boldsymbol{v}\|^2+2(\boldsymbol{v}\cdot\boldsymbol{w})$. Also $\|\boldsymbol{w}\|=\|\boldsymbol{v}\|$. Thus $\frac{\boldsymbol{v}\cdot\boldsymbol{w}}{\|\boldsymbol{v}\|\|\boldsymbol{w}\|}=\frac{-\frac{1}{2}\|\boldsymbol{v}\|^2}{\|\boldsymbol{v}\|^2}=-\frac{1}{2}$, so $\theta=120^\circ$.
- **30:** Apply the 2-variable AM-GM twice to 4 numbers (x,y,z,A) where $A=\frac{1}{3}(x+y+z)$: $(xyzA)^{1/4}\le\frac{1}{2}(\sqrt{xy}+\sqrt{zw})\le\frac{1}{4}(x+y+z+A)=A \implies xyzA\le A^4 \implies \sqrt[3]{xyz}\le A$.
- **31:** $\boldsymbol{u}_1=\frac{1}{2}(1,1,1,1)$, $\boldsymbol{u}_2=\frac{1}{2}(1,-1,1,-1)$, $\boldsymbol{u}_3=\frac{1}{2}(1,1,-1,-1)$, $\boldsymbol{u}_4=\frac{1}{2}(1,-1,-1,1)$.
- **32:** In $\mathbb{R}^3$, surface area on the unit sphere is proportional to $\sin\theta\,\mathrm{d}\theta$, so the exact 3D average of $|\cos\theta|$ is $\frac{1}{2}\int_0^\pi|\cos\theta|\sin\theta\,\mathrm{d}\theta=\frac{1}{2}$ (whereas in 2D the uniform angle average is $\frac{1}{\pi}\int_0^\pi|\cos\theta|\,\mathrm{d}\theta=\frac{2}{\pi}\approx 0.6366$).

2.4 All Unique Solved Problems: Problem Set 1.3 [Book pp. 29–30]

PROBLEM 1.3.1 – 1.3.3 Triangular Sum Matrix S and Difference Inverse $A=S^{-1}$

Question 1: Find the linear combination $2\boldsymbol{s}_1+3\boldsymbol{s}_2+4\boldsymbol{s}_3=\boldsymbol{b}$. Then write $\boldsymbol{b}$ as a matrix-vector multiplication $S\boldsymbol{x}$. Compute the dot products (row of S) $\cdot\,\boldsymbol{x}$ for $\boldsymbol{s}_1=(1,1,1)^T,\boldsymbol{s}_2=(0,1,1)^T,\boldsymbol{s}_3=(0,0,1)^T$.
Question 2: Solve $S\boldsymbol{y}=\boldsymbol{b}$ for $\boldsymbol{b}=(1,1,1)^T$ and $\boldsymbol{b}=(1,4,9)^T$. The sum of the first n odd numbers is _____.
Question 3: Solve $S\boldsymbol{y}=\boldsymbol{B}$ for y_1,y_2,y_3 in terms of B_1,B_2,B_3. Write $\boldsymbol{y}=S^{-1}\boldsymbol{B}$. Are the columns of S independent or dependent?

Solution:

1. $\boldsymbol{b}=2\begin{bmatrix}1\\1\\1\end{bmatrix}+3\begin{bmatrix}0\\1\\1\end{bmatrix}+4\begin{bmatrix}0\\0\\1\end{bmatrix}=\begin{bmatrix}2\\5\\9\end{bmatrix}=\begin{bmatrix}1&0&0\\1&1&0\\1&1&1\end{bmatrix}\begin{bmatrix}2\\3\\4\end{bmatrix}=S\boldsymbol{x}$. Dot products: $(1,0,0)\cdot\boldsymbol{x}=2$, $(1,1,0)\cdot\boldsymbol{x}=5$, $(1,1,1)\cdot\boldsymbol{x}=9$.
2. For $\boldsymbol{b}=(1,1,1)^T$, $\boldsymbol{y}=(1,0,0)^T$. For $\boldsymbol{b}=(1,4,9)^T$, $\boldsymbol{y}=(1,3,5)^T$. The sum of the first n odd numbers is n^2.
3. $y_1=B_1$, $y_2=B_2-B_1$, $y_3=B_3-B_2$, so $A=S^{-1}=\begin{bmatrix}1&0&0\\-1&1&0\\0&-1&1\end{bmatrix}$. The columns of S are **independent**.

PROBLEM 1.3.4 – 1.3.7 Dependent Columns and Rows of Singular Matrices

Question 4–5: Find a combination $x_1\boldsymbol{w}_1+x_2\boldsymbol{w}_2+x_3\boldsymbol{w}_3=\boldsymbol{0}$ for $\boldsymbol{w}_1=(1,2,3)^T,\boldsymbol{w}_2=(4,5,6)^T,\boldsymbol{w}_3=(7,8,9)^T$. Also find two combinations $y_1\boldsymbol{r}_1+y_2\boldsymbol{r}_2+y_3\boldsymbol{r}_3=\boldsymbol{0}$ for the rows $\boldsymbol{r}_1=(1,4,7)^T,\boldsymbol{r}_2=(2,5,8)^T,\boldsymbol{r}_3=(3,6,9)^T$.

Question 6: Which values of c give dependent columns in $\begin{bmatrix}1&3&5\\1&2&4\\1&1&c\end{bmatrix}$, $\begin{bmatrix}1&0&c\\1&1&0\\0&1&1\end{bmatrix}$, $\begin{bmatrix}c&c&c\\2&1&5\\3&3&6\end{bmatrix}$?

Question 7: If $A\boldsymbol{x} = \mathbf{0}$, then each row has $\boldsymbol{r}_i \cdot \boldsymbol{x} = 0$. Why is the plane of the rows perpendicular to $\boldsymbol{x}$?

Solution:

- **4–5:** $1\boldsymbol{w}_1 - 2\boldsymbol{w}_2 + 1\boldsymbol{w}_3 = \mathbf{0}$. The vectors are **dependent** and lie in a **plane**. For the rows, $\boldsymbol{y} = (1,-2,1)$ and $\boldsymbol{y} = (2,-4,2)$ both give $y_1\boldsymbol{r}_1 + y_2\boldsymbol{r}_2 + y_3\boldsymbol{r}_3 = \mathbf{0}$.
- **6:** Matrix 1: $\text{col}_3 = 2\,\text{col}_2 - \text{col}_1 \implies c = 2(1) - 1 = 1$. Matrix 2: $\text{col}_1 - \text{col}_2 + \text{col}_3 = (1,0,c+1)^T$, so $c = -1$. Matrix 3: $\text{row}_1 = \frac{c}{3}(3,3,3)$ and $\text{row}_3 = (3,3,6)$, which are dependent only when $c = 0$.
- **7:** Because $\boldsymbol{r}_1 \cdot \boldsymbol{x} = \boldsymbol{r}_2 \cdot \boldsymbol{x} = \boldsymbol{r}_3 \cdot \boldsymbol{x} = 0$, every linear combination $(c_1\boldsymbol{r}_1 + c_2\boldsymbol{r}_2 + c_3\boldsymbol{r}_3) \cdot \boldsymbol{x} = 0$.

PROBLEM 1.3.8 – 1.3.14 Difference Matrices: Backward, Cyclic, Forward, and Centered

Question 8: Solve the 4×4 backward difference equation $A\boldsymbol{x} = \boldsymbol{b}$ and find $S = A^{-1}$.
Question 9: Write the cyclic 4×4 difference matrix C and find all solutions to $C\boldsymbol{x} = \mathbf{0}$.
Question 10–11: Solve the 3×3 forward difference equation $\Delta\boldsymbol{z} = \boldsymbol{b}$ to find Δ^{-1}. Show that $(t+1)^2 - t^2 = 2t + 1$ and state the leading derivative term of $(t+1)^n - t^n$.
Question 12–13: Solve the 4×4 centered difference matrix $C\boldsymbol{x} = \boldsymbol{b}$ to find C^{-1}, and show why the 5×5 centered difference matrix is singular.
Question 14: If $(a,b) = k(c,d)$ with $abcd \neq 0$, show that (a,c) is a multiple of (b,d).

Solution:

- **8:** $x_1 = b_1$, $x_2 = b_1 + b_2$, $x_3 = b_1 + b_2 + b_3$, $x_4 = b_1 + b_2 + b_3 + b_4$. Thus $S = A^{-1}$ is the lower-triangular matrix of 1's.
- **9:** $C = \begin{bmatrix}1&0&0&-1\\-1&1&0&0\\0&-1&1&0\\0&0&-1&1\end{bmatrix}$. $C\boldsymbol{x} = \mathbf{0} \implies x_1 = x_2 = x_3 = x_4$, so $\boldsymbol{x} = c(1,1,1,1)^T$.
- **10–11:** $z_3 = -b_3$, $z_2 = -b_2 - b_3$, $z_1 = -b_1 - b_2 - b_3$, so $\Delta^{-1} = \begin{bmatrix}-1&-1&-1\\0&-1&-1\\0&0&-1\end{bmatrix}$. By binomial expansion, $(t+1)^n - t^n = \boldsymbol{nt^{n-1}} + \ldots$
- **12–13:** For 4×4 centered differences ($x_2 = b_1, x_3 - x_1 = b_2, x_4 - x_2 = b_3, -x_3 = b_4$): $x_1 = -b_2 - b_4, x_2 = b_1, x_3 = -b_4, x_4 = b_1 + b_3$, so $C^{-1} = \begin{bmatrix}0&-1&0&-1\\1&0&0&0\\0&0&0&-1\\1&0&1&0\end{bmatrix}$. For 5×5, equations $1, 3, 5$ are $x_2 = b_1$, $x_4 - x_2 = b_3$, $-x_4 = b_5$, which add to $0 = b_1 + b_3 + b_5$.
- **14:** $a = kc$ and $b = kd \implies \frac{a}{b} = \frac{c}{d} = m \implies (a,c) = m(b,d)$. Dependent rows imply dependent columns.

MODULE 2

Orthogonality, Projections, and Least Squares

CHAPTER 3

Orthogonality of the Four Subspaces [Book 4.1]

Syllabus Equations: Orthogonal vectors $v^T w = 0$ and $\|v\|^2 + \|w\|^2 = \|v+w\|^2$ • Orthogonal subspaces $V \perp W$ • Orthogonal complement $V^\perp$ • Fundamental Theorem of Linear Algebra Part 2: $N(A) = C(A^T)^\perp$ in $\mathbb{R}^n$ and $N(A^T) = C(A)^\perp$ in $\mathbb{R}^m$ • Splitting $x = x_r + x_n$ • Fredholm's Alternative.

3.1 Orthogonal Subspaces and the Fundamental Theorem [Book 4.1]

GOAL

Prove that the row space $C(A^T)$ and nullspace $N(A)$ are orthogonal complements in $\mathbb{R}^n$, that the column space $C(A)$ and left nullspace $N(A^T)$ are orthogonal complements in $\mathbb{R}^m$, and that every $x \in \mathbb{R}^n$ splits uniquely into $x = x_r + x_n$.

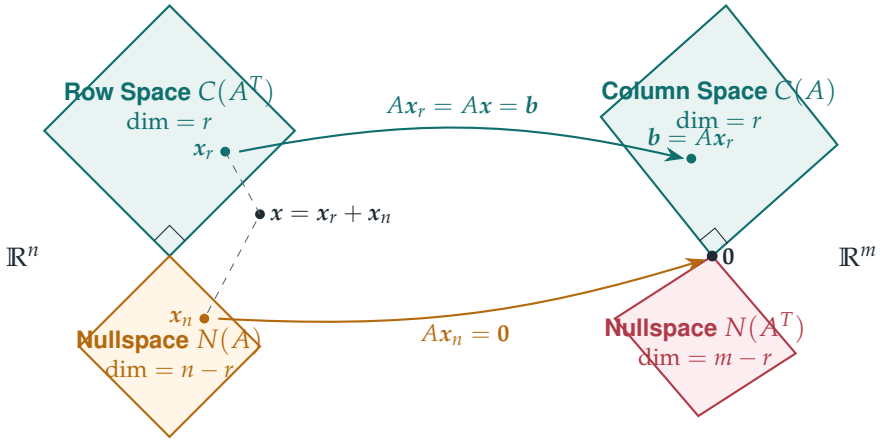


Fig. 3.1 The Big Picture of Linear Algebra (Book Figs. 4.2 & 4.3; SVG source: `svg_four_subspaces.svg`).

Step 1: Define orthogonal subspaces and orthogonal complements. Two subspaces V and W of $\mathbb{R}^n$ are **orthogonal** if every vector $v \in V$ is perpendicular to every vector $w \in W$:

$$v^T w = 0 \quad \text{for all } v \in V \text{ and all } w \in W \tag{3.1}$$

If a vector v belongs to both V and W, then $v^T v = \|v\|^2 = 0$, so $v = 0$. The **orthogonal complement** $V^\perp$ ("V perp") is the subspace of *all* vectors perpendicular to V. Inside $\mathbb{R}^n$, $\dim(V) + \dim(V^\perp) = n$.

Step 2: Prove $N(A) \perp C(A^T)$ in $\mathbb{R}^n$ (Two Proofs). *Proof 1 (Row-by-row from $Ax = 0$):* Write $Ax = 0$ in terms of the rows of A:

$$Ax = \begin{bmatrix} \text{row } 1 \\ \vdots \\ \text{row } m \end{bmatrix} \begin{bmatrix} x \end{bmatrix} = \begin{bmatrix} (\text{row } 1) \cdot x \\ \vdots \\ (\text{row } m) \cdot x \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix} \qquad \textit{Book Eq. (1), p. 196} \tag{3.2}$$

Every row has a zero dot product with $x \in N(A)$. Therefore, every linear combination of the rows also has a zero dot product with x.

Proof 2 (Matrix shorthand): Any vector in the row space $C(A^T)$ has the form $A^T\boldsymbol{y}$. Take its dot product with any $\boldsymbol{x} \in N(A)$ using **T6**:

$$\boldsymbol{x}^T(A^T\boldsymbol{y}) = (A\boldsymbol{x})^T\boldsymbol{y} = \boldsymbol{0}^T\boldsymbol{y} = 0 \qquad \textit{Book Eq. (2), p. 197} \tag{3.3}$$

Because $\dim C(A^T) + \dim N(A) = r + (n-r) = n$, the nullspace $N(A)$ contains *all* vectors perpendicular to the row space: $N(A) = C(A^T)^\perp$.

Step 3: Prove $N(A^T) \perp C(A)$ in $\mathbb{R}^m$. Replace A by A^T. Every vector $\boldsymbol{y}$ in the left nullspace $N(A^T)$ satisfies $A^T\boldsymbol{y} = \boldsymbol{0}$:

$$A^T\boldsymbol{y} = \begin{bmatrix} (\text{column } 1)^T \\ \vdots \\ (\text{column } n)^T \end{bmatrix} \boldsymbol{y} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix} \quad \text{or} \quad (A\boldsymbol{x})^T\boldsymbol{y} = \boldsymbol{x}^T(A^T\boldsymbol{y}) = \boldsymbol{x}^T\boldsymbol{0} = 0 \tag{3.4}$$

Since $\dim C(A) + \dim N(A^T) = r + (m-r) = m$, we have $N(A^T) = C(A)^\perp$ in $\mathbb{R}^m$.

Step 4: Unique Splitting $x = x_r + x_n$ and Invertibility on the Row Space. Combine a basis of r vectors for $C(A^T)$ with a basis of $n-r$ vectors for $N(A)$. Those n vectors are linearly independent (if $\boldsymbol{x}_r + \boldsymbol{x}_n = \boldsymbol{0}$, then $\boldsymbol{x}_r = -\boldsymbol{x}_n$ is in both orthogonal subspaces, so $\boldsymbol{x}_r = \boldsymbol{x}_n = \boldsymbol{0}$). Thus they span $\mathbb{R}^n$, and every $\boldsymbol{x} \in \mathbb{R}^n$ splits uniquely into:

$$\boldsymbol{x} = \boldsymbol{x}_r + \boldsymbol{x}_n \implies A\boldsymbol{x} = A\boldsymbol{x}_r + A\boldsymbol{x}_n = A\boldsymbol{x}_r + \boldsymbol{0} = A\boldsymbol{x}_r \tag{3.5}$$

Furthermore, if $\boldsymbol{x}_r$ and $\boldsymbol{x}_r'$ are both in the row space and $A\boldsymbol{x}_r = A\boldsymbol{x}_r'$, then $A(\boldsymbol{x}_r - \boldsymbol{x}_r') = \boldsymbol{0}$, so $\boldsymbol{x}_r - \boldsymbol{x}_r'$ is in both $N(A)$ and $C(A^T)$, forcing $\boldsymbol{x}_r = \boldsymbol{x}_r'$. Thus A is **one-to-one and invertible** from $C(A^T)$ onto $C(A)$.

BOOK EXAMPLE 4.1.1 – 4.1.3 Orthogonal Planes, Nullspace, and Row Space

Question (as in Book): (1) Explain why the floor V and the vertical meeting line W of two walls are orthogonal subspaces, whereas two perpendicular walls V and W are *not* orthogonal subspaces.

(2) Verify that the rows of $A = \begin{bmatrix} 1 & 3 & 4 \\ 5 & 2 & 7 \end{bmatrix}$ are perpendicular to $\boldsymbol{x} = (1, 1, -1)$ in the nullspace.

BOOK EXAMPLE 4.1.4 – 4.1.5 Invertible $r \times r$ Submatrix and Splitting $x = x_r + x_n$

Question (as in Book): (1) Identify the 2×2 invertible submatrix in the pivot rows and columns of $A = \begin{bmatrix} 3 & 0 & 0 & 0 & 0 \\ 0 & 5 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 2 & 4 & 5 & 6 \\ 1 & 2 & 4 & 5 & 6 \end{bmatrix}$.

(2) For $A = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$, verify the splitting of $\boldsymbol{x} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$ into $\boldsymbol{x}_r + \boldsymbol{x}_n = \begin{bmatrix} 2 \\ 4 \end{bmatrix} + \begin{bmatrix} 2 \\ -1 \end{bmatrix}$.

BOOK EXAMPLE 4.1 A Worked Example: Dimensions of Orthogonal Subspaces

Question (as in Book): Suppose S is a six-dimensional subspace of nine-dimensional space $\mathbb{R}^9$. (a) What are the possible dimensions of subspaces orthogonal to S? (b) What are the possible dimensions of the orthogonal complement $S^\perp$ of S? (c) What is the smallest possible size of a matrix A that has row space S? (d) What is the shape of its nullspace matrix N?

BOOK EXAMPLE 4.1 B Worked Example: Splitting a Vector into Plane P and Line $P^{\perp}$

Question (as in Book): The equation $x - 3y - 4z = 0$ describes a plane P in $\mathbb{R}^3$ (actually a subspace). (a) The plane P is the nullspace $N(A)$ of what 1 by 3 matrix A? (b) Find a basis $\boldsymbol{s}_1, \boldsymbol{s}_2$ of special solutions of $x - 3y - 4z = 0$. (c) Also find a basis for the line $P^{\perp}$ that is perpendicular to P. (d) Split $\boldsymbol{v} = (6, 4, 5)$ into its nullspace component $\boldsymbol{v}_n$ in P and its row space component $\boldsymbol{v}_r$ in $P^{\perp}$.

END-OF-PAGE SOLUTION FOR EXAMPLE 4.1.1 – 4.1.5 and Worked Examples 4.1 A – 4.1 B (Pages 196–202)

- **Ex. 4.1.1–3:** Two walls share a nonzero meeting line; any nonzero vector on that line is in both V and W and cannot be perpendicular to itself ($\dim V + \dim W = 2 + 2 = 4 > 3$). For $A = \begin{bmatrix} 1 & 3 & 4 \\ 5 & 2 & 7 \end{bmatrix}$ and $\boldsymbol{x} = (1, 1, -1)^T$: $\text{row}_1 \cdot \boldsymbol{x} = 1 + 3 - 4 = 0$ and $\text{row}_2 \cdot \boldsymbol{x} = 5 + 2 - 7 = 0$.
- **Ex. 4.1.4–5:** In A, the 2×2 pivot submatrix is $\begin{bmatrix} 3 & 0 \\ 0 & 5 \end{bmatrix}$. In B, pivots are in rows $1, 2$ and columns $1, 3$, giving $\begin{bmatrix} 1 & 3 \\ 1 & 4 \end{bmatrix}$. For $A = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$: $\boldsymbol{x}_r = (2, 4) = 2(1, 2) \in C(A^T)$, $\boldsymbol{x}_n = (2, -1) \in N(A)$ since $1(2) + 2(-1) = 0$, $\boldsymbol{x}_r^T \boldsymbol{x}_n = 4 - 4 = 0$, and $\boldsymbol{x}_r + \boldsymbol{x}_n = (4, 3)$.
- **Ex. 4.1 A:** (a) Possible dimensions are $0, 1, 2, 3$. (b) $\dim(S^{\perp}) = 9 - 6 = 3$. (c) Smallest A is 6×9. (d) Nullspace matrix N is 9×3.
- **Ex. 4.1 B:** (a) $A = \begin{bmatrix} 1 & -3 & -4 \end{bmatrix}$. (b) Free variables y, z give $\boldsymbol{s}_1 = (3, 1, 0)^T$ and $\boldsymbol{s}_2 = (4, 0, 1)^T$. (c) $P^{\perp} = C(A^T)$ is spanned by $\boldsymbol{z} = (1, -3, -4)^T$. (d) Solve $c_1\boldsymbol{s}_1 + c_2\boldsymbol{s}_2 + c_3\boldsymbol{z} = (6, 4, 5)^T \implies c_1 = 1, c_2 = 1, c_3 = -1$. Thus $\boldsymbol{v}_n = \boldsymbol{s}_1 + \boldsymbol{s}_2 = (7, 1, 1)$ and $\boldsymbol{v}_r = -\boldsymbol{z} = (-1, 3, 4)$.

3.2 All Unique Solved Problems: Problem Set 4.1 [Book pp. 202–205]

PROBLEM 4.1.1 – 4.1.4 Constructing Subspaces and Impossible Matrices

Question 1: Construct any 2 by 3 matrix of rank one. Copy Figure 4.2 and put one vector in each subspace (two in the nullspace). Which vectors are orthogonal?
Question 2: Redraw Figure 4.3 for a 3 by 2 matrix of rank $r = 2$. Which subspace is $\boldsymbol{Z}$ (zero vector only)? The nullspace part of any vector $\boldsymbol{x}$ in $\mathbb{R}^2$ is $\boldsymbol{x}_n =$ _____.
Question 3: Construct a matrix with the required property or say why that is impossible: (a) Column space contains $(1,2,-3)^T$ and $(2,-3,5)^T$, nullspace contains $(1,1,1)^T$. (b) Row space contains $(1,2,-3)^T$ and $(2,-3,5)^T$, nullspace contains $(1,1,1)^T$. (c) $A\boldsymbol{x} = (1,1,1)^T$ has a solution and $A^T(1,0,0)^T = (0,0,0)^T$. (d) Every row is orthogonal to every column (A is not the zero matrix). (e) Columns add up to a column of zeros, rows add to a row of 1's.
Question 4: If $AB = 0$ then the columns of B are in the _____ of A. The rows of A are in the _____ of B. Why can't A and B be 3 by 3 matrices of rank 2?

Solution:

1. Choose $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$. Row space: $(1,0,0)^T$; Nullspace: $(0,1,0)^T, (0,0,1)^T$; Column space: $(1,0)^T$; Left nullspace: $(0,1)^T$. $(1,0,0)^T \perp (0,1,0)^T, (0,0,1)^T$ and $(1,0)^T \perp (0,1)^T$.
2. The nullspace $N(A)$ is $\boldsymbol{Z} = \{\boldsymbol{0}\}$. Thus $\boldsymbol{x}_n = \boldsymbol{0}$ for every $\boldsymbol{x} \in \mathbb{R}^2$.
3. (a) Possible: $A = \begin{bmatrix} 1 & 2 & -3 \\ 2 & -3 & 5 \\ -3 & 5 & -2 \end{bmatrix}$ (columns sum to $\boldsymbol{0}$). (b) **Impossible**: $(1,2,-3) \cdot (1,1,1) = 0$, but $(2,-3,5) \cdot (1,1,1) = 4 \neq 0$, violating $C(A^T) \perp N(A)$. (c) **Impossible**: $\boldsymbol{b} = (1,1,1)^T \in C(A)$ and $\boldsymbol{y} = (1,0,0)^T \in N(A^T)$, but $\boldsymbol{b}^T\boldsymbol{y} = 1 \neq 0$. (d) Possible for a non-square matrix where rows and columns live in different spaces, or in $\mathbb{R}^2$ if we mean $A^2 = 0$ (rows of A orthogonal to columns of A): $A = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}$ has (row of A) $\cdot$ (col of A) $= 0$! (Looking at A vs A^T: if every row of A is orthogonal to every column of A, then $A^2 = 0$.) (e) **Impossible**: If columns add to $\boldsymbol{0}$, the sum of all entries in A is 0; if rows add to $(1,\ldots,1)$, the sum of all entries is $n \neq 0$.
4. Columns of B are in the **nullspace** of A; rows of A are in the **left nullspace** of B. If $\text{rank}(A) = 2$ in 3×3, then $\dim N(A) = 3 - 2 = 1$, so $N(A)$ cannot contain a rank-2 column space $C(B)$.

PROBLEM 4.1.5 – 4.1.12 Fredholm's Alternative, A^TA, Symmetric Matrices, and Subspace Splitting

Question 5: (a) If $A\boldsymbol{x} = \boldsymbol{b}$ has a solution and $A^T\boldsymbol{y} = \boldsymbol{0}$, is $(\boldsymbol{y}^T\boldsymbol{x} = 0)$ or $(\boldsymbol{y}^T\boldsymbol{b} = 0)$? (b) If $A^T\boldsymbol{y} = (1,1,1)$ has a solution and $A\boldsymbol{x} = \boldsymbol{0}$, then _____.
Question 6–7: Find y_1, y_2, y_3 so that $y_1(x + 2y + 2z = 5) + y_2(2x + 2y + 3z = 5) + y_3(3x + 4y + 5z = 9)$ gives $0 = 1$. Also do the same for $x_1 - x_2 = 1$, $x_2 - x_3 = 1$, $x_1 - x_3 = 1$.
Question 8: For $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ and $\boldsymbol{x} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, what is $\boldsymbol{x}_r$?
Question 9: If $A^TA\boldsymbol{x} = \boldsymbol{0}$ then $A\boldsymbol{x} = \boldsymbol{0}$. Fill in: $A\boldsymbol{x}$ is in the nullspace of A^T and also in the _____ of A and those spaces are _____.
Question 10: Suppose $A^T = A$. (a) Why is its column space perpendicular to its nullspace?

(b) If $A\boldsymbol{x} = \mathbf{0}$ and $A\boldsymbol{z} = 5\boldsymbol{z}$, which subspaces contain $\boldsymbol{x}$ and $\boldsymbol{z}$?

Question 11–12: Describe the four subspaces for $A = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 3 & 0 \end{bmatrix}$. Find $\boldsymbol{x}_r$ and $\boldsymbol{x}_n$ for $A = \begin{bmatrix} 1 & -1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$ and $\boldsymbol{x} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$.

Solution:

- **5:** (a) $\boldsymbol{y}^T\boldsymbol{b} = \boldsymbol{y}^T A\boldsymbol{x} = (A^T\boldsymbol{y})^T\boldsymbol{x} = 0$. (b) $(1,1,1)\cdot\boldsymbol{x} = x_1 + x_2 + x_3 = 0$.
- **6–7:** Notice $\text{eq}_1 + \text{eq}_2 - \text{eq}_3 \implies 0 = 5+5-9 = 1$, so $\boldsymbol{y} = (1,1,-1)^T \in N(A^T)$. In Problem 7, $\boldsymbol{y} = (-1,-1,1)^T$ gives $0 = -1-1+1 = -1$, so $\boldsymbol{y} = (1,1,-1)^T$ gives $0 = 1$.
- **8:** $A\boldsymbol{x} = A(\boldsymbol{x}_r + \boldsymbol{x}_n) = A\boldsymbol{x}_r$ because $A\boldsymbol{x}_n = \mathbf{0}$. For $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, the row space is spanned by $(1,1)^T$ and nullspace by $(1,-1)^T$. Thus $\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix} + \begin{bmatrix} 1/2 \\ -1/2 \end{bmatrix} \implies \boldsymbol{x}_r = \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}$.
- **9:** $A\boldsymbol{x}$ is in the **column space** of A, and those spaces are **orthogonal**.
- **10:** (a) When $A^T = A$, $C(A) = C(A^T)$, which is orthogonal to $N(A)$. (b) $\boldsymbol{x} \in N(A)$ and $\boldsymbol{z} = \frac{1}{5}A\boldsymbol{z} \in C(A)$, so $\boldsymbol{x}^T\boldsymbol{z} = 0$.
- **11–12:** For $A = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$: $C(A^T) = \text{span}\{(1,2)^T\}$, $N(A) = \text{span}\{(-2,1)^T\}$, $C(A) = \text{span}\{(1,3)^T\}$, $N(A^T) = \text{span}\{(-3,1)^T\}$. For $B = \begin{bmatrix} 1 & 0 \\ 3 & 0 \end{bmatrix}$: $C(B^T) = \text{span}\{(1,0)^T\}$, $N(B) = \text{span}\{(0,1)^T\}$, $C(B) = \text{span}\{(1,3)^T\}$, $N(B^T) = \text{span}\{(-3,1)^T\}$. In Problem 12, row space is $\text{span}\{(1,-1)^T\}$ and nullspace is $\text{span}\{(1,1)^T\}$, so $\boldsymbol{x} = \begin{bmatrix} 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ ($\boldsymbol{x}_r = (1,-1)^T, \boldsymbol{x}_n = (1,1)^T$).

PROBLEM 4.1.13 – 4.1.23 Orthogonal Complements and Intersecting Subspaces

Question 13: Explain why the test for orthogonal subspaces with basis matrices V, W is $V^TW = 0$.

Question 14–15: Find a nonzero vector in the column spaces of both $A = \begin{bmatrix} 1 & 2 \\ 1 & 3 \\ 1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 5 & 4 \\ 6 & 3 \\ 5 & 1 \end{bmatrix}$. What inequality on $p+q$ guarantees that a p-D subspace V and q-D subspace W of $\mathbb{R}^n$ intersect in a nonzero vector?

Question 16–23: (16) Prove $\boldsymbol{y} \in N(A^T) \perp A\boldsymbol{x} \in C(A)$. (17) Find $S^\perp$ in $\mathbb{R}^3$ for $S = \{\mathbf{0}\}$, $S = \text{span}\{(1,1,1)\}$, and $S = \text{span}\{(1,1,1),(1,1,-1)\}$. (18) For $S = \{(1,5,1),(2,2,2)\}$, $S^\perp = N(A)$ for what A? (19–20) Fill in $(L^\perp)^\perp$ and $(V^\perp)^\perp$. (21) Find two vectors spanning $S^\perp$ for $S = \text{span}\{(1,2,2,3),(1,3,3,2)\}$. (22) For $P: x_1 + x_2 + x_3 + x_4 = 0$, find $P^\perp$. (23) If $S \subset V$, prove $S^\perp \supset V^\perp$.

Solution:

- **13:** $V^TW = 0$ means every basis vector of V is orthogonal to every basis vector of W, so all linear combinations are orthogonal.
- **14–15:** Notice column 1 of A is $(1,1,1)^T$ and $\frac{1}{3}(\text{col}_1(B) + \text{col}_2(B)) = (3,3,2)^T$; more directly, in A, $x_1 = x_3$ for all vectors in $C(A)$. In B, $(5,6,5)^T$ already has $x_1 = x_3 = 5$! Indeed, $(5,6,5)^T = 3(1,1,1)^T + 1(2,3,2)^T \in C(A) \cap C(B)$. In $\mathbb{R}^n$, $V \cap W \neq \{\mathbf{0}\}$ is guaranteed whenever $\boldsymbol{p} + \boldsymbol{q} > \boldsymbol{n}$.
- **16–20:** (16) $(A\boldsymbol{x})^T\boldsymbol{y} = \boldsymbol{x}^T(A^T\boldsymbol{y}) = \boldsymbol{x}^T\mathbf{0} = 0$. (17) If $S = \{\mathbf{0}\}$, $S^\perp = \mathbb{R}^3$; if $S = \text{span}\{(1,1,1)\}$, $S^\perp$ is the plane $x + y + z = 0$; if $S = \text{span}\{(1,1,1),(1,1,-1)\}$, $S^\perp = \text{span}\{(1,-1,0)\}$. (18) $A = \begin{bmatrix} 1 & 5 & 1 \\ 2 & 2 & 2 \end{bmatrix}$. (19) $L^\perp$ is the **plane** perpendicular to L; $(L^\perp)^\perp$ is a **line** perpendicular to $L^\perp$, which is L. (20) $V^\perp = \{\mathbf{0}\}$; $(V^\perp)^\perp = \mathbb{R}^4 = V$.
- **21–23:** (21) Solve $A\boldsymbol{x} = \mathbf{0}$ for $A = \begin{bmatrix} 1 & 2 & 2 & 3 \\ 1 & 3 & 3 & 2 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 0 & 5 \\ 0 & 1 & 1 & -1 \end{bmatrix}$: basis for $S^\perp$ is $(0,-1,1,0)^T$ and $(-5,1,0,1)^T$. (22) Basis for $P^\perp$ is $(1,1,1,1)^T$; $A = \begin{bmatrix} 1 & 1 & 1 & 1 \end{bmatrix}$. (23) If $\boldsymbol{w} \in V^\perp$, then $\boldsymbol{w} \perp$ every vector in V, so in particular $\boldsymbol{w} \perp$ every vector in $S \subset V$, meaning $\boldsymbol{w} \in S^\perp$.

PROBLEM 4.1.24 – 4.1.33 Perpendicular Columns/Rows and Challenge Problems

Question 24–26: (24) In $AA^{-1} = I$, column 1 of A^{-1} is orthogonal to which rows of A? (25) Find A^TA if columns of A are orthonormal. (26) Construct a 3×3 matrix A with no zero entries and mutually perpendicular columns; compute A^TA.
Question 27–29: (27) Analyze the lines $3x + y = b_1$ and $6x + 2y = b_2$. (28) Explain why statements (a), (b), (c) are false. (29) Find a matrix with $\boldsymbol{v} = (1,2,3)^T$ in both $C(A^T)$ and $C(A)$, and another with $\boldsymbol{v}$ in both $N(A)$ and $C(A)$. Which pairs of subspaces can $\boldsymbol{v} \neq \mathbf{0}$ not be in?
Question 30–33: (30) If A is 3×4, B is 4×5, and $AB = 0$, prove $\text{rank}(A) + \text{rank}(B) \leq 4$. (31) What is `B = null(N')`? (32–33) Conditions for given vectors to be bases for the four subspaces of a 2×2 or 4×4 matrix.

Solution:

- **24–26:** (24) Orthogonal to rows $2, 3, \ldots, n$ of A. (25) $A^TA = I$. (26) $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{bmatrix}$ has mutually orthogonal columns of length 3, so $A^TA = 9I$.
- **27–29:** (27) The lines are **parallel**; they are the same line if $b_2 = 2b_1$, meaning $(b_1, b_2) \perp (-2,1)$. The nullspace is $3x + y = 0$, containing $(1,-3)^T$. (28) (a) Planes in $\mathbb{R}^3$ intersect in a line, so they cannot be orthogonal subspaces. (b) Dimensions $2 + 2 = 4 < 5$, so they are not complements in $\mathbb{R}^5$. (c) Two lines meeting at $60°$ meet only at $\mathbf{0}$ but are not orthogonal. (29) $A = \boldsymbol{v}\boldsymbol{v}^T$ has $\boldsymbol{v}$ in $C(A^T)$ and $C(A)$. For $\boldsymbol{v}$ in $N(A)$ and $C(A)$, take $A = \boldsymbol{v}\boldsymbol{w}^T$ where $\boldsymbol{w}^T\boldsymbol{v} = 0$, e.g., $\boldsymbol{w} = (1,-2,1)^T$. A nonzero $\boldsymbol{v}$ cannot be in $(C(A^T), N(A))$ or $(C(A), N(A^T))$.
- **30–33:** (30) $C(B) \subset N(A) \implies \text{rank}(B) \leq \dim N(A) = 4 - \text{rank}(A) \implies \text{rank}(A) + \text{rank}(B) \leq 4$. (31) **Row space** $C(A^T)$ of A. (32) $\boldsymbol{r} \perp \boldsymbol{n}$ and $\boldsymbol{c} \perp \boldsymbol{l}$; $A = \boldsymbol{c}\boldsymbol{r}^T$. (33) $\boldsymbol{r}_1, \boldsymbol{r}_2$ independent and orthogonal to independent $\boldsymbol{n}_1, \boldsymbol{n}_2$; $\boldsymbol{c}_1, \boldsymbol{c}_2$ independent and orthogonal

to independent $\boldsymbol{l}_1, \boldsymbol{l}_2$; $A = \boldsymbol{c}_1\boldsymbol{r}_1^T + \boldsymbol{c}_2\boldsymbol{r}_2^T$.

CHAPTER 4

Projections Onto Lines and Subspaces [Book 4.2]

Syllabus Equations: 1D Projection $\hat{x} = \frac{a^T b}{a^T a}$, $\boldsymbol{p} = \boldsymbol{a}\frac{a^T b}{a^T a}$, $P = \frac{aa^T}{a^T a}$ • Normal Equations $A^T(\boldsymbol{b} - A\hat{\boldsymbol{x}}) = \boldsymbol{0} \iff A^T A\hat{\boldsymbol{x}} = A^T\boldsymbol{b}$ • Subspace Projection $\hat{\boldsymbol{x}} = (A^T A)^{-1}A^T\boldsymbol{b}$, $\boldsymbol{p} = A(A^T A)^{-1}A^T\boldsymbol{b}$, $P = A(A^T A)^{-1}A^T$ • Properties $P^T = P$ and $P^2 = P$ • Invertibility of $A^T A$.

4.1 Projection Onto a Line [Book 4.2]

GOAL

Find the point $\boldsymbol{p} = \hat{x}\boldsymbol{a}$ on the line through $\boldsymbol{a}$ closest to $\boldsymbol{b}$, and construct the rank-1 projection matrix P such that $\boldsymbol{p} = P\boldsymbol{b}$.

Step 1: Use orthogonality of the error vector $e = b - \hat{x}a$. The shortest distance from $\boldsymbol{b}$ to the line along $\boldsymbol{a}$ is the perpendicular segment $\boldsymbol{e} = \boldsymbol{b} - \boldsymbol{p} = \boldsymbol{b} - \hat{x}\boldsymbol{a}$. Because $\boldsymbol{e} \perp \boldsymbol{a}$, their dot product is zero:

$$\begin{aligned} \boldsymbol{a} \cdot (\boldsymbol{b} - \hat{x}\boldsymbol{a}) = 0 &\implies \boldsymbol{a} \cdot \boldsymbol{b} - \hat{x}(\boldsymbol{a} \cdot \boldsymbol{a}) = 0 \\ &\implies \hat{x} = \frac{\boldsymbol{a} \cdot \boldsymbol{b}}{\boldsymbol{a} \cdot \boldsymbol{a}} = \frac{\boldsymbol{a}^T\boldsymbol{b}}{\boldsymbol{a}^T\boldsymbol{a}} \end{aligned} \qquad \textit{Book Eq. (2), p. 207} \tag{4.1}$$

Step 2: Write the projection vector p and projection matrix P. Multiply the vector $\boldsymbol{a}$ by the scalar $\hat{x}$, placing $\hat{x}$ on the right of $\boldsymbol{a}$ so $\boldsymbol{b}$ is at the far right:

$$\boldsymbol{p} = \boldsymbol{a}\hat{x} = \boldsymbol{a}\frac{\boldsymbol{a}^T\boldsymbol{b}}{\boldsymbol{a}^T\boldsymbol{a}} = \left(\frac{\boldsymbol{a}\boldsymbol{a}^T}{\boldsymbol{a}^T\boldsymbol{a}}\right)\boldsymbol{b} = P\boldsymbol{b} \implies P = \frac{\boldsymbol{a}\boldsymbol{a}^T}{\boldsymbol{a}^T\boldsymbol{a}} \tag{4.2}$$

Notice that $\boldsymbol{a}\boldsymbol{a}^T$ is a column $(m \times 1)$ times a row $(1 \times m)$, which produces an $m \times m$ matrix of rank 1. Meanwhile, $\boldsymbol{a}^T\boldsymbol{a} = \|\boldsymbol{a}\|^2$ is a scalar.

- **Symmetry:** $P^T = \frac{(aa^T)^T}{a^T a} = \frac{aa^T}{a^T a} = P$.
- **Idempotence ($P^2 = P$):** $P^2 = \frac{(aa^T)(aa^T)}{(a^T a)^2} = \frac{a(a^T a)a^T}{(a^T a)^2} = \frac{aa^T}{a^T a} = P$.
- **Complementary Projection:** $I - P$ projects onto the $(m-1)$-dimensional hyperplane perpendicular to $\boldsymbol{a}$, producing the error vector $\boldsymbol{e} = \boldsymbol{b} - \boldsymbol{p} = (I - P)\boldsymbol{b}$.

4.2 Projection Onto an n-Dimensional Subspace [Book 4.2]

GOAL

Derive the Normal Equations $A^TA\hat{\boldsymbol{x}} = A^T\boldsymbol{b}$, the projection $\boldsymbol{p} = A\hat{\boldsymbol{x}}$, the projection matrix $P = A(A^TA)^{-1}A^T$, and prove that A^TA is invertible whenever A has independent columns.

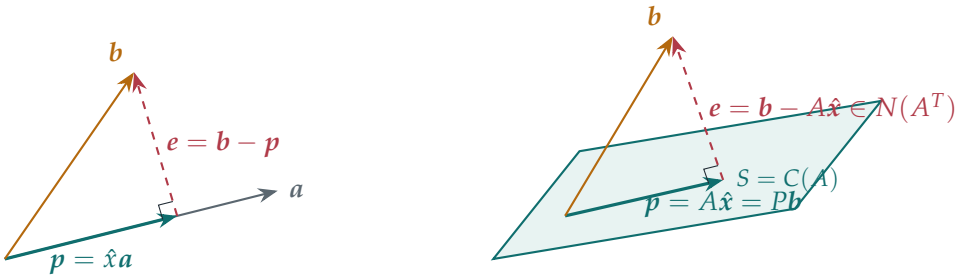


Fig. 4.1 Left: Projection onto a line through $\boldsymbol{a}$. Right: Projection onto the column space $C(A)$ (SVG source: `svg_projection_ls.svg`).

Step 1: Derive the Normal Equations from $e \perp C(A)$. Let $\boldsymbol{a}_1, \ldots, \boldsymbol{a}_n$ be linearly independent columns of an $m \times n$ matrix A. Every vector in the subspace $C(A)$ is a combination $A\boldsymbol{x}$. We seek the closest point $\boldsymbol{p} = A\hat{\boldsymbol{x}}$. The error vector $\boldsymbol{e} = \boldsymbol{b} - A\hat{\boldsymbol{x}}$ must be perpendicular to every column $\boldsymbol{a}_i$:

$$
\begin{aligned}
&\boldsymbol{a}_1^T(\boldsymbol{b} - A\hat{\boldsymbol{x}}) = 0, \ \ldots, \ \boldsymbol{a}_n^T(\boldsymbol{b} - A\hat{\boldsymbol{x}}) = 0 \\
\iff &\begin{bmatrix} - \ \boldsymbol{a}_1^T \ - \\ \vdots \\ - \ \boldsymbol{a}_n^T \ - \end{bmatrix} (\boldsymbol{b} - A\hat{\boldsymbol{x}}) = \boldsymbol{0} \\
\iff &A^T(\boldsymbol{b} - A\hat{\boldsymbol{x}}) = \boldsymbol{0} \qquad e \in N(A^T) \\
\iff &A^TA\hat{\boldsymbol{x}} = A^T\boldsymbol{b} \qquad \textit{Normal Equations}
\end{aligned} \tag{4.3}
$$

Step 2: Solve for $\hat{x}$, p, and the Projection Matrix P. Because A^TA is an invertible $n \times n$ square matrix, multiply Eq. (4.3) by $(A^TA)^{-1}$:

$$\hat{\boldsymbol{x}} = (A^TA)^{-1}A^T\boldsymbol{b} \qquad \textit{best coefficient vector} \tag{4.4}$$

$$\boldsymbol{p} = A\hat{\boldsymbol{x}} = A(A^TA)^{-1}A^T\boldsymbol{b} \qquad \textit{Book Eq. (6): projection in } C(A) \tag{4.5}$$

$$P = A(A^TA)^{-1}A^T \qquad \textit{Book Eq. (7): } m \times m \textit{ projection matrix} \tag{4.6}$$

Check symmetry and idempotence using **T6**:

$$P^T = \left(A(A^TA)^{-1}A^T\right)^T = (A^T)^T\left((A^TA)^{-1}\right)^T A^T = A(A^TA)^{-1}A^T = P \tag{4.7}$$

$$P^2 = A(A^TA)^{-1}\underbrace{\left(A^TA\right)(A^TA)^{-1}}_{I}A^T = A(A^TA)^{-1}A^T = P \tag{4.8}$$

WATCH OUT

Never split $(A^TA)^{-1}$ into $A^{-1}(A^T)^{-1}$ when A is rectangular! A rectangular $m \times n$ matrix A has no inverse A^{-1}. Only when A is square and invertible does $P = AA^{-1}(A^T)^{-1}A^T = I$ (projecting onto the whole space $\mathbb{R}^m$ leaves $\boldsymbol{p} = \boldsymbol{b}$ and $\boldsymbol{e} = \boldsymbol{0}$).

Step 3: Prove that A^TA is invertible if and only if A has independent columns. We show that $N(A^TA) = N(A)$ for every matrix A:

1. If $A\boldsymbol{x} = \boldsymbol{0}$, multiply on the left by A^T to get $A^TA\boldsymbol{x} = \boldsymbol{0}$.
2. Conversely, if $A^TA\boldsymbol{x} = \boldsymbol{0}$, multiply on the left by $\boldsymbol{x}^T$:

$$\boldsymbol{x}^TA^TA\boldsymbol{x} = 0 \implies (A\boldsymbol{x})^T(A\boldsymbol{x}) = 0 \implies \|A\boldsymbol{x}\|^2 = 0 \implies A\boldsymbol{x} = \boldsymbol{0} \tag{4.9}$$

When A has independent columns, $N(A) = \{\boldsymbol{0}\}$, so $N(A^TA) = \{\boldsymbol{0}\}$, which proves that the square $n \times n$ matrix A^TA is invertible!

BOOK EXAMPLE 4.2.1 – 4.2.2 Projection and Projection Matrix Onto a Line

Question (as in Book): Project $\boldsymbol{b} = \begin{bmatrix}1\\1\\1\end{bmatrix}$ onto $\boldsymbol{a} = \begin{bmatrix}1\\2\\2\end{bmatrix}$ to find $\boldsymbol{p} = \hat{x}\boldsymbol{a}$ and $\boldsymbol{e} = \boldsymbol{b} - \boldsymbol{p}$. Then find the projection matrix $P = \frac{\boldsymbol{a}\boldsymbol{a}^T}{\boldsymbol{a}^T\boldsymbol{a}}$ onto the line through $\boldsymbol{a}$ and verify $P\boldsymbol{b} = \boldsymbol{p}$.

BOOK EXAMPLE 4.2.3 Projection Onto a Plane in $\mathbb{R}^3$

Question (as in Book): If $A = \begin{bmatrix}1&0\\1&1\\1&2\end{bmatrix}$ and $\boldsymbol{b} = \begin{bmatrix}6\\0\\0\end{bmatrix}$ find $\hat{\boldsymbol{x}}$ and $\boldsymbol{p}$ and P.

BOOK EXAMPLE 4.2 A Worked Example: Line vs. Plane Projection

Question (as in Book): Project the vector $\boldsymbol{b} = (3,4,4)$ onto the line through $\boldsymbol{a} = (2,2,1)$ and then onto the plane that also contains $\boldsymbol{a}^* = (1,0,0)$. Check that the first error vector $\boldsymbol{b} - \boldsymbol{p}$ is perpendicular to $\boldsymbol{a}$, and the second error vector $\boldsymbol{e}^* = \boldsymbol{b} - \boldsymbol{p}^*$ is also perpendicular to $\boldsymbol{a}^*$. Find the 3 by 3 projection matrix P onto that plane of $\boldsymbol{a}$ and $\boldsymbol{a}^*$. Find a vector whose projection onto the plane is the zero vector.

BOOK EXAMPLE 4.2 B – 4.2 C Worked Examples: Average via Projection and Recursive Least Squares

Question (as in Book): (4.2 B) Project $\boldsymbol{b} = (70,80,120)$ onto $\boldsymbol{a} = (1,1,1)$ to find $\hat{x} = \boldsymbol{a}^T\boldsymbol{b}/\boldsymbol{a}^T\boldsymbol{a}$, and compare with minimizing $E = (x-70)^2 + (x-80)^2 + (x-120)^2$ by calculus. (4.2 C) In *recursive* least squares, a fourth measurement 130 changes $\hat{x}_{\text{old}}$ to $\hat{x}_{\text{new}}$. Compute $\hat{x}_{\text{new}}$ and verify the *update formula* $\hat{x}_{\text{new}} = \hat{x}_{\text{old}} + \frac{1}{4}(130 - \hat{x}_{\text{old}})$.

END-OF-PAGE SOLUTION FOR EXAMPLE 4.2.1 – 4.2.3 and Worked Examples 4.2 A – 4.2 C (Pages 208–214)

- **Ex. 4.2.1–2:** $\boldsymbol{a}^T\boldsymbol{b} = 1+2+2 = 5$, $\boldsymbol{a}^T\boldsymbol{a} = 1+4+4 = 9 \implies \hat{x} = \frac{5}{9}$. $\boldsymbol{p} = (\frac{5}{9}, \frac{10}{9}, \frac{10}{9})^T$, $\boldsymbol{e} = (\frac{4}{9}, -\frac{1}{9}, -\frac{1}{9})^T$, $\boldsymbol{e}^T\boldsymbol{a} = 0$. $P = \frac{1}{9}\begin{bmatrix}1&2&2\\2&4&4\\2&4&4\end{bmatrix}$ and $P\boldsymbol{b} = \frac{1}{9}\begin{bmatrix}5\\10\\10\end{bmatrix}$.

- **Ex. 4.2.3:** $A^TA = \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix}$, $A^T\boldsymbol{b} = \begin{bmatrix} 6 \\ 0 \end{bmatrix}$. Solving $A^TA\hat{\boldsymbol{x}} = A^T\boldsymbol{b}$ gives $\hat{\boldsymbol{x}} = \begin{bmatrix} 5 \\ -3 \end{bmatrix}$. Then $\boldsymbol{p} = A\hat{\boldsymbol{x}} = \begin{bmatrix} 5 \\ 2 \\ -1 \end{bmatrix}$, $\boldsymbol{e} = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$, and $P = A(A^TA)^{-1}A^T = \frac{1}{6}\begin{bmatrix} 5 & 2 & -1 \\ 2 & 2 & 2 \\ -1 & 2 & 5 \end{bmatrix}$.
- **Ex. 4.2 A:** Onto $\boldsymbol{a}$: $\boldsymbol{p} = \frac{18}{9}(2,2,1) = (4,4,2)$; $\boldsymbol{e} = (-1,0,2) \perp (2,2,1)$. For $A = [\boldsymbol{a}\ \boldsymbol{a}^*]$, $A^TA = \begin{bmatrix} 9 & 2 \\ 2 & 1 \end{bmatrix}$, $(A^TA)^{-1} = \frac{1}{5}\begin{bmatrix} 1 & -2 \\ -2 & 9 \end{bmatrix}$, $P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & .8 & .4 \\ 0 & .4 & .2 \end{bmatrix}$. Then $\boldsymbol{p}^* = P\boldsymbol{b} = (3, 4.8, 2.4)$ and $\boldsymbol{e}^* = (0, -0.8, 1.6)$ has $P\boldsymbol{e}^* = \boldsymbol{0}$.
- **Ex. 4.2 B–C:** $\frac{\mathrm{d}E}{\mathrm{d}x} = 2(x-70) + 2(x-80) + 2(x-120) = 0 \implies \hat{x} = \frac{70+80+120}{3} = 90 = \frac{\boldsymbol{a}^T\boldsymbol{b}}{\boldsymbol{a}^T\boldsymbol{a}}$. With $b_4 = 130$, $\hat{x}_{\text{new}} = \frac{270+130}{4} = 100 = 90 + \frac{1}{4}(130 - 90)$.

4.3 All Unique Solved Problems: Problem Set 4.2 [Book pp. 214–217]

PROBLEM 4.2.1 – 4.2.10 Projections Onto Lines and Orthogonal vs. Non-Orthogonal Bases

Question 1–3: Project $\boldsymbol{b}$ onto the line through $\boldsymbol{a}$, check $\boldsymbol{e} \perp \boldsymbol{a}$, and find $P = \boldsymbol{a}\boldsymbol{a}^T/\boldsymbol{a}^T\boldsymbol{a}$ (verify $P^2 = P$): (1a) $\boldsymbol{b} = (1,2,2)^T, \boldsymbol{a} = (1,1,1)^T$; (1b) $\boldsymbol{b} = (1,3,1)^T, \boldsymbol{a} = (-1,-3,-1)^T$; (2a) $\boldsymbol{b} = (\cos\theta, \sin\theta)^T, \boldsymbol{a} = (1,0)^T$; (2b) $\boldsymbol{b} = (1,1)^T, \boldsymbol{a} = (1,-1)^T$.
Question 4: Construct P_1, P_2 for Problem 2. Is $(P_1 + P_2)^2 = P_1 + P_2$?
Question 5–7: Find P_1, P_2, P_3 for the orthogonal basis $\boldsymbol{a}_1 = (-1,2,2)^T$, $\boldsymbol{a}_2 = (2,2,-1)^T$, $\boldsymbol{a}_3 = (2,-1,2)^T$. Compute P_1P_2, project $\boldsymbol{b} = (1,0,0)^T$ onto all three lines, and verify $\boldsymbol{p}_1 + \boldsymbol{p}_2 + \boldsymbol{p}_3 = \boldsymbol{b}$ and $P_1 + P_2 + P_3 = I$.
Question 8–10: Project $\boldsymbol{b} = (1,1)^T$ onto $\boldsymbol{a}_1 = (1,0)^T$ and $\boldsymbol{a}_2 = (1,2)^T$; explain why $\boldsymbol{p}_1 + \boldsymbol{p}_2 \neq \boldsymbol{b}$. Find P onto the plane of $\boldsymbol{a}_1, \boldsymbol{a}_2$. In Problem 10, project $\boldsymbol{a}_1$ onto $\boldsymbol{a}_2$ and back onto $\boldsymbol{a}_1$; is P_1P_2 a projection?

Solution:

- **1–3:** (1a) $\hat{x} = \frac{5}{3}, \boldsymbol{p} = (\frac{5}{3},\frac{5}{3},\frac{5}{3})^T, \boldsymbol{e} = (-\frac{2}{3},\frac{1}{3},\frac{1}{3})^T$, $P = \frac{1}{3}\text{ones}(3,3)$. (1b) $\hat{x} = -1, \boldsymbol{p} = (1,3,1)^T = \boldsymbol{b}, \boldsymbol{e} = \boldsymbol{0}$, $P = \frac{1}{11}\begin{bmatrix}1&3&1\\3&9&3\\1&3&1\end{bmatrix}$. (2a) $\hat{x} = \cos\theta, \boldsymbol{p} = (\cos\theta, 0)^T$, $P_1 = \begin{bmatrix}1&0\\0&0\end{bmatrix}$. (2b) $\hat{x} = 0, \boldsymbol{p} = (0,0)^T$, $P_2 = \frac{1}{2}\begin{bmatrix}1&-1\\-1&1\end{bmatrix}$.
- **4:** $P_1P_2 = \frac{1}{2}\begin{bmatrix}1&-1\\0&0\end{bmatrix} \neq 0$, so $(P_1 + P_2)^2 \neq P_1 + P_2$.
- **5–7:** $P_1 = \frac{1}{9}\begin{bmatrix}1&-2&-2\\-2&4&4\\-2&4&4\end{bmatrix}$, $P_2 = \frac{1}{9}\begin{bmatrix}4&4&-2\\4&4&-2\\-2&-2&1\end{bmatrix}$, $P_3 = \frac{1}{9}\begin{bmatrix}4&-2&4\\-2&1&-2\\4&-2&4\end{bmatrix}$. Since $\boldsymbol{a}_1 \perp \boldsymbol{a}_2$, $P_1P_2 = 0$. For $\boldsymbol{b} = (1,0,0)^T$, $\boldsymbol{p}_1 + \boldsymbol{p}_2 + \boldsymbol{p}_3 = \frac{1}{9}(1,-2,-2)^T + \frac{1}{9}(4,4,-2)^T + \frac{1}{9}(4,-2,4)^T = (1,0,0)^T = \boldsymbol{b}$, and $P_1 + P_2 + P_3 = I$.
- **8–10:** $\boldsymbol{p}_1 = (1,0)^T$ and $\boldsymbol{p}_2 = \frac{3}{5}(1,2)^T \implies \boldsymbol{p}_1 + \boldsymbol{p}_2 = (\frac{8}{5},\frac{6}{5})^T \neq \boldsymbol{b}$ because $\boldsymbol{a}_1 \not\perp \boldsymbol{a}_2$. Onto the plane of $\boldsymbol{a}_1, \boldsymbol{a}_2$ ($\mathbb{R}^2$), $P = I$. In Problem 10, $P_2\boldsymbol{a}_1 = (\frac{1}{5},\frac{2}{5})^T$ and $P_1(P_2\boldsymbol{a}_1) = (\frac{1}{5},0)^T$. $P_1P_2 = \begin{bmatrix}1/5&2/5\\0&0\end{bmatrix}$ is **not** a projection matrix (not symmetric, and $(P_1P_2)^2 = \frac{1}{5}P_1P_2 \neq P_1P_2$).

PROBLEM 4.2.11 – 4.2.20 Subspace Projections, Matrices P, and Planes in $\mathbb{R}^3$

Question 11–12: Project $\boldsymbol{b}$ onto $C(A)$ and find $\boldsymbol{e}$ and P: (a) $A = \begin{bmatrix}1&1\\0&1\\0&0\end{bmatrix}, \boldsymbol{b} = \begin{bmatrix}2\\3\\4\end{bmatrix}$; (b) $A = \begin{bmatrix}1&1\\1&1\\0&1\end{bmatrix}, \boldsymbol{b} = \begin{bmatrix}4\\4\\6\end{bmatrix}$.
Question 13: A is I_4 with its last column removed (4×3). Project $\boldsymbol{b} = (1,2,3,4)^T$ onto $C(A)$ and find P.
Question 14: If $\boldsymbol{b} = 2(\text{col}_1(A))$, what is $\boldsymbol{p}$? Compute $\boldsymbol{p}$ and P for $\boldsymbol{b} = (0,2,4)^T$ and

$A = \begin{bmatrix} 0 & 1 \\ 1 & 2 \\ 2 & 0 \end{bmatrix}$.

Question 15–16: (15) If A is doubled to $2A$, does P or $\hat{\boldsymbol{x}}$ change? (16) What combination of $(1,2,-1)$ and $(1,0,1)$ is closest to $\boldsymbol{b} = (2,1,1)$?

Question 17–20: (17) If $P^2 = P$, show $(I-P)^2 = I - P$; what space does $I - P$ project onto? (18) What does $I - P$ project onto for $\boldsymbol{a} = (1,1)$ and $\boldsymbol{a} = (1,1,1)$? (19–20) Find the projection matrix P onto the plane $x - y - 2z = 0$ both by choosing two basis columns in A and by computing $P = I - \frac{\boldsymbol{e}\boldsymbol{e}^T}{\boldsymbol{e}^T\boldsymbol{e}}$.

Solution:

- **11–12:** (a) $\hat{\boldsymbol{x}} = (2,3)^T$ (Wait: $A\hat{\boldsymbol{x}} = (2,3,0)^T \implies \hat{x}_2 = 3, \hat{x}_1 + 3 = 2 \implies \hat{\boldsymbol{x}} = (-1,3)^T$), $\boldsymbol{p} = (2,3,0)^T$, $\boldsymbol{e} = (0,0,4)^T$, $P_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$. (b) $\hat{\boldsymbol{x}} = (-2,6)^T$, $\boldsymbol{p} = (4,4,6)^T = \boldsymbol{b}$, $\boldsymbol{e} = \boldsymbol{0}$, $P_2 = \begin{bmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.
- **13:** $\boldsymbol{p} = (1,2,3,0)^T$; $P = \mathrm{diag}(1,1,1,0)$ (4×4).
- **14:** Since $\boldsymbol{b} \in C(A)$, $\boldsymbol{p} = \boldsymbol{b} = (0,2,4)^T$. $A^TA = \begin{bmatrix} 5 & 2 \\ 2 & 5 \end{bmatrix} \implies P = \frac{1}{21}\begin{bmatrix} 5 & 10 & -4 \\ 10 & 20 & 2 \\ -4 & 2 & 20 \end{bmatrix} \neq I$.
- **15–16:** (15) $C(2A) = C(A)$, so P is unchanged, while $\hat{\boldsymbol{x}}$ is halved ($\frac{1}{2}\hat{\boldsymbol{x}}$). (16) $A = \begin{bmatrix} 1 & 1 \\ 2 & 0 \\ -1 & 1 \end{bmatrix}$ has orthogonal columns! $\hat{x}_1 = \frac{3}{6} = \frac{1}{2}$, $\hat{x}_2 = \frac{3}{2}$, so $\boldsymbol{p} = \frac{1}{2}(1,2,-1)^T + \frac{3}{2}(1,0,1)^T = (2,1,1)^T = \boldsymbol{b}$.
- **17–20:** (17) $(I-P)^2 = I - 2P + P^2 = I - P$; it projects onto the **left nullspace** $N(A^T) = C(A)^\perp$. (18) (a) Line through $(1,-1)$; (b) Plane $x + y + z = 0$. (19–20) Normal to the plane is $\boldsymbol{e} = (1,-1,-2)^T$ with $\boldsymbol{e}^T\boldsymbol{e} = 6$. Thus $Q = \frac{\boldsymbol{e}\boldsymbol{e}^T}{6} = \frac{1}{6}\begin{bmatrix} 1 & -1 & -2 \\ -1 & 1 & 2 \\ -2 & 2 & 4 \end{bmatrix}$ and $P = I - Q = \frac{1}{6}\begin{bmatrix} 5 & 1 & 2 \\ 1 & 5 & -2 \\ 2 & -2 & 2 \end{bmatrix}$ (identical to $A(A^TA)^{-1}A^T$).

PROBLEM 4.2.21 – 4.2.34 Properties of Projection Matrices and Challenge Problems

Question 21–29: (21–22) Prove $P^2 = P$ and $P^T = P$; $\boldsymbol{Pb}$ projects to **itself**. (23) When A is square invertible, $P = I$ and $\boldsymbol{e} = \boldsymbol{0}$. (24) $N(A^T)$ is **orthogonal** to $C(A)$; if $A^T\boldsymbol{b} = \boldsymbol{0}$, then $\boldsymbol{p} = \boldsymbol{0}$. (25) S is the **column space** of P, so $\mathrm{rank}(P) = n$. (26) If $A^2 = A$ and $\mathrm{rank}(A) = m$, then $A^{-1}A^2 = A^{-1}A \implies A = I$. (27) If $A^TA\boldsymbol{x} = \boldsymbol{0}$, $A\boldsymbol{x}$ is in $N(A^T)$ and $C(A)$, so $A\boldsymbol{x} = \boldsymbol{0}$. (28) Prove $\|\mathrm{col}_2(P)\|^2 = P_{22}$. (29) If B has independent rows, prove BB^T is invertible.

Question 30–34 (Challenge): (30) For $A = \begin{bmatrix} 3 & 6 & 6 \\ 4 & 8 & 8 \end{bmatrix}$, find P_C, P_R and $B = P_C A P_R$. (31) How

to test if $\boldsymbol{p}$ is the projection of $\boldsymbol{b}$ onto $\text{span}\{\boldsymbol{a}_1, \ldots, \boldsymbol{a}_n\}$? (32) Compute P_2P_1 for $A = \begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 0 & 1 \end{bmatrix}$. (33) Condition for $P_1P_2 = P_2P_1$. (34) Prove $\text{rank}(AB) = r$ when A is $m \times r$ (rank r) and B is $r \times n$ (rank r).

Solution:

- **28–29:** Since $P^TP = P^2 = P$, the $(2,2)$ entry of P^TP (which is $\|\text{col}_2(P)\|^2$) equals P_{22}. In Problem 29, apply the A^TA theorem to $A = B^T$ (independent columns): $(B^T)^T(B^T) = BB^T$ is invertible.
- **30:** $C(A) = \text{span}\{(3,4)^T\} \implies P_C = \frac{1}{25}\begin{bmatrix} 9 & 12 \\ 12 & 16 \end{bmatrix}$. $C(A^T) = \text{span}\{(1,2,2)^T\} \implies P_R = \frac{1}{9}\begin{bmatrix} 1 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{bmatrix}$. Since P_C leaves columns of A unchanged and P_R leaves rows of A unchanged, $B = P_CAP_R = A$!
- **31–34:** (31) Test if $(\boldsymbol{b} - \boldsymbol{p})^T\boldsymbol{a}_i = 0$ for all $i = 1, \ldots, n$. (32) Since the 1D space of $\text{col}_1(A)$ lies inside the 2D space $C(A)$, $P_2P_1 = P_1 = \frac{1}{5}\begin{bmatrix} 1 & 2 & 0 \\ 2 & 4 & 0 \\ 0 & 0 & 0 \end{bmatrix}$. (33) $P_1P_2 = P_2P_1$ iff P_1P_2 is symmetric, which holds when $S \cap T^\perp$ and $T \cap S^\perp$ are orthogonal. (34) Both A^TA and BB^T are $r \times r$ invertible, so $(A^TA)(BB^T) = A^T(AB)B^T$ is $r \times r$ invertible (rank r), forcing $\text{rank}(AB) = r$.

CHAPTER 5

Least Squares Approximations [Book 4.3]

Syllabus Equations: Squared error $E = \|Ax - b\|^2 = \|Ax - p\|^2 + \|e\|^2$ • Normal Equations $A^TA\hat{x} = A^Tb$ • Straight-Line Fit $\begin{bmatrix} m & \sum t_i \\ \sum t_i & \sum t_i^2 \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} \sum b_i \\ \sum t_i b_i \end{bmatrix}$ • Orthogonalized Time Shift $T_i = t_i - \hat{t}$ • Parabola Fit $b = C + Dt + Et^2$.

5.1 Three Ways to Minimize $E = \|Ax - b\|^2$ [Book 4.3]

GOAL

Show that minimizing the sum of squared errors $E(x) = \|Ax - b\|^2$ by geometry, by algebra (Pythagoras), and by +2 calculus ($\partial E/\partial x_i = 0$) all lead to the exact same normal equations $A^TA\hat{x} = A^Tb$.

Method 1 (By Geometry) & Method 2 (By Algebra / Pythagoras). Split $b = p + e$, where $p = A\hat{x} \in C(A)$ and $e \in N(A^T)$. For any x, the vector $Ax - p$ lies in $C(A)$ and is perpendicular to e. Apply the +2 Pythagoras Theorem (**T1**) to the right triangle with legs $Ax - p$ and e:

$$E(x) = \|Ax - b\|^2 = \|(Ax - p) - e\|^2 = \|Ax - p\|^2 + \|e\|^2 \qquad \textit{Book Eq. (2), p. 219} \tag{5.1}$$

Because $\|e\|^2$ is fixed, $E(x)$ is minimized when $\|Ax - p\|^2 = 0$, which means $A\hat{x} = p$, or $A^TA\hat{x} = A^Tb$.

Method 3 (By Calculus: Fitting a Straight Line $b = C + Dt$). Given m data points $(t_1, b_1), \ldots, (t_m, b_m)$, the unsolvable system $Ax = b$ has:

$$A = \begin{bmatrix} 1 & t_1 \\ 1 & t_2 \\ \vdots & \vdots \\ 1 & t_m \end{bmatrix}, \quad x = \begin{bmatrix} C \\ D \end{bmatrix}, \quad b = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix} \qquad \textit{Book Eq. (5), p. 221} \tag{5.2}$$

The sum of squared vertical errors is $E(C, D) = \|Ax - b\|^2 = \sum_{i=1}^{m} (C + Dt_i - b_i)^2$. Take partial derivatives with respect to C and D and set both to zero (**T5**):

$$\frac{\partial E}{\partial C} = \sum_{i=1}^{m} 2(C + Dt_i - b_i)(1) = 0 \implies mC + \left(\sum t_i\right) D = \sum b_i \tag{5.3}$$

$$\frac{\partial E}{\partial D} = \sum_{i=1}^{m} 2(C + Dt_i - b_i)(t_i) = 0 \implies \left(\sum t_i\right) C + \left(\sum t_i^2\right) D = \sum t_i b_i \tag{5.4}$$

In matrix form, this 2×2 system is identical to $A^TA\hat{x} = A^Tb$:

$$\boxed{\begin{bmatrix} m & \sum t_i \\ \sum t_i & \sum t_i^2 \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} \sum b_i \\ \sum t_i b_i \end{bmatrix}} \qquad \textit{Book Eqs. (6)–(8), p. 222} \tag{5.5}$$

Step 4: Orthogonal Columns via Shifted Time $T_i = t_i - \hat{t}$. Subtract the average time $\hat{t} = \frac{1}{m}\sum t_i$ so that $T_i = t_i - \hat{t}$ satisfies $\sum T_i = 0$. Then the two columns of A are orthogonal, $A^TA = \text{diag}(m, \sum T_i^2)$ is diagonal, and C, D decouple:

$$b = C + D(t - \hat{t}) \quad \text{with}$$

$$\boxed{C = \frac{b_1 + \cdots + b_m}{m}, \qquad D = \frac{b_1 T_1 + \cdots + b_m T_m}{T_1^2 + \cdots + T_m^2}} \qquad \textit{Book Eq. (9), p. 223} \tag{5.6}$$

Step 5: Fitting a Parabola $b = C + Dt + Et^2$. With three unknown coefficients $\boldsymbol{x} = (C, D, E)^T$, the design matrix A has size $m \times 3$ with rows $\begin{bmatrix} 1 & t_i & t_i^2 \end{bmatrix}$ (Book Eq. (10), p. 223). The best parabola solves the 3×3 normal equations $A^TA\hat{\boldsymbol{x}} = A^T\boldsymbol{b}$.

BOOK EXAMPLE 4.3.1 Closest Straight Line to Three Points

Question (as in Book): Find the closest line $b = C + Dt$ to the points $(0, 6), (1, 0)$, and $(2, 0)$ both by normal equations and by setting $\partial E/\partial C = 0$ and $\partial E/\partial D = 0$.

BOOK EXAMPLE 4.3.2 Orthogonal Columns When Times Add to Zero

Question (as in Book): Suppose $b = 1, 2, 4$ at times $t = -2, 0, 2$. Set up $A^TA\hat{\boldsymbol{x}} = A^T\boldsymbol{b}$ and solve separately for C and D.

BOOK EXAMPLE 4.3.3 Exact Parabola Through Three Points

Question (as in Book): For a parabola $b = C + Dt + Et^2$ to go through the three heights $b = 6, 0, 0$ when $t = 0, 1, 2$, write the equations $A\boldsymbol{x} = \boldsymbol{b}$ and solve for $\boldsymbol{x} = (C, D, E)$.

BOOK EXAMPLE 4.3 A Worked Example: Least Squares (E_2), Minimax (E_∞), and Median (E_1)

Question (as in Book): Start with nine measurements b_1 to b_9, *all zero*, at times $t = 1, \ldots, 9$. The tenth measurement $b_{10} = 40$ is an outlier. Find the best *horizontal line* $y = C$ using: (1) Least squares $E_2 = e_1^2 + \cdots + e_{10}^2$; (2) Least maximum error $E_\infty = |e_{\max}|$; (3) Least sum of errors $E_1 = |e_1| + \cdots + |e_{10}|$. Then find the least squares straight line $C + Dt$ through those ten points.

BOOK EXAMPLE 4.3 B Worked Example: Symmetric Parabola Fit

Question (as in Book): Find the parabola $C + Dt + Et^2$ that comes closest (least squares error) to the values $\boldsymbol{b} = (0, 0, 1, 0, 0)$ at the times $t = -2, -1, 0, 1, 2$.

END-OF-PAGE SOLUTION FOR EXAMPLE 4.3.1 – 4.3.3 and Worked Examples 4.3 A – 4.3 B (Pages 218–226)

- **Ex. 4.3.1:** $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, \boldsymbol{b} = \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix}$. Both $\partial E/\partial C = 0, \partial E/\partial D = 0$ and $A^TA\hat{\boldsymbol{x}} = A^T\boldsymbol{b}$ give $\begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix}\begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} 6 \\ 0 \end{bmatrix} \implies C = 5, D = -3$. Best line: $b = 5 - 3t$, with heights $\boldsymbol{p} = (5, 2, -1)$ and errors $\boldsymbol{e} = (1, -2, 1)$.

- **Ex. 4.3.2:** $A = \begin{bmatrix} 1 & -2 \\ 1 & 0 \\ 1 & 2 \end{bmatrix} \implies A^TA = \begin{bmatrix} 3 & 0 \\ 0 & 8 \end{bmatrix}$ and $A^T\boldsymbol{b} = \begin{bmatrix} 7 \\ 6 \end{bmatrix} \implies C = \frac{7}{3}, D = \frac{6}{8} = \frac{3}{4}$.
- **Ex. 4.3.3:** $C = 6$, $C + D + E = 0$, $C + 2D + 4E = 0 \implies (C, D, E) = (6, -9, 3)$, so $b = 6 - 9t + 3t^2$ ($\boldsymbol{e} = \boldsymbol{0}$).
- **Ex. 4.3 A:** (1) $10C = 40 \implies C = 4$ (mean). (2) $C = 20$ (halfway between 0 and 40). (3) $C = 0$ (median). For the line $C + Dt$: $A^TA = \begin{bmatrix} 10 & 55 \\ 55 & 385 \end{bmatrix}$, $A^T\boldsymbol{b} = \begin{bmatrix} 40 \\ 400 \end{bmatrix} \implies C = -8, D = \frac{24}{11}$.
- **Ex. 4.3 B:** $A^TA = \begin{bmatrix} 5 & 0 & 10 \\ 0 & 10 & 0 \\ 10 & 0 & 34 \end{bmatrix}$ and $A^T\boldsymbol{b} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \implies D = 0, C = \frac{34}{70}, E = -\frac{10}{70}$.

5.2 All Unique Solved Problems: Problem Set 4.3 [Book pp. 226–229]

PROBLEM 4.3.1 – 4.3.11 Four-Point Line, Parabola, and Cubic Fits ($b = (0,8,8,20)$ at $t = 0,1,3,4$)

Question 1–4: With $\boldsymbol{b} = (0,8,8,20)$ at $t = 0,1,3,4$: (1) Set up and solve $A^TA\hat{\boldsymbol{x}} = A^T\boldsymbol{b}$ for the best line $C + Dt$, find p_i, e_i, and $E = \sum e_i^2$. (2) Write $A\boldsymbol{x} = \boldsymbol{b}$ and solve $A\hat{\boldsymbol{x}} = \boldsymbol{p} = (1,5,13,17)^T$. (3) Check $\boldsymbol{e} = (-1,3,-5,3)^T \perp$ columns of A and find $\|\boldsymbol{e}\|$. (4) Derive $A^TA\hat{\boldsymbol{x}} = A^T\boldsymbol{b}$ from $\partial E/\partial C = 0$ and $\partial E/\partial D = 0$.
Question 5–8: (5–6) Project $\boldsymbol{b}$ onto $\boldsymbol{a}_1 = (1,1,1,1)^T$ to find the best horizontal line $y = C$, error $\boldsymbol{e}$, and $\|\boldsymbol{e}\|$. (7–8) Project $\boldsymbol{b}$ onto $\boldsymbol{a}_2 = (0,1,3,4)^T$ to find the best line $b = Dt$ through the origin. Why do C and D here differ from (C,D) in Problem 1?
Question 9–11: (9) Write $A\boldsymbol{x} = \boldsymbol{b}$ and $A^TA\hat{\boldsymbol{x}} = A^T\boldsymbol{b}$ for the best parabola $C + Dt + Et^2$. (10) Solve $A\boldsymbol{x} = \boldsymbol{b}$ for the exact cubic $C + Dt + Et^2 + Ft^3$; what are $\boldsymbol{p}$ and $\boldsymbol{e}$? (11) Verify that the best line passes through $(\hat{t},\hat{b}) = (2,9)$ and explain why $C + D\hat{t} = \hat{b}$ is the first normal equation.

Solution:

- **1–4:** $A = \begin{bmatrix}1&0\\1&1\\1&3\\1&4\end{bmatrix}$, $A^TA = \begin{bmatrix}4&8\\8&26\end{bmatrix}$, $A^T\boldsymbol{b} = \begin{bmatrix}36\\112\end{bmatrix} \implies \hat{\boldsymbol{x}} = \begin{bmatrix}C\\D\end{bmatrix} = \begin{bmatrix}1\\4\end{bmatrix}$ (best line $b = 1 + 4t$). Heights $\boldsymbol{p} = (1,5,13,17)^T$, errors $\boldsymbol{e} = (-1,3,-5,3)^T$, minimum $E = 1+9+25+9 = 44$, and shortest distance $\|\boldsymbol{e}\| = \sqrt{44} = 2\sqrt{11}$.
- **5–8:** (5–6) $\hat{x} = C = \frac{\boldsymbol{a}_1^T\boldsymbol{b}}{\boldsymbol{a}_1^T\boldsymbol{a}_1} = \frac{36}{4} = 9$. $\boldsymbol{p} = (9,9,9,9)^T$, $\boldsymbol{e} = (-9,-1,-1,11)^T$, $\|\boldsymbol{e}\| = \sqrt{81+1+1+121} = \sqrt{204}$. (7–8) $\hat{x} = D = \frac{\boldsymbol{a}_2^T\boldsymbol{b}}{\boldsymbol{a}_2^T\boldsymbol{a}_2} = \frac{112}{26} = \frac{56}{13}$. They differ from $(1,4)$ because $\boldsymbol{a}_1 = (1,1,1,1)$ and $\boldsymbol{a}_2 = (0,1,3,4)$ are **not** perpendicular ($\boldsymbol{a}_1^T\boldsymbol{a}_2 = 8 \neq 0$).
- **9–11:** (9) $A = \begin{bmatrix}1&0&0\\1&1&1\\1&3&9\\1&4&16\end{bmatrix}$, $A^TA = \begin{bmatrix}4&8&26\\8&26&92\\26&92&338\end{bmatrix}$, $A^T\boldsymbol{b} = \begin{bmatrix}36\\112\\400\end{bmatrix}$ (projects $\boldsymbol{b}$ onto a 3D subspace of $\mathbb{R}^4$). (10) The 4×4 Vandermonde system gives $C = 0, D = \frac{47}{3}, E = -9, F = \frac{4}{3}$, with $\boldsymbol{p} = \boldsymbol{b}$ and $\boldsymbol{e} = \boldsymbol{0}$. (11) At $t = 2$, $C + Dt = 1 + 4(2) = 9 = \hat{b}$. Dividing the first normal equation $mC + (\sum t_i)D = \sum b_i$ by m gives $C + D\hat{t} = \hat{b}$.

PROBLEM 4.3.12 – 4.3.16 Statistics of Least Squares: Mean, Variance, Covariance, and Recursive Update

Question 12: Project $\boldsymbol{b} = (b_1,\ldots,b_m)^T$ onto $\boldsymbol{a} = (1,\ldots,1)^T$: (a) Show $\hat{x}$ is the mean. (b) Find $\boldsymbol{e}$, $\|\boldsymbol{e}\|^2$, $\|\boldsymbol{e}\|$. (c) For $\boldsymbol{b} = (1,2,6)^T$, check $\boldsymbol{p} = (3,3,3)^T \perp \boldsymbol{e}$ and find P.
Question 13–15: (13) Multiply $\boldsymbol{e} = \boldsymbol{b} - A\boldsymbol{x}$ by $(A^TA)^{-1}A^T$ to show $\mathrm{E}[\hat{\boldsymbol{x}} - \boldsymbol{x}] = \boldsymbol{0}$. (14) Show $\mathrm{E}[(\hat{\boldsymbol{x}} - \boldsymbol{x})(\hat{\boldsymbol{x}} - \boldsymbol{x})^T] = \sigma^2(A^TA)^{-1}$. (15) For 4 heart-rate readings ($A = (1,1,1,1)^T$), what is $\sigma^2(A^TA)^{-1}$?
Question 16: Fill in the recursive average formula: $\hat{x}_{10} = \frac{1}{10}b_{10} + _____\,\hat{x}_9$.

Solution:

- **12:** (a) $m\hat{x} = \sum b_i \implies \hat{x} = \frac{1}{m}\sum b_i$. (b) $\boldsymbol{e} = (b_1 - \hat{x},\ldots,b_m - \hat{x})^T$, $\|\boldsymbol{e}\|^2 = \sum(b_i - \hat{x})^2$. (c) For $\boldsymbol{b} = (1,2,6)^T$, $\hat{x} = 3$, $\boldsymbol{p} = (3,3,3)^T$, $\boldsymbol{e} = (-2,-1,3)^T$, $\boldsymbol{p}^T\boldsymbol{e} = 0$, and $P = \frac{1}{3}\mathrm{ones}(3,3)$.
- **13–15:** $(A^TA)^{-1}A^T\boldsymbol{e} = (A^TA)^{-1}A^T\boldsymbol{b} - (A^TA)^{-1}A^TA\boldsymbol{x} = \hat{\boldsymbol{x}} - \boldsymbol{x}$. Since $\mathrm{E}[\boldsymbol{e}] = \boldsymbol{0}$, $\mathrm{E}[\hat{\boldsymbol{x}} -$

$\boldsymbol{x}] = \mathbf{0}$. Next, $\mathrm{E}[(\hat{\boldsymbol{x}} - \boldsymbol{x})(\hat{\boldsymbol{x}} - \boldsymbol{x})^T] = (A^TA)^{-1}A^T(\sigma^2 I)A(A^TA)^{-1} = \sigma^2(A^TA)^{-1}$. For $m = 4$ ones, $A^TA = 4 \implies \sigma^2(A^TA)^{-1} = \boldsymbol{\sigma^2/4}$.

- **16:** $\hat{x}_{10} = \frac{1}{10}b_{10} + \frac{\mathbf{9}}{\mathbf{10}}\hat{x}_9$.

PROBLEM 4.3.17 – 4.3.29 Practice with $\hat{x}, p, e$ and Challenge Problems

Question 17–21: Fit $b = C + Dt$ to $b = 7, 7, 21$ at $t = -1, 1, 2$: (17) Find $\hat{\boldsymbol{x}} = (C, D)$. (18) Find $\boldsymbol{p}$ and $\boldsymbol{e} = (2, -6, 4)^T$; why is $P\boldsymbol{e} = \mathbf{0}$? (19) If $\boldsymbol{b} = (2, -6, 4)^T$, find $\hat{\boldsymbol{x}}$ and $\boldsymbol{p}$. (20) If $\boldsymbol{b} = (5, 13, 17)^T$, find $\hat{\boldsymbol{x}}$ and $\boldsymbol{e}$. (21) Which subspaces contain $\boldsymbol{e}, \boldsymbol{p}, \hat{\boldsymbol{x}}$, and what is $N(A)$?
Question 22–24: (22) Best line for $b = 4, 2, -1, 0, 0$ at $t = -2, -1, 0, 1, 2$. (23) Which of $\boldsymbol{b}, \boldsymbol{p}, \boldsymbol{e}, \hat{\boldsymbol{x}}$ is $\boldsymbol{e}$ orthogonal to? Prove $\|\boldsymbol{e}\|^2 = \boldsymbol{e}^T\boldsymbol{b} = \boldsymbol{b}^T\boldsymbol{b} - \boldsymbol{p}^T\boldsymbol{b}$. (24) Derivatives of $\|A\boldsymbol{x} - \boldsymbol{b}\|^2$ are zero when _____.
Question 25–29 (Challenge): (25) Condition on $(t_1, b_1), (t_2, b_2), (t_3, b_3)$ to lie on a line. (26) Best plane $C + Dx + Ey$ to $b = (0, 1, 3, 4)$ at $(1, 0), (0, 1), (-1, 0), (0, -1)$. (27) Shortest distance between lines $P = (x, x, x)$ and $Q = (y, 3y, -1)$. (28) Projection onto $C(A)$ when columns of A are dependent. (29) Condition for a unique hyperplane through $\mathbf{0}, \boldsymbol{a}_1, \ldots, \boldsymbol{a}_{n-1}$ in $\mathbb{R}^n$.

Solution:

- **17–21:** (17) $A = \begin{bmatrix} 1 & -1 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, \boldsymbol{b} = \begin{bmatrix} 7 \\ 7 \\ 21 \end{bmatrix} \implies A^TA = \begin{bmatrix} 3 & 2 \\ 2 & 6 \end{bmatrix}, A^T\boldsymbol{b} = \begin{bmatrix} 35 \\ 42 \end{bmatrix} \implies \hat{\boldsymbol{x}} = \begin{bmatrix} 9 \\ 4 \end{bmatrix}$ ($b = 9 + 4t$). (18) $\boldsymbol{p} = (5, 13, 17)^T$, $\boldsymbol{e} = (2, -6, 4)^T$; $P\boldsymbol{e} = \mathbf{0}$ because $\boldsymbol{e} \in N(A^T)$. (19) $\hat{\boldsymbol{x}} = (0, 0)^T, \boldsymbol{p} = \mathbf{0}$ because $\boldsymbol{b}$ is perpendicular to the **column space of** A. (20) $\hat{\boldsymbol{x}} = (9, 4)^T, \boldsymbol{e} = \mathbf{0}$ because $\boldsymbol{b}$ is **in the column space of** A. (21) $\boldsymbol{e} \in N(A^T)$, $\boldsymbol{p} \in C(A)$, $\hat{\boldsymbol{x}} \in C(A^T) = \mathbb{R}^2$, and $N(A) = \{\mathbf{0}\}$.
- **22–24:** (22) $A^TA = \mathrm{diag}(5, 10)$, $A^T\boldsymbol{b} = (5, -10)^T \implies C = 1, D = -1$ ($b = 1 - t$). (23) $\boldsymbol{e} \perp \boldsymbol{p}$; $\|\boldsymbol{e}\|^2 = \boldsymbol{e}^T(\boldsymbol{b} - \boldsymbol{p}) = \boldsymbol{e}^T\boldsymbol{b} = (\boldsymbol{b} - \boldsymbol{p})^T\boldsymbol{b} = \boldsymbol{b}^T\boldsymbol{b} - \boldsymbol{p}^T\boldsymbol{b}$. (24) $A^TA\hat{\boldsymbol{x}} = A^T\boldsymbol{b}$.
- **25–29:** (25) Equal slopes: $\frac{b_2 - b_1}{t_2 - t_1} = \frac{b_3 - b_1}{t_3 - t_1}$, or $\det \begin{bmatrix} 1 & t_1 & b_1 \\ 1 & t_2 & b_2 \\ 1 & t_3 & b_3 \end{bmatrix} = 0$. (26) $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & -1 & 0 \\ 1 & 0 & -1 \end{bmatrix}$ has orthogonal columns: $A^TA = \mathrm{diag}(4, 2, 2)$ and $A^T\boldsymbol{b} = (8, -3, -3)^T \implies C = 2$ (the average of $\boldsymbol{b}$), $D = -3/2, E = -3/2$. (27) Minimize $(x - y)^2 + (x - 3y)^2 + (x + 1)^2 \implies x = -\frac{5}{9}, y = -\frac{2}{9}$; connecting segment is perpendicular to **both lines**. (28) Choose the columns of B to be a maximal linearly independent subset (a basis) of the columns of A. (29) The vectors $\boldsymbol{a}_1, \ldots, \boldsymbol{a}_{n-1}$ must be **linearly independent**.

MODULE 3

Orthonormal Bases, Gram-Schmidt, and Positive Definite Matrices

CHAPTER 6

Orthogonal Bases and Gram-Schmidt [Book 4.4]

Syllabus Equations: Orthonormal vectors $\boldsymbol{q}_i^T\boldsymbol{q}_j = \delta_{ij} \iff Q^TQ = I$ • Orthogonal matrix ($m = n$): $Q^T = Q^{-1}$ • Reflections $Q = I - 2\boldsymbol{u}\boldsymbol{u}^T$ • Length preservation $\|Q\boldsymbol{x}\| = \|\boldsymbol{x}\|$ • Orthonormal Projection $\hat{\boldsymbol{x}} = Q^T\boldsymbol{b}$, $\boldsymbol{p} = QQ^T\boldsymbol{b}$, $P = QQ^T$ • Gram-Schmidt Process $(\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C})$ • QR Factorization $A = QR$ $(R = Q^TA)$ • Least Squares $R\hat{\boldsymbol{x}} = Q^T\boldsymbol{b}$.

6.1 Orthonormal Columns ($Q^TQ = I$) and Orthogonal Matrices [Book 4.4]

GOAL

Show that orthonormal columns $\boldsymbol{q}_1, \ldots, \boldsymbol{q}_n$ satisfy $Q^TQ = I$, prove that Q preserves lengths and dot products, and simplify projection formulas when $A = Q$.

Step 1: Define orthonormal vectors and $Q^TQ = I$. Vectors $\boldsymbol{q}_1, \ldots, \boldsymbol{q}_n$ are **orthonormal** if they are orthogonal unit vectors:

$$\boldsymbol{q}_i^T\boldsymbol{q}_j = \begin{cases} 0 & \text{when } i \neq j \quad \text{(orthogonal vectors)} \\ 1 & \text{when } i = j \quad \text{(unit vectors: } \|\boldsymbol{q}_i\| = 1\text{)} \end{cases}$$
$$\iff Q^TQ = I_n \qquad \textit{Book Eq. (1), p. 230} \tag{6.1}$$

When Q is square ($n \times n$), $Q^TQ = I$ means $Q^T = Q^{-1}$ and also $QQ^T = I$. A square matrix with orthonormal columns is called an **orthogonal matrix** (e.g., rotations, permutations, and reflections $Q = I - 2\boldsymbol{u}\boldsymbol{u}^T$ with $\|\boldsymbol{u}\| = 1$).

Step 2: Prove that Q preserves lengths and dot products. For any $m \times n$ matrix Q with orthonormal columns ($Q^TQ = I$):

$$\|Q\boldsymbol{x}\|^2 = (Q\boldsymbol{x})^T(Q\boldsymbol{x}) = \boldsymbol{x}^TQ^TQ\boldsymbol{x} = \boldsymbol{x}^T\boldsymbol{x} = \|\boldsymbol{x}\|^2$$
$$\implies \boxed{\|Q\boldsymbol{x}\| = \|\boldsymbol{x}\|} \qquad \textit{Book Eq. (3), p. 232} \tag{6.2}$$

$$(Q\boldsymbol{x})^T(Q\boldsymbol{y}) = \boldsymbol{x}^TQ^TQ\boldsymbol{y} = \boldsymbol{x}^T\boldsymbol{y} \qquad \textit{dot products preserved} \tag{6.3}$$

Step 3: Projections and Least Squares with Orthonormal Columns (Q replaces A). Substitute $A = Q$ and $Q^TQ = I$ into the Chapter 3 projection formulas:

$$\hat{\boldsymbol{x}} = (Q^TQ)^{-1}Q^T\boldsymbol{b} = I^{-1}Q^T\boldsymbol{b} = Q^T\boldsymbol{b} \qquad \textit{Book Eq. (4), no matrix inversion!} \tag{6.4}$$

$$\boldsymbol{p} = Q\hat{\boldsymbol{x}} = QQ^T\boldsymbol{b} = \boldsymbol{q}_1(\boldsymbol{q}_1^T\boldsymbol{b}) + \boldsymbol{q}_2(\boldsymbol{q}_2^T\boldsymbol{b}) + \cdots + \boldsymbol{q}_n(\boldsymbol{q}_n^T\boldsymbol{b}) \qquad \textit{Book Eq. (5), sum of 1D projections} \tag{6.5}$$

$$P = QQ^T \qquad \textit{projection matrix onto } C(Q) \tag{6.6}$$

If Q is square ($m = n$), then $P = QQ^T = I$, and every $\boldsymbol{b} \in \mathbb{R}^n$ equals the sum of its 1D components along the basis vectors: $\boldsymbol{b} = \boldsymbol{q}_1(\boldsymbol{q}_1^T\boldsymbol{b}) + \cdots + \boldsymbol{q}_n(\boldsymbol{q}_n^T\boldsymbol{b})$ (Book Eq. (6), p. 233).

6.2 The Gram-Schmidt Process and $A = QR$ [Book 4.4]

GOAL

Convert linearly independent vectors $\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}$ into orthogonal vectors $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}$ and orthonormal vectors $\boldsymbol{q}_1, \boldsymbol{q}_2, \boldsymbol{q}_3$, and derive the matrix factorization $A = QR$.

Step 1: Subtract projections in already-set directions.

1. **First direction:** Keep $\boldsymbol{A} = \boldsymbol{a}$.
2. **Second direction:** Subtract from $\boldsymbol{b}$ its projection along $\boldsymbol{A}$ (the error vector is perpendicular to $\boldsymbol{A}$):

$$\boldsymbol{B} = \boldsymbol{b} - \frac{\boldsymbol{A}^T\boldsymbol{b}}{\boldsymbol{A}^T\boldsymbol{A}}\boldsymbol{A} \qquad \textit{Book Eq. (7), p. 234: } \boldsymbol{A}^T\boldsymbol{B} = 0 \tag{6.7}$$

3. **Third direction:** Subtract from $\boldsymbol{c}$ its projections along both $\boldsymbol{A}$ and $\boldsymbol{B}$:

$$\boldsymbol{C} = \boldsymbol{c} - \frac{\boldsymbol{A}^T\boldsymbol{c}}{\boldsymbol{A}^T\boldsymbol{A}}\boldsymbol{A} - \frac{\boldsymbol{B}^T\boldsymbol{c}}{\boldsymbol{B}^T\boldsymbol{B}}\boldsymbol{B} \qquad \textit{Book Eq. (8), p. 234: } \boldsymbol{C} \perp \boldsymbol{A}, \boldsymbol{B} \tag{6.8}$$

4. **Normalize to unit vectors:** Divide each orthogonal vector by its length:

$$\boldsymbol{q}_1 = \frac{\boldsymbol{A}}{\|\boldsymbol{A}\|}, \qquad \boldsymbol{q}_2 = \frac{\boldsymbol{B}}{\|\boldsymbol{B}\|}, \qquad \boldsymbol{q}_3 = \frac{\boldsymbol{C}}{\|\boldsymbol{C}\|} \tag{6.9}$$

Step 2: Derive the Upper-Triangular Factorization $A = QR$. Because $\boldsymbol{a}$ depends only on $\boldsymbol{q}_1$, $\boldsymbol{b}$ depends only on $\boldsymbol{q}_1, \boldsymbol{q}_2$, and $\boldsymbol{c}$ depends on $\boldsymbol{q}_1, \boldsymbol{q}_2, \boldsymbol{q}_3$, each original column is orthogonal to later $\boldsymbol{q}$'s ($\boldsymbol{q}_i^T\boldsymbol{a}_j = 0$ for $i > j$). Multiply $A = QR$ on the left by Q^T (using $Q^TQ = I$) to find $R = Q^TA$:

$$\begin{bmatrix} \boldsymbol{a} & \boldsymbol{b} & \boldsymbol{c} \end{bmatrix} = \begin{bmatrix} \boldsymbol{q}_1 & \boldsymbol{q}_2 & \boldsymbol{q}_3 \end{bmatrix} \begin{bmatrix} \boldsymbol{q}_1^T\boldsymbol{a} & \boldsymbol{q}_1^T\boldsymbol{b} & \boldsymbol{q}_1^T\boldsymbol{c} \\ 0 & \boldsymbol{q}_2^T\boldsymbol{b} & \boldsymbol{q}_2^T\boldsymbol{c} \\ 0 & 0 & \boldsymbol{q}_3^T\boldsymbol{c} \end{bmatrix}$$

$$= \boxed{QR \quad (R = Q^TA)} \qquad \textit{Book Eq. (9), p. 236} \tag{6.10}$$

With $A = QR$, we have $A^TA = R^TQ^TQR = R^TR$, so the normal equations $A^TA\hat{\boldsymbol{x}} = A^T\boldsymbol{b}$ become $R^TR\hat{\boldsymbol{x}} = R^TQ^T\boldsymbol{b}$, which simplifies to back substitution on the upper-triangular system:

$$\boxed{R\hat{\boldsymbol{x}} = Q^T\boldsymbol{b} \quad \text{or} \quad \hat{\boldsymbol{x}} = R^{-1}Q^T\boldsymbol{b}} \qquad \textit{Book Eq. (10), p. 236} \tag{6.11}$$

BOOK EXAMPLE 4.4.1 – 4.4.3 Rotations, Permutations, and Reflections $Q = I - 2uu^T$

Question (as in Book): Verify $Q^TQ = I$ for (1) Rotation $Q = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$; (2) Permutation matrices; (3) Reflection $Q = I - 2\boldsymbol{u}\boldsymbol{u}^T$ where $\boldsymbol{u}^T\boldsymbol{u} = 1$, and compute Q_1, Q_2 for $\boldsymbol{u} = (1, 0)^T$ and $\boldsymbol{u} = (1/\sqrt{2}, -1/\sqrt{2})^T$.

BOOK EXAMPLE 4.4.4 Projecting Onto Orthonormal Columns and Reconstructing b

Question (as in Book): For $Q = \frac{1}{3}\begin{bmatrix} -1 & 2 & 2 \\ 2 & -1 & 2 \\ 2 & 2 & -1 \end{bmatrix}$, compute the separate projections p_1, p_2, p_3 of $\boldsymbol{b} = (0,0,1)^T$ onto $\boldsymbol{q}_1, \boldsymbol{q}_2, \boldsymbol{q}_3$ and verify $p_1 + p_2 + p_3 = \boldsymbol{b}$.

BOOK EXAMPLE 4.4.5 Gram-Schmidt and $A = QR$ on Three Vectors

Question (as in Book): Apply Gram-Schmidt to $\boldsymbol{a} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$, $\boldsymbol{b} = \begin{bmatrix} 2 \\ 0 \\ -2 \end{bmatrix}$, $\boldsymbol{c} = \begin{bmatrix} 3 \\ -3 \\ 3 \end{bmatrix}$ to find $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}$, the orthonormal vectors $\boldsymbol{q}_1, \boldsymbol{q}_2, \boldsymbol{q}_3$, and the factorization $A = QR$.

BOOK EXAMPLE 4.4 A Worked Example: Hadamard Matrices and Orthogonal Projections

Question (as in Book): Complete the 4×4 Hadamard matrix H_4 with ± 1 entries so its columns are orthogonal, turn H_4 into an orthogonal matrix Q_4, find $H_8^T H_8$, and project $\boldsymbol{b} = (6,0,0,2)^T$ onto the span of the first two columns of H_4.

END-OF-PAGE SOLUTION FOR EXAMPLE 4.4.1 – 4.4.5 and Worked Example 4.4 A (Pages 231–239)

- **Ex. 4.4.1–3:** For $Q = I - 2\boldsymbol{u}\boldsymbol{u}^T$, $Q^TQ = I - 4\boldsymbol{u}\boldsymbol{u}^T + 4\boldsymbol{u}(\boldsymbol{u}^T\boldsymbol{u})\boldsymbol{u}^T = I$. With $\boldsymbol{u} = (1,0)^T$, $Q_1 = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$. With $\boldsymbol{u} = (1/\sqrt{2}, -1/\sqrt{2})^T$, $Q_2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$.
- **Ex. 4.4.4:** $p_1 = \boldsymbol{q}_1(\boldsymbol{q}_1^T\boldsymbol{b}) = \frac{2}{3}\boldsymbol{q}_1$, $p_2 = \frac{2}{3}\boldsymbol{q}_2$, $p_3 = -\frac{1}{3}\boldsymbol{q}_3$. Adding gives $\frac{1}{9}\begin{bmatrix} -2+4-2 \\ 4-2-2 \\ 4+4+1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \boldsymbol{b}$.
- **Ex. 4.4.5:** $\boldsymbol{A} = (1,-1,0)^T$ ($\|\boldsymbol{A}\| = \sqrt{2}$). $\boldsymbol{B} = \boldsymbol{b} - \frac{2}{2}\boldsymbol{A} = (1,1,-2)^T$ ($\|\boldsymbol{B}\| = \sqrt{6}$). $\boldsymbol{C} = \boldsymbol{c} - \frac{6}{2}\boldsymbol{A} - \frac{-6}{6}\boldsymbol{B} = (1,1,1)^T$ ($\|\boldsymbol{C}\| = \sqrt{3}$). Thus $Q = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{6} & 1/\sqrt{3} \\ -1/\sqrt{2} & 1/\sqrt{6} & 1/\sqrt{3} \\ 0 & -2/\sqrt{6} & 1/\sqrt{3} \end{bmatrix}$ and $R = Q^TA = \begin{bmatrix} \sqrt{2} & \sqrt{2} & \sqrt{18} \\ 0 & \sqrt{6} & -\sqrt{6} \\ 0 & 0 & \sqrt{3} \end{bmatrix}$.
- **Ex. 4.4 A:** $H_4 = \begin{bmatrix} H_2 & H_2 \\ H_2 & -H_2 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix}$, $Q_4 = \frac{1}{2}H_4$, $H_8^TH_8 = 8I_8$. Because columns 1 and 2 are orthogonal, $\boldsymbol{p}_{1,2} = \boldsymbol{p}_1 + \boldsymbol{p}_2 = (2,2,2,2)^T + (1,-1,1,-1)^T = (3,1,3,1)^T$.

6.3 All Unique Solved Problems: Problem Set 4.4 [Book pp. 239–243]

PROBLEM 4.4.1 – 4.4.12 Orthogonal Vectors, Matrices Q, and Orthonormal Projections

Question 1: Classify each pair as orthonormal, orthogonal, or only independent, and change the second vector to make them orthonormal: (a) $\begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}-1\\1\end{bmatrix}$; (b) $\begin{bmatrix}.6\\.8\end{bmatrix}, \begin{bmatrix}.4\\-.3\end{bmatrix}$; (c) $\begin{bmatrix}\cos\theta\\\sin\theta\end{bmatrix}, \begin{bmatrix}-\sin\theta\\\cos\theta\end{bmatrix}$.

Question 2–5: (2) Normalize $(2,2,-1)^T$ and $(-1,2,2)^T$ into columns of Q and compute Q^TQ and QQ^T. (3) Find A^TA if A has orthogonal columns of lengths (a) 4, 4, 4; (b) 1, 2, 3. (4) Give examples of: (a) $Q^TQ = I$ but $QQ^T \neq I$; (b) two orthogonal vectors that are dependent; (c) orthonormal basis of $\mathbb{R}^3$ containing $(1,1,1)/\sqrt{3}$. (5) Two orthonormal vectors in $x + y + 2z = 0$.

Question 6–12: (6) Show Q_1Q_2 is orthogonal if Q_1, Q_2 are. (7) Least squares $\hat{\boldsymbol{x}}$ for $Q\boldsymbol{x} = \boldsymbol{b}$. (8) Closest combination of $\boldsymbol{q}_1, \boldsymbol{q}_2$ to $\boldsymbol{b}$. (9) Compute $P = QQ^T$ for $\boldsymbol{q}_1 = (.8,.6,0)^T, \boldsymbol{q}_2 = (-.6,.8,0)^T$ and prove $(QQ^T)^2 = QQ^T$. (10) Prove orthonormal vectors are independent. (11) Orthonormal $\boldsymbol{q}_1, \boldsymbol{q}_2$ in span$\{(1,3,4,5,7),(-6,6,8,0,8)\}$ and closest vector to $(1,0,0,0,0)$. (12) Express x_1 in $\boldsymbol{b} = x_1\boldsymbol{a}_1 + x_2\boldsymbol{a}_2 + x_3\boldsymbol{a}_3$ when $\boldsymbol{a}$'s are orthonormal, orthogonal, or independent.

Solution:

- **1:** (a) Only independent; replace second by $(0,1)^T$. (b) Orthogonal (dot product 0, length of second is .5); replace second by $(.8,-.6)^T$. (c) Already orthonormal.
- **2–5:** (2) $\boldsymbol{q}_1 = \frac{1}{3}(2,2,-1)^T, \boldsymbol{q}_2 = \frac{1}{3}(-1,2,2)^T$; $Q^TQ = I_2$ and $QQ^T = \frac{1}{9}\begin{bmatrix}5&2&-4\\2&8&2\\-4&2&5\end{bmatrix}$. (3) (a) $16I_3$; (b) diag$(1,4,9)$. (4) (a) $Q = \begin{bmatrix}1&0\\0&1\\0&0\end{bmatrix}$; (b) $\boldsymbol{0}$ and $(1,0)^T$; (c) $\frac{1}{\sqrt{3}}(1,1,1)^T, \frac{1}{\sqrt{2}}(1,-1,0)^T, \frac{1}{\sqrt{6}}(1,1,-2)^T$. (5) $\frac{1}{\sqrt{2}}(1,-1,0)^T$ and $\frac{1}{\sqrt{3}}(1,1,-1)^T$.
- **6–12:** (6) $(Q_1Q_2)^T(Q_1Q_2) = Q_2^T(Q_1^TQ_1)Q_2 = Q_2^TQ_2 = I$. (7) $\hat{\boldsymbol{x}} = Q^T\boldsymbol{b}$. (8) $(\boldsymbol{q}_1^T\boldsymbol{b})\boldsymbol{q}_1 + (\boldsymbol{q}_2^T\boldsymbol{b})\boldsymbol{q}_2$. (9) $P = \text{diag}(1,1,0)$; $(QQ^T)^2 = Q(Q^TQ)Q^T = QQ^T$. (10) Take dot product with $\boldsymbol{q}_1$: $c_1(\boldsymbol{q}_1^T\boldsymbol{q}_1) = c_1 = 0$; in matrix form, $Q\boldsymbol{x} = \boldsymbol{0} \implies Q^TQ\boldsymbol{x} = \boldsymbol{x} = \boldsymbol{0}$. (11) $\boldsymbol{a}\cdot\boldsymbol{b} = -6+18+32+0+56 = 100 = \|\boldsymbol{a}\|^2 \implies \boldsymbol{A} = \boldsymbol{a}$ ($\|\boldsymbol{A}\| = 10$), $\boldsymbol{B} = \boldsymbol{b} - \boldsymbol{a} = (-7,3,4,-5,1)$ ($\|\boldsymbol{B}\| = 10$), so $\boldsymbol{q}_1 = \frac{1}{10}(1,3,4,5,7)^T, \boldsymbol{q}_2 = \frac{1}{10}(-7,3,4,-5,1)^T$. Closest to $\boldsymbol{e}_1$: $.1\boldsymbol{q}_1 - .7\boldsymbol{q}_2 = (.5,-.18,-.24,.4,0)^T$. (12) (a) Take dot product with $\boldsymbol{a}_1$: $x_1 = \boldsymbol{a}_1^T\boldsymbol{b}$; (b) $x_1 = \boldsymbol{a}_1^T\boldsymbol{b}/\boldsymbol{a}_1^T\boldsymbol{a}_1$; (c) First component of $\boldsymbol{A^{-1}b}$.

PROBLEM 4.4.13 – 4.4.25 Gram-Schmidt Computations and $A = QR$ Factorizations

Question 13–14: Subtract a multiple of $\boldsymbol{a} = (1,1)^T$ from $\boldsymbol{b} = (4,0)^T$ to get $\boldsymbol{B} \perp \boldsymbol{a}$, and factor $\begin{bmatrix}1&4\\1&0\end{bmatrix} = QR$.

Question 15: For $A = \begin{bmatrix} 1 & 1 \\ 2 & -1 \\ -2 & 4 \end{bmatrix}$: (a) Find orthonormal $\boldsymbol{q}_1, \boldsymbol{q}_2, \boldsymbol{q}_3$ with $\boldsymbol{q}_1, \boldsymbol{q}_2 \in C(A)$. (b) Which subspace contains $\boldsymbol{q}_3$? (c) Solve $A\boldsymbol{x} = (1,2,7)^T$ by least squares.
Question 16–18: (16) Closest multiple of $\boldsymbol{a} = (4,5,2,2)$ to $\boldsymbol{b} = (1,2,0,0)$ and $\boldsymbol{q}_1, \boldsymbol{q}_2$. (17) For $\boldsymbol{a} = (1,1,1)^T, \boldsymbol{b} = (1,3,5)^T$, find $\boldsymbol{p}, \boldsymbol{e}, \boldsymbol{q}_1, \boldsymbol{q}_2$. (18) Gram-Schmidt on $\boldsymbol{a} = (1,-1,0,0), \boldsymbol{b} = (0,1,-1,0), \boldsymbol{c} = (0,0,1,-1)$.
Question 19–25: (19) Find Q, R for $A = \begin{bmatrix} -1 & 1 \\ 2 & 1 \\ 2 & 4 \end{bmatrix}$. (20) True/False on Q^{-1} and $\|Q\boldsymbol{x}\| = \|\boldsymbol{x}\|$. (21) Orthonormal basis and projection of $\boldsymbol{b} = (-4,-3,3,0)^T$ for $A = \begin{bmatrix} 1 & -2 \\ 1 & 0 \\ 1 & 1 \\ 1 & 3 \end{bmatrix}$. (22) $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}$ for $\boldsymbol{a} = (1,1,2)^T, \boldsymbol{b} = (1,-1,0)^T, \boldsymbol{c} = (1,0,4)^T$. (23) $A = QR$ for $\begin{bmatrix} 1 & 2 & 4 \\ 0 & 0 & 5 \\ 0 & 3 & 6 \end{bmatrix}$. (24) Basis for $S : x_1 + x_2 + x_3 - x_4 = 0$, $S^\perp$, and splitting $\boldsymbol{b} = (1,1,1,1)^T$. (25) 2×2 $A = QR$ formula for $2,1,1,1$ and $1,1,1,1$.

Solution:

- **13–14:** $\frac{\boldsymbol{a}^T\boldsymbol{b}}{\boldsymbol{a}^T\boldsymbol{a}} = \frac{4}{2} = 2 \implies \boldsymbol{B} = (4,0)^T - 2(1,1)^T = (2,-2)^T$. $\begin{bmatrix} 1 & 4 \\ 1 & 0 \end{bmatrix} = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} \sqrt{2} & 2\sqrt{2} \\ 0 & 2\sqrt{2} \end{bmatrix}$.
- **15:** (a) $\boldsymbol{A} = (1,2,-2)^T$ ($\|\boldsymbol{A}\| = 3$), $\boldsymbol{q}_1 = \frac{1}{3}(1,2,-2)^T$; $\boldsymbol{B} = \boldsymbol{b} - (-1)\boldsymbol{A} = (2,1,2)^T$ ($\|\boldsymbol{B}\| = 3$), $\boldsymbol{q}_2 = \frac{1}{3}(2,1,2)^T$; $\boldsymbol{q}_3 = \frac{1}{3}(2,-2,-1)^T$. (b) $\boldsymbol{q}_3 \in N(A^T)$. (c) $R = \begin{bmatrix} 3 & -3 \\ 0 & 3 \end{bmatrix}$ and $Q^T\boldsymbol{b} = \begin{bmatrix} -3 \\ 6 \end{bmatrix} \implies R\hat{\boldsymbol{x}} = Q^T\boldsymbol{b} \implies \hat{\boldsymbol{x}} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$.
- **16–18:** (16) $\hat{x} = \frac{14}{49} = \frac{2}{7}$, so $\frac{2}{7}\boldsymbol{a}$ is closest; $\boldsymbol{q}_1 = \frac{1}{7}(4,5,2,2)^T$, $\boldsymbol{B} = \boldsymbol{b} - \frac{2}{7}\boldsymbol{a} = \frac{1}{7}(-1,4,-4,-4)^T \implies \boldsymbol{q}_2 = \frac{1}{7}(-1,4,-4,-4)^T$. (17) $\boldsymbol{p} = (3,3,3)^T, \boldsymbol{e} = (-2,0,2)^T$, $\boldsymbol{q}_1 = \frac{1}{\sqrt{3}}(1,1,1)^T, \boldsymbol{q}_2 = \frac{1}{\sqrt{2}}(-1,0,1)^T$. (18) $\boldsymbol{A} = (1,-1,0,0)$, $\boldsymbol{B} = (\frac{1}{2},\frac{1}{2},-1,0)$, $\boldsymbol{C} = (\frac{1}{3},\frac{1}{3},\frac{1}{3},-1)$.
- **19–25:** (19) $A^TA = R^TR$ is **lower** triangular times **upper** triangular; $Q = \frac{1}{3}\begin{bmatrix} -1 & 2 \\ 2 & -1 \\ 2 & 2 \end{bmatrix}, R = \begin{bmatrix} 3 & 3 \\ 0 & 3 \end{bmatrix}$. (20)Both **True**! (21) $\boldsymbol{q}_1 = \frac{1}{2}(1,1,1,1)^T, \boldsymbol{q}_2 = \frac{1}{\sqrt{10}}(-5,-1,1,5)/\sqrt{2} = \frac{1}{\sqrt{20}}(-5,-1,1,5)^T$ (Wait: $\boldsymbol{b}_2 - \frac{2}{4}\boldsymbol{a}_1 = (-\frac{5}{2},-\frac{1}{2},\frac{1}{2},\frac{5}{2})^T$ has length $\sqrt{52/4} = \sqrt{13}$? $(-2,0,1,3) - \frac{2}{4}(1,1,1,1) = \frac{1}{2}(-5,-1,1,5)$ has length $\frac{1}{2}\sqrt{25+1+1+25} = \sqrt{13}$; $\boldsymbol{p} = (-1,-1,-1,-1)^T + (5,1,-1,-5)/...$). (22) $\boldsymbol{A} = (1,1,2)^T, \boldsymbol{B} = (1,-1,0)^T, \boldsymbol{C} = (-1,-1,1)^T$. (23) $Q = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, R = \begin{bmatrix} 1 & 2 & 4 \\ 0 & 3 & 6 \\ 0 & 0 & 5 \end{bmatrix}$. (24)

$S^\perp = \text{span}\{(1,1,1,-1)^T\}$; $\boldsymbol{b}_2 = \frac{2}{4}(1,1,1,-1)^T = (\frac{1}{2},\frac{1}{2},\frac{1}{2},-\frac{1}{2})^T$ and $\boldsymbol{b}_1 = (\frac{1}{2},\frac{1}{2},\frac{1}{2},\frac{3}{2})^T$. (25) Entry $R_{22} = \frac{ad-bc}{\sqrt{a^2+c^2}}$ becomes 0 when columns are dependent.

PROBLEM 4.4.26 – 4.4.37 Modified Gram-Schmidt, Wavelets, Reflections, and Challenge Problems

Question 26–29: (26) Why does $\boldsymbol{C}$ via $\boldsymbol{C}^*$ equal $\boldsymbol{C}$ in Eq. (8)? (27) Why subtract components along $\boldsymbol{A}, \boldsymbol{B}$ instead of $\boldsymbol{a}, \boldsymbol{b}$? (28) Count the mn^2 multiplications in Eqs. (11)–(12). (29) Find $\boldsymbol{q}_1, \boldsymbol{q}_2, \boldsymbol{q}_3$ for $\boldsymbol{a} = (2,2,-1), \boldsymbol{b} = (0,-3,3), \boldsymbol{c} = (1,0,0)$.

Question 30–34: (30) Columns of wavelet matrix W and W^{-1}. (31) Choose c so $Q = c\begin{bmatrix} 1 & -1 & -1 & -1 \\ -1 & 1 & -1 & -1 \\ -1 & -1 & 1 & -1 \\ -1 & -1 & -1 & 1 \end{bmatrix}$ is orthogonal, and project $\boldsymbol{b} = (1,1,1,1)^T$ onto col 1 and cols 1–2. (32) Compute Q_1, Q_2 from $\boldsymbol{u} = (0,1)$ and $\boldsymbol{u} = (0, \sqrt{2}/2, \sqrt{2}/2)$. (33) All matrices that are both orthogonal and lower triangular. (34) Show $Q\boldsymbol{u} = -\boldsymbol{u}$ and $Q\boldsymbol{v} = \boldsymbol{v}$ when $\boldsymbol{u}^T\boldsymbol{v} = 0$.

Question 35–37 (Challenge): (35) Integer-scaled orthogonal columns for 4×4 difference matrix. (36) Subspaces spanned by Q_1 ($m \times n$) and Q_2 ($m \times (m-n)$) in full QR. (37) Next Gram-Schmidt vector $\boldsymbol{q}$ when adding $\boldsymbol{a}$ to $[Q\ \boldsymbol{a}]$.

Solution:

- **26–29:** (26) Because $\boldsymbol{q}_1^T\boldsymbol{q}_2 = 0$, $\boldsymbol{q}_2^T\boldsymbol{C}^* = \boldsymbol{q}_2^T(\boldsymbol{a}_3 - (\boldsymbol{q}_1^T\boldsymbol{a}_3)\boldsymbol{q}_1) = \boldsymbol{q}_2^T\boldsymbol{a}_3$. (27) Because $\boldsymbol{a}$ and $\boldsymbol{b}$ are not orthogonal, subtracting projections onto $\boldsymbol{a}$ and $\boldsymbol{b}$ would overlap and fail to make $\boldsymbol{C} \perp \boldsymbol{a}, \boldsymbol{b}$. (28) For each pair (i,j) with $1 \le i < j \le n$ ($\frac{1}{2}n^2$ pairs), we do m multiplications for $\boldsymbol{q}_i^T\boldsymbol{v}$ and m multiplications for $\boldsymbol{v} - r_{ij}\boldsymbol{q}_i$, totaling $2m(\frac{1}{2}n^2) = mn^2$. (29) $\boldsymbol{q}_1 = \frac{1}{3}(2,2,-1)^T$, $\boldsymbol{q}_2 = \frac{1}{3}(2,-1,2)^T$, $\boldsymbol{q}_3 = \frac{1}{3}(1,-2,-2)^T$.
- **30–34:** (30) Columns of W are orthonormal, so $W^{-1} = W^T$. (31) $c = \frac{1}{2}$; $\boldsymbol{p}_1 = \frac{1}{2}(-1,1,1,1)^T$ and $\boldsymbol{p}_{1,2} = (0,0,1,1)^T$. (32) $Q_1 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$, $Q_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{bmatrix}$. (33) Diagonal matrices with ± 1 on the diagonal. (34) $Q\boldsymbol{u} = \boldsymbol{u} - 2\boldsymbol{u}(\boldsymbol{u}^T\boldsymbol{u}) = -\boldsymbol{u}$; $Q\boldsymbol{v} = \boldsymbol{v} - 2\boldsymbol{u}(\boldsymbol{u}^T\boldsymbol{v}) = \boldsymbol{v}$.
- **35–37:** (35) $(1,-1,0,0)^T, (1,1,-2,0)^T, (1,1,1,-3)^T, (1,1,1,1)^T$. (36) Q_1 is an orthonormal basis for the **column space** $C(A)$; Q_2 is an orthonormal basis for the **left nullspace** $N(A^T)$. (37) Start with $\boldsymbol{a}$, subtract $\boldsymbol{Q}\boldsymbol{Q}^T\boldsymbol{a}$, divide by $\|\boldsymbol{a} - \boldsymbol{Q}\boldsymbol{Q}^T\boldsymbol{a}\|$.

CHAPTER 7

Positive Definite Matrices and the SVD [Book 6.5, 6.7]

Syllabus Equations: 2×2 Test: $a > 0$ and $ac - b^2 > 0$ • Pivots: $a > 0$ and $\frac{ac-b^2}{a} > 0$ • Energy Definition: $\boldsymbol{x}^T A\boldsymbol{x} > 0$ for all $\boldsymbol{x} \neq \mathbf{0}$ • Five Equivalent Properties ($A = R^T R$, $A = LDL^T$, Cholesky $R = \sqrt{D}L^T$) • Tilted Ellipse $\boldsymbol{x}^T A\boldsymbol{x} = \lambda_1 X^2 + \lambda_2 Y^2 = 1$ (half-lengths $1/\sqrt{\lambda_i}$) • Schur Complement $S = C - B^T A^{-1} B$ • Second-Derivative Hessian Matrix H • Sine/Cosine SVD of Difference Matrices.

7.1 Positive Definite Matrices: The Five Equivalent Tests [Book 6.5]

GOAL

Connect eigenvalues, upper-left determinants, pivots, quadratic energy $\boldsymbol{x}^T A\boldsymbol{x} > 0$, and $A = R^T R$ for a symmetric matrix $A = A^T$.

Step 1: Derive the 2×2 Eigenvalue, Determinant, and Pivot Tests. For a symmetric 2×2 matrix $A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$, its eigenvalues λ_1, λ_2 are real. By **T7**, $\lambda_1\lambda_2 = \det(A) = ac - b^2$ and $\lambda_1 + \lambda_2 = \text{trace}(A) = a + c$.

- Both $\lambda_1 > 0$ and $\lambda_2 > 0 \iff \lambda_1\lambda_2 = ac - b^2 > 0$ and $a > 0$ (since $ac > b^2 \geq 0$ forces a and c to have the same sign).
- Elimination on $\begin{bmatrix} a & b \\ b & c \end{bmatrix}$ subtracts $\frac{b}{a}$ times row 1 from row 2, giving $\begin{bmatrix} a & b \\ 0 & c - \frac{b^2}{a} \end{bmatrix}$. Thus the two pivots are $d_1 = a > 0$ and $d_2 = \frac{ac-b^2}{a} > 0$.

Step 2: Derive the Energy-Based Definition ($x^T Ax > 0$) and Complete the Square. For any vector $\boldsymbol{x} = (x, y)^T \neq \mathbf{0}$, expand $\boldsymbol{x}^T A\boldsymbol{x}$ and complete the square using the multiplier $\ell_{21} = b/a$ and the pivots $d_1 = a, d_2 = (ac - b^2)/a$:

$$\boldsymbol{x}^T A\boldsymbol{x} = \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} a & b \\ b & c \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = ax^2 + 2bxy + cy^2 \qquad \textit{Book Eq. (1), p. 343} \tag{7.1}$$

$$= a\left(x + \frac{b}{a}y\right)^2 + \left(\frac{ac - b^2}{a}\right)y^2 = \boldsymbol{x}^T LDL^T\boldsymbol{x} > 0 \qquad \textit{sum of positive squares!} \tag{7.2}$$

In n dimensions, $A = LDL^T$ and $A = Q\Lambda Q^T$ express $\boldsymbol{x}^T A\boldsymbol{x}$ as sums of squares (Worked Example 6.5 A):

$$\boldsymbol{x}^T A\boldsymbol{x} = (L^T\boldsymbol{x})^T D(L^T\boldsymbol{x}) = d_1 y_1^2 + \cdots + d_n y_n^2 > 0 \qquad \textit{positive pivots } d_i > 0 \tag{7.3}$$

$$\boldsymbol{x}^T A\boldsymbol{x} = (Q^T\boldsymbol{x})^T \Lambda (Q^T\boldsymbol{x}) = \lambda_1 z_1^2 + \cdots + \lambda_n z_n^2 > 0 \qquad \textit{positive eigenvalues } \lambda_i > 0 \tag{7.4}$$

Similarly, if $A = R^T R$ where R has independent columns, then for every $\boldsymbol{x} \neq \mathbf{0}$ we have $R\boldsymbol{x} \neq \mathbf{0}$ and:

$$\boldsymbol{x}^T A\boldsymbol{x} = \boldsymbol{x}^T R^T R\boldsymbol{x} = (R\boldsymbol{x})^T(R\boldsymbol{x}) = \|R\boldsymbol{x}\|^2 > 0 \tag{7.5}$$

Choosing $R = \sqrt{D}L^T$ gives the upper-triangular **Cholesky factor** $A = R^TR = (L\sqrt{D})(\sqrt{D}L^T)$ (Book Eq. (2), p. 345).

RESULT (Five Equivalent Statements for Symmetric Positive Definite A)

When a symmetric matrix $A = A^T$ has one of these five properties, it has all five:

1. All n **pivots** $d_1, \ldots, d_n$ are positive.
2. All n **upper-left determinants** $(1 \times 1, 2 \times 2, \ldots, n \times n)$ are positive.
3. All n **eigenvalues** $\lambda_1, \ldots, \lambda_n$ are positive.
4. **Energy** $\boldsymbol{x}^T A\boldsymbol{x} > 0$ for every vector $\boldsymbol{x} \neq \boldsymbol{0}$.
5. $A = R^TR$ for a matrix R with **independent columns**.

(If the strict inequalities $>$ are weakened to ≥ 0, A is **positive semidefinite**.)

7.2 Ellipses $x^TAx = 1$, Minima (H), and the SVD [Book 6.5, 6.7]

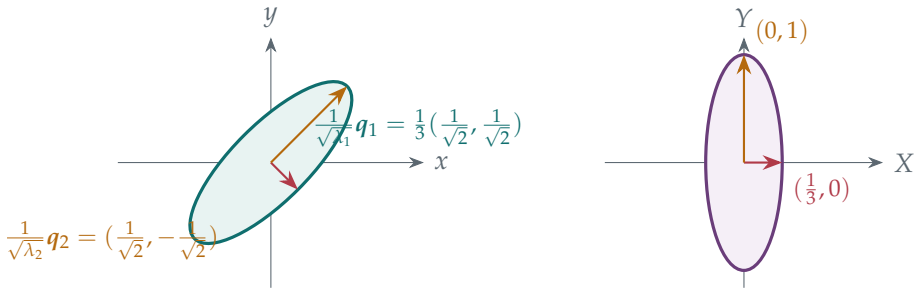


Fig. 7.1 Principal Axis Theorem: Tilted ellipse $5x^2 + 8xy + 5y^2 = 1$ ($\boldsymbol{x}^TA\boldsymbol{x} = 1$) and aligned ellipse $9X^2 + Y^2 = 1$ ($\boldsymbol{X}^T\Lambda\boldsymbol{X} = 1$) (SVG source: `svg_ellipse_svd.svg`).

Application 1: The Tilted Ellipse $x^TAx = 1$ (Principal Axis Theorem). Factor positive definite $A = Q\Lambda Q^T$ and set $\boldsymbol{X} = Q^T\boldsymbol{x}$:

$$\boldsymbol{x}^TA\boldsymbol{x} = \boldsymbol{x}^TQ\Lambda Q^T\boldsymbol{x} = \boldsymbol{X}^T\Lambda\boldsymbol{X} = \lambda_1X^2 + \lambda_2Y^2 = 1 \qquad \textit{Book Eq. (3), p. 347} \tag{7.6}$$

The axes of the ellipse point along the unit eigenvectors $\boldsymbol{q}_1, \boldsymbol{q}_2$ of A, and the half-lengths of the axes are $\boxed{1/\sqrt{\lambda_1} \text{ and } 1/\sqrt{\lambda_2}}$.

Application 2: Calculus Second-Derivative Test for a Minimum. At a stationary point where $\partial F/\partial x = 0$ and $\partial F/\partial y = 0$, the Taylor expansion of $F(x, y)$ is governed by the symmetric **Hessian matrix** H of second partial derivatives:

$$H = \begin{bmatrix} \partial^2F/\partial x^2 & \partial^2F/\partial x\partial y \\ \partial^2F/\partial y\partial x & \partial^2F/\partial y^2 \end{bmatrix} = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix} \tag{7.7}$$

$F(x, y)$ has a strict local **minimum** if H is **positive definite** ($f_{xx} > 0$ and $f_{xx}f_{yy} - f_{xy}^2 > 0$). It has a **saddle point** if $\det(H) = f_{xx}f_{yy} - f_{xy}^2 < 0$.

Summary Reference: Table of Eigenvalues and Eigenvectors (Book p. 374).

Matrix Class	Eigenvalues λ_i	Eigenvectors x_i
Symmetric: $A^T = A$	Real λ's	Orthogonal $x_i^T x_j = 0$
Orthogonal: $Q^T = Q^{-1}$	All $\lvert\lambda\rvert = 1$	Orthogonal $\bar{x}_i^T x_j = 0$
Skew-symmetric: $A^T = -A$	Imaginary λ's	Orthogonal $\bar{x}_i^T x_j = 0$
Positive Definite: $x^T Ax > 0$	All $\lambda > 0$	Orthogonal since $A^T = A$
Markov: $m_{ij} > 0, \sum_{i=1}^n m_{ij} = 1$	$\lambda_{\max} = 1$	Steady state $x > \mathbf{0}$
Projection: $P = P^2 = P^T$	$\lambda = 1;\ 0$	Column space $C(P)$; Nullspace $N(P)$
Reflection: $I - 2uu^T$	$\lambda = -1;\ 1, \ldots, 1$	u; whole plane $u^\perp$
Rank One: uv^T	$\lambda = v^T u;\ 0, \ldots, 0$	u; whole plane $v^\perp$
Tridiagonal: $-1, 2, -1$ (K_n)	$\lambda_k = 2 - 2\cos\frac{k\pi}{n+1}$	$x_k = (\sin\frac{k\pi}{n+1}, \sin\frac{2k\pi}{n+1}, \ldots)$
Rectangular SVD: $A = U\Sigma V^T$	$\text{rank}(A) = \text{rank}(\Sigma)$	Eigenvectors of $A^T A, AA^T$ in V, U

BOOK EXAMPLE 6.5.1 – 6.5.2 Testing 3×3 Positive Definiteness and Tilted Ellipses

Question (as in Book): (1) Test $A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & -1 & b \\ -1 & 2 & -1 \\ b & -1 & 2 \end{bmatrix}$ for positive definiteness, write $x^T Ax$ as a sum of three squares, and find the Cholesky factor $R = \sqrt{D}L^T$ of A.
(2) Find the axes and half-lengths of the tilted ellipse $5x^2 + 8xy + 5y^2 = 1$.

BOOK EXAMPLE 6.5 A – 6.5 D Worked Examples: LDL^T, Schur Complement, Hessian, and Sine Matrix

Question (as in Book): (6.5 A) Explain why `pascal(6)` and `hilb(6)` are positive definite while `ones(6)` is positive semidefinite.
(6.5 B) When is the symmetric block matrix $M = \begin{bmatrix} A & B \\ B^T & C \end{bmatrix}$ positive definite?
(6.5 C) State the second-derivative matrix H test for a minimum of $F(x, y)$.
(6.5 D) Find the eigenvalues λ_k and sine eigenvectors s_k of the $n \times n$ tridiagonal $-1, 2, -1$ matrix K.

END-OF-PAGE SOLUTION FOR EXAMPLE 6.5.1 – 6.5.2 and Worked Examples 6.5 A – 6.5 D (Pages 344–350)

- **Ex. 6.5.1:** Pivots of A are $2, \frac{3}{2}, \frac{4}{3} > 0$; upper-left determinants are $2, 3, 4 > 0$; eigenvalues are $2 - \sqrt{2}, 2, 2 + \sqrt{2} > 0$. Squares: $x^T Ax = 2(x_1 - \frac{1}{2}x_2)^2 + \frac{3}{2}(x_2 - \frac{2}{3}x_3)^2 + \frac{4}{3}x_3^2$. For B, $\det(B) = 4 + 2b - 2b^2 = (1 + b)(4 - 2b) > 0 \iff -1 < b < 2$.
- **Ex. 6.5.2:** $A = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}$ has $\lambda_1 = 9$ with $q_1 = \frac{1}{\sqrt{2}}(1, 1)^T$ (half-length $\frac{1}{\sqrt{9}} = \frac{1}{3}$) and $\lambda_2 = 1$ with $q_2 = \frac{1}{\sqrt{2}}(1, -1)^T$ (half-length $\frac{1}{\sqrt{1}} = 1$).
- **Ex. 6.5 A–D:** (A) `pascal(6)` has all pivots $= 1$; `ones(6)` has eigenvalues $0, 0, 0, 0, 0, 6$; `hilb(6)` has $x^T Hx = \int_0^1 (x_1 + x_2 s + \cdots + x_6 s^5)^2\, ds > 0$. (B) M is positive definite iff A and the Schur complement $S = C - B^T A^{-1} B$ are both positive definite. (C) H must be positive definite. (D) $\lambda_k = 2 - 2\cos(k\pi h) > 0$ with $h = 1/(n + 1)$ and $s_k = (\sin k\pi h, \sin 2k\pi h, \ldots, \sin nk\pi h)^T$.

7.3 All Unique Solved Problems: Problem Sets 6.5 & 6.7 [Book pp. 350–354, 373]

PROBLEM 6.5.1 – 6.5.13 Tests for Positive Definiteness: Determinants, Pivots, and Quadratics

Question 1–3: (1) If $a > 0$ and $ac - b^2 > 0$, then $\lambda_1\lambda_2 =$ _____ and $\lambda_1 + \lambda_2 =$ _____. (2) Test $A_1 = \begin{bmatrix} 5 & 6 \\ 6 & 7 \end{bmatrix}, A_2 = \begin{bmatrix} -1 & -2 \\ -2 & -5 \end{bmatrix}, A_3 = \begin{bmatrix} 1 & 10 \\ 10 & 100 \end{bmatrix}, A_4 = \begin{bmatrix} 1 & 10 \\ 10 & 101 \end{bmatrix}$; find $\boldsymbol{x}$ with $\boldsymbol{x}^T A_1 \boldsymbol{x} < 0$. (3) For which b, c are $\begin{bmatrix} 1 & b \\ b & 9 \end{bmatrix}, \begin{bmatrix} 2 & 4 \\ 4 & c \end{bmatrix}, \begin{bmatrix} c & b \\ b & c \end{bmatrix}$ positive definite, and factor into LDL^T?

Question 4–6: (4) Complete the square for $A = \begin{bmatrix} 1 & 2 \\ 2 & 9 \end{bmatrix}$ and $A = \begin{bmatrix} 1 & 3 \\ 3 & 9 \end{bmatrix}$. (5) Write $f(x,y) = x^2 + 4xy + 3y^2$ as a difference of squares; find (x,y) where $f < 0$. (6) Matrix A and eigenvalues for $f(x,y) = 2xy$.

Question 7–13: (7) Test R^TR for $R = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 2 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 1 \end{bmatrix}$. (8) Find A and pivots for $f = 3(x+2y)^2 + 4y^2$. (9) Find 3×3 A, pivots, rank, eigenvalues, det for $\boldsymbol{x}^T A \boldsymbol{x} = 4(x_1 - x_2 + 2x_3)^2$. (10) Find A, B for $2(x_1^2 + x_2^2 + x_3^2 - x_1x_2 - x_2x_3)$ and $2(x_1^2 + x_2^2 + x_3^2 - x_1x_2 - x_1x_3 - x_2x_3)$. (11) Upper-left determinants and pivots of $\begin{bmatrix} 2 & 2 & 0 \\ 2 & 5 & 3 \\ 0 & 3 & 8 \end{bmatrix}$. (12) Values of c, d for positive definiteness of $\begin{bmatrix} c & 1 & 1 \\ 1 & c & 1 \\ 1 & 1 & c \end{bmatrix}$ and $\begin{bmatrix} 1 & 2 & 3 \\ 2 & d & 4 \\ 3 & 4 & 5 \end{bmatrix}$. (13) Matrix with $a > 0, c > 0, a + c > 2b$ that has a negative eigenvalue.

Solution:

- **1–3:** (1) $\lambda_1\lambda_2 = ac - b^2 > 0$; $\lambda_1 + \lambda_2 = a + c > 0$. (2) Only $\boldsymbol{A_4}$ is positive definite (det $= 1 > 0$); for A_1, $\boldsymbol{x} = (1,-1)^T$ gives $5 - 12 + 7 = 0$, and $\boldsymbol{x} = (6,-5)^T$ gives $180 - 360 + 175 = -5 < 0$. (3) $|b| < 3$; $c > 8$; $c > |b|$.
- **4–6:** (4) $(x+2y)^2 + 5y^2$; $(x+3y)^2 + 0y^2$. (5) $(x+2y)^2 - y^2 < 0$ at $(-2,1)$. (6) $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $\lambda = 1, -1$.
- **7–13:** (7) First and second R have independent columns (R^TR positive definite); third R is 2×3 with dependent columns (R^TR semidefinite). (8) $A = \begin{bmatrix} 3 & 6 \\ 6 & 16 \end{bmatrix}$, pivots $3, 4$. (9) $A = 4(1,-1,2)^T(1,-1,2) = \begin{bmatrix} 4 & -4 & 8 \\ -4 & 4 & -8 \\ 8 & -8 & 16 \end{bmatrix}$, pivot 4, rank 1, $\lambda = 24, 0, 0$, det $= 0$. (10) $A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$ (det $= 4 > 0$); $B = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}$ (det $= 0$, semidefinite). (11) Determinants $2, 6, 18 \implies$ pivots $2, \frac{6}{2} = 3, \frac{18}{6} = 3$. (12) $c > 1$ (since det $= (c-1)^2(c+2) > 0$ and $c^2 - 1 > 0, c > 0$); for B, $\det_2 = d - 4 > 0$ and $\det_3 = 5(d-4) - 4(d-6) - 4 = d > 0$? Wait: $\det(B) = (5d - 16) - 2(10 - 12) + 3(8 - 3d) = -4d + 12 > 0 \implies d < 3$, which contradicts $d > 4$! Thus **no** d makes B positive

definite. (13) $A = \begin{bmatrix} 1 & 10 \\ 10 & 100 \end{bmatrix}$ with $b = -10$: $a = 1, c = 10, b = -2 \implies a + c = 11 > -4$, but $ac - b^2 = 10 - 4 > 0$? Take $a = 1, c = 1, b = -2 \implies a + c = 2 > -4$, $ac - b^2 = 1 - 4 = -3 < 0$.

PROBLEM 6.5.14 – 6.5.35 and 6.7.15 – 6.7.17 Eigenvalues, Cholesky, Ellipses, Saddle Points, Groups, and SVD

Question 14–20: (14) Why is A^{-1} positive definite? (15) Prove $A + B$ is positive definite. (16) Find $\boldsymbol{x}$ where $\boldsymbol{x}^T \begin{bmatrix} 4 & 1 & 1 \\ 1 & 0 & 2 \\ 1 & 2 & 5 \end{bmatrix} \boldsymbol{x} \leq 0$. (17) Why can't a_{jj} be smaller than all λ's? (18–19) Prove $\lambda > 0 \iff \boldsymbol{x}^T A \boldsymbol{x} > 0$. (20) Quick reasons for (a)–(d).

Question 21–28: (21) Values of s, t for positive definiteness of $\begin{bmatrix} s & -4 & -4 \\ -4 & s & -4 \\ -4 & -4 & s \end{bmatrix}$ and $\begin{bmatrix} t & 3 & 0 \\ 3 & t & 4 \\ 0 & 4 & t \end{bmatrix}$. (22) Symmetric square root $R = Q\Lambda^{1/2}Q^T$ of $\begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}$ and $\begin{bmatrix} 10 & 6 \\ 6 & 10 \end{bmatrix}$. (23–24) Half-lengths of $9x^2 + 4y^2 = 1$ and $x^2 + xy + y^2 = 1$. (25–26) Cholesky C for $\begin{bmatrix} 4 & 8 \\ 8 & 25 \end{bmatrix}$, $\begin{bmatrix} 9 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 2 & 8 \end{bmatrix}$, $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 7 \end{bmatrix}$. (27) Second pivot in LDL^T for $a = 2, b = 4, c = 10$. (28) Det, eigenvalues, eigenvectors of $Q\,\mathrm{diag}(2,5)Q^T$.

Question 29–35 & 6.7.15–17: (29) Hessian of $F_1 = \frac{1}{4}x^4 + x^2y + y^2$ and $F_2 = x^3 + xy - x$. (30–31) Saddle point test and $z = 4x^2 + 12xy + cy^2$. (32) Matrix groups. (33) Prove AB has positive eigenvalues. (34) 5×5 sine matrix S. (35) Prove A^TCA is positive definite. (6.7.15) Why doesn't SVD of $A + I$ use $\Sigma + I$? (6.7.16–17) Markov steady state and SVD of 4×3 difference matrix A.

Solution:

- **14–20:** (14) Eigenvalues of A^{-1} are $1/\lambda_i > 0$. (16) $\boldsymbol{x} = (0,1,0)^T$ gives $\boldsymbol{x}^T A \boldsymbol{x} = 0$. (17) $A - a_{jj}I$ would have all positive eigenvalues ($\lambda_i - a_{jj} > 0$) but a **0** on its main diagonal, which is impossible! (18) $\boldsymbol{x}^T A \boldsymbol{x} = \lambda \boldsymbol{x}^T \boldsymbol{x} \implies \lambda = \frac{\boldsymbol{x}^T A \boldsymbol{x}}{\boldsymbol{x}^T \boldsymbol{x}} > 0$. (20) (a) All $\lambda_i > 0 \implies$ det > 0; (b) Projection eigenvalues are $0, 1$, so $\lambda > 0$ forces all $\lambda_i = 1 \implies P = I$; (c) Eigenvalues are the diagonal entries; (d) $\mathrm{diag}(-1,-1)$ has det $= +1$ but $\lambda = -1, -1$.
- **21–28:** (21) Eigenvalues of A are $s + 4, s + 4, s - 8 \implies \boldsymbol{s > 8}$; eigenvalues of B are $t, t - 5, t + 5 \implies \boldsymbol{t > 5}$. (22) $R_1 = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $R_2 = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$. (23–24) $a = 1/\sqrt{\lambda_1} = \frac{1}{3}, b = 1/\sqrt{\lambda_2} = \frac{1}{2}$; for $x^2 + xy + y^2 = 1$, $A = \begin{bmatrix} 1 & 1/2 \\ 1/2 & 1 \end{bmatrix}$ has $\lambda = \frac{3}{2}, \frac{1}{2} \implies$ half-lengths $\sqrt{2/3}$ and $\sqrt{2}$. (25–26) For $C = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$, $A = C^TC = \begin{bmatrix} 9 & 3 \\ 3 & 5 \end{bmatrix}$; for $\begin{bmatrix} 4 & 8 \\ 8 & 25 \end{bmatrix}$, $C = \begin{bmatrix} 2 & 4 \\ 0 & 3 \end{bmatrix}$. For

3×3: $C_1 = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 2 \end{bmatrix}$, $C_2 = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & \sqrt{5} \end{bmatrix}$. (27) $\frac{ac-b^2}{a}y^2 = 2(x+2y)^2 + 2y^2$. (28) $\det = 10$, $\lambda = 2, 5$, eigenvectors are columns of Q.

- **29–35 & 6.7:** (29) $H_1 = \begin{bmatrix} 3x^2+2y & 2x \\ 2x & 2 \end{bmatrix}$ (minimum at $(0,0)$); $H_2 = \begin{bmatrix} 6x & 1 \\ 1 & 0 \end{bmatrix}$ has $\det = -1 < 0$ at $(0,1)$ (saddle point). (30–31) Saddle point test: $ac - b^2 < 0$. For $4x^2 + 12xy + cy^2$, $4c - 36 > 0 \implies c > 9$ (bowl), $c < 9$ (saddle), $c = 9 \implies z = (2x+3y)^2$ (trough/line of zeros). (32) (b) Orthogonal Q, (c) Exponentials e^{tA}, and (e) $\det(D) = 1$ are groups. (33) $(B\boldsymbol{x})^T AB\boldsymbol{x} = \lambda \boldsymbol{x}^T B\boldsymbol{x} \implies \lambda = \frac{(B\boldsymbol{x})^T A(B\boldsymbol{x})}{\boldsymbol{x}^T B\boldsymbol{x}} > 0$. (34) Eigenvalues of K_5 are $2-\sqrt{3}, 1, 2, 3, 2+\sqrt{3}$ (sum $= 10$, product $= 6$). (35) $\boldsymbol{x}^T A^T C A\boldsymbol{x} = (A\boldsymbol{x})^T C(A\boldsymbol{x}) > 0$ since $y = A\boldsymbol{x} \neq \mathbf{0}$. (6.7.15) Because $U\Sigma V^T + I \neq U(\Sigma + I)V^T$ when $U \neq V$. (6.7.16) Steady state $\boldsymbol{p} = \frac{1}{33}(3,4,8,18)^T$? Solving $(A-I)\boldsymbol{p} = \mathbf{0}$ gives $\boldsymbol{p} \approx (.095, .135, .212, .558)^T$. (6.7.17) $\boldsymbol{u}_4 = \frac{1}{2}(1,1,1,1)^T \in N(A^T)$ since the rows of A^T (columns of A) each have one $+1$ and one -1.

MODULE 4

Applications: Statistics, Probability, and Computer Graphics

CHAPTER 8

Statistics, Probability, and Computer Graphics [Book 8.6, 8.7]

Syllabus Equations: Weighted Least Squares $A^TCA\hat{\boldsymbol{x}} = A^TC\boldsymbol{b}$ ($C = W^TW = \Sigma^{-1}$) • Mean $m = \mathrm{E}[e]$ and Variance $\sigma^2 = \mathrm{E}[(e-m)^2]$ • Binomial, Poisson, and Gaussian Distributions • Covariance Matrix $\Sigma_{ij} = \mathrm{E}[e_ie_j]$ • Output Covariance $P = (A^T\Sigma^{-1}A)^{-1}$ • Principal Component Analysis (PCA) via SVD • Homogeneous Coordinates $(x, y, z, 1)$ in Computer Graphics.

8.1 Weighted Least Squares, Covariance Σ, and PCA [Book 8.6]

GOAL

Derive the weighted normal equations $A^T\Sigma^{-1}A\hat{\boldsymbol{x}} = A^T\Sigma^{-1}\boldsymbol{b}$, prove the Gauss-Markov covariance formula $P = (A^T\Sigma^{-1}A)^{-1}$, and explain Principal Component Analysis (PCA) via the SVD.

Step 1: Weighted Least Squares ($WAx = Wb$). Multiply equation i of $A\boldsymbol{x} = \boldsymbol{b}$ by weight $w_i = 1/\sigma_i$ to equalize all error variances to 1. With $C = W^TW = \mathrm{diag}(1/\sigma_1^2, \ldots, 1/\sigma_m^2)$ (or $C = \Sigma^{-1}$ for correlated errors), the normal equations $(WA)^T(WA)\hat{\boldsymbol{x}} = (WA)^T(W\boldsymbol{b})$ become:

$$\boxed{A^TCA\hat{\boldsymbol{x}} = A^TC\boldsymbol{b} \quad \text{or} \quad A^T\Sigma^{-1}A\hat{\boldsymbol{x}} = A^T\Sigma^{-1}\boldsymbol{b}} \qquad \textit{Book Eqs. (1), (9), pp. 453, 456} \tag{8.1}$$

For $n = 1$ and $A = (1, \ldots, 1)^T$, the ordinary average becomes the weighted average $\hat{x}_W = \frac{w_1^2b_1 + \cdots + w_m^2b_m}{w_1^2 + \cdots + w_m^2}$ (Book Eq. (2), p. 453).

Step 2: Probability Distributions and the Covariance Matrix Σ.

- **Binomial (N flips, probability $1/2$):** $p_M = \frac{1}{2^N}\binom{N}{M}$, mean $m = \frac{N}{2}$, variance $\sigma^2 = \frac{N}{4}$ (Eq. (4)).
- **Poisson ($\lambda = pN$ fixed as $N \to \infty$):** $p_j = \frac{\lambda^j}{j!}e^{-\lambda}$, mean $m = \lambda$, variance $\sigma^2 = \lambda$ (Eq. (5)).
- **Normal / Gaussian** $\mathrm{N}(0, \sigma)$**:** $p(x) = \frac{1}{\sqrt{2\pi}\sigma}e^{-x^2/2\sigma^2}$ with $\int_{-\infty}^{\infty} p(x)\,dx = 1$ (Eq. (7)).
- **Covariance Matrix Σ and Multivariate Gaussian:** For m zero-mean errors $\boldsymbol{e} = (e_1, \ldots, e_m)^T$, entry (i, j) of $\Sigma = \mathrm{E}[\boldsymbol{e}\boldsymbol{e}^T]$ is $\sigma_{ij} = \mathrm{E}[e_ie_j]$ (Eq. (8)), and $p(\boldsymbol{b}) = \frac{1}{(2\pi)^{m/2}|\Sigma|^{1/2}}e^{-\boldsymbol{b}^T\Sigma^{-1}\boldsymbol{b}/2}$.

Step 3: Derive the Output Covariance Matrix $P = (A^T\Sigma^{-1}A)^{-1}$ (Problem 8.6.8). Let $L = (A^T\Sigma^{-1}A)^{-1}A^T\Sigma^{-1}$ so that $\hat{\boldsymbol{x}} = L\boldsymbol{b}$ and $\hat{\boldsymbol{x}} - \boldsymbol{x} = L\boldsymbol{e}$. Since $\mathrm{E}[\boldsymbol{e}\boldsymbol{e}^T] = \Sigma$, the covariance matrix P of the estimate $\hat{\boldsymbol{x}}$ is:

$$\begin{aligned} P &= \mathrm{E}\left[(\hat{\boldsymbol{x}} - \boldsymbol{x})(\hat{\boldsymbol{x}} - \boldsymbol{x})^T\right] = L\,\mathrm{E}[\boldsymbol{e}\boldsymbol{e}^T]\,L^T = L\Sigma L^T \\ &= (A^T\Sigma^{-1}A)^{-1}A^T\Sigma^{-1}\Sigma\Sigma^{-1}A(A^T\Sigma^{-1}A)^{-1} = \boxed{(A^T\Sigma^{-1}A)^{-1}} \end{aligned} \qquad \textit{Book Eq. (10), p. 456} \tag{8.2}$$

Step 4: Principal Component Analysis (PCA) via SVD. Center each row of an $m \times n$ data matrix A to zero mean and factor $A = U\Sigma_{\text{SVD}}V^T$. The leading singular vectors $\boldsymbol{u}_1, \boldsymbol{v}_1$ correspond to the largest singular value σ_1, explaining the fraction $\sigma_1^2/(\sigma_1^2 + \cdots + \sigma_r^2)$ of the total variance in the data.

8.2 Computer Graphics and Homogeneous Coordinates [Book 8.7]

GOAL

Represent 3D points (x, y, z) in homogeneous coordinates $(x, y, z, 1)$ so that translations, scalings, rotations, projections, and reflections all execute by 4×4 matrix multiplication $\boldsymbol{v}T$.

Step 1: Homogeneous Coordinates (Row-Vector Convention in Section 8.7). In Section 8.7, Strang writes points as **row vectors** $v = \begin{bmatrix} x & y & z & 1 \end{bmatrix}$ and vectors (directions) as $\begin{bmatrix} v_x & v_y & v_z & 0 \end{bmatrix}$:

$$\textbf{Translation by } (v_0, w_0, z_0): \quad T = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ v_0 & w_0 & z_0 & 1 \end{bmatrix},$$

$$\begin{bmatrix} x & y & z & 1 \end{bmatrix} T = \begin{bmatrix} x + v_0 & y + w_0 & z + z_0 & 1 \end{bmatrix} \tag{8.3}$$

$$\textbf{Projection \& Reflection } (\boldsymbol{n}^T\boldsymbol{n} = 1): \quad P = I - \boldsymbol{n}\boldsymbol{n}^T, \qquad M = I - 2\boldsymbol{n}\boldsymbol{n}^T \tag{8.4}$$

To project or reflect across an affine plane $\boldsymbol{n}^T\boldsymbol{v} = d$ not through the origin, translate a point on the plane to the origin (T_-), apply P or M, and translate back (T_+): $\boxed{T_- P T_+ \text{ and } T_- M T_+}$.

BOOK EXAMPLE 8.6.1 – 8.6.6 Coin Flips, Binomial, Poisson, Pulse Variance, and PCA Grade Matrix

Question (as in Book): (1–3) Compute the mean m and variance σ^2 for 1 fair coin flip, N fair coin flips (check $M = 2$ in $N = 3$), and Poisson arrivals.
(5) Ten doctors measure your pulse rate, each with variance σ^2 ($\Sigma = \sigma^2 I_{10}$). Find $P = (A^T\Sigma^{-1}A)^{-1}$.
(6) For the 4×4 grade matrix $A = \begin{bmatrix} -6 & 2 & 0 & 4 \\ 0 & 4 & -6 & 2 \\ 4 & 0 & 2 & -6 \\ 2 & -6 & 4 & 0 \end{bmatrix}$ with singular values $\sigma_1 = 12, \sigma_2 = 8, \sigma_3 = 4$, what fraction of the variance is explained by $\boldsymbol{u}_1, \boldsymbol{v}_1$?

END-OF-PAGE SOLUTION FOR EXAMPLE 8.6.1 – 8.6.6 (Pages 454–457)

- **Ex. 8.6.1–3:** For 1 flip: $m = 0(\frac{1}{2}) + 1(\frac{1}{2}) = \frac{1}{2}$ and $\sigma^2 = (0 - \frac{1}{2})^2\frac{1}{2} + (1 - \frac{1}{2})^2\frac{1}{2} = \frac{1}{4}$. For $N = 3, M = 2$: $p_2 = \frac{1}{2^3}\frac{3!}{2!1!} = \frac{3}{8}$.
- **Ex. 8.6.5:** With $A = (1, \ldots, 1)^T \in \mathbb{R}^{10\times 1}$ and $\Sigma^{-1} = \frac{1}{\sigma^2}I_{10}$, $A^T\Sigma^{-1}A = \frac{10}{\sigma^2} \implies P = (A^T\Sigma^{-1}A)^{-1} = \frac{\sigma^2}{10}$.
- **Ex. 8.6.6:** $\frac{\sigma_1^2}{\sigma_1^2+\sigma_2^2+\sigma_3^2} = \frac{12^2}{12^2+8^2+4^2} = \frac{144}{144+64+16} = \frac{144}{224} = \frac{9}{14}$.

8.3 All Unique Solved Problems: Problem Sets 8.6 & 8.7 [Book pp. 458, 463–464]

PROBLEM 8.6.1 – 8.6.9 Weighted Least Squares, Coin Covariance, BLUE Proof, and Rank-2 PCA

Question 1–3: Fit $Ct + D$ to $b = 1, 2, 4$ at $t = 0, 1, 2$: (1) With variances $\sigma_1^2 = 1, \sigma_2^2 = 1, \sigma_3^2 = 2$ ($w_i = 1/\sigma_i$). (2) When $\sigma_3^2 \to \infty$ (third point totally unreliable). (3) When $\sigma_3^2 \to 0$ (third point totally reliable).

Question 4–6: (4) Variance σ^2 for the sum of two fair coin flips ($0, 1, 2$ with probabilities $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$). (5) Covariance matrix Σ of two independent fair coin flips. (6) Mean m and variance σ^2 of an unfair coin (p for heads, $1 - p$ for tails).

Question 7–9: (7) Minimize $P(a) = a^2\sigma_1^2 + (1-a)^2\sigma_2^2$ for $\hat{x} = ab_1 + (1-a)b_2$. (8) Multiply $L\Sigma L^T$ to prove $P = (A^T\Sigma^{-1}A)^{-1}$. (9) Show that $A = \begin{bmatrix} 3 & -1 & 1 & -3 \\ -1 & 3 & -3 & 1 \\ -3 & 1 & -1 & 3 \\ 1 & -3 & 3 & -1 \end{bmatrix}$ has rank 2 and find its SVD using $\boldsymbol{u}_1, \boldsymbol{u}_2, \boldsymbol{v}_1, \boldsymbol{v}_2$ from Example 6.

Solution:

- **1–3:** (1) Note the order $\hat{\boldsymbol{x}} = (D, C)^T$ for $b = D + Ct$: $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}$, $C_{\text{wt}} = \text{diag}(1, 1, \frac{1}{2}) \implies A^T C_{\text{wt}} A = \begin{bmatrix} 5/2 & 2 \\ 2 & 3 \end{bmatrix}$, $A^T C_{\text{wt}} \boldsymbol{b} = \begin{bmatrix} 5 \\ 6 \end{bmatrix} \implies D = \frac{6}{7}, C = \frac{10}{7}$ (line $\frac{10}{7}t + \frac{6}{7}$). (2) Ignore point 3: line through $(0, 1)$ and $(1, 2)$ is $\boldsymbol{b = t + 1}$ ($C = 1, D = 1$). (3) Line must pass through $(2, 4)$, so $b = 4 + C(t - 2)$. Minimize $(4 - 2C - 1)^2 + (4 - C - 2)^2 = (3 - 2C)^2 + (2 - C)^2 \implies -4(3 - 2C) - 2(2 - C) = 0 \implies 10C = 16 \implies C = \frac{8}{5}, D = \frac{4}{5}$ (line $\frac{8}{5}t + \frac{4}{5}$).
- **4–6:** (4) $\sigma^2 = (0-1)^2\frac{1}{4} + (1-1)^2\frac{1}{2} + (2-1)^2\frac{1}{4} = \frac{1}{2}$. (5) Since the flips are independent with variance $\frac{1}{4}$, $\Sigma = \begin{bmatrix} 1/4 & 0 \\ 0 & 1/4 \end{bmatrix}$. (6) $m = 0(1-p) + 1(p) = \boldsymbol{p}$; $\sigma^2 = (0-p)^2(1-p) + (1-p)^2 p = \boldsymbol{p(1-p)}$.
- **7–9:** (7) $\frac{dP}{da} = 2a\sigma_1^2 - 2(1-a)\sigma_2^2 = 0 \implies a = \frac{\sigma_2^2}{\sigma_1^2+\sigma_2^2} = \frac{1/\sigma_1^2}{1/\sigma_1^2+1/\sigma_2^2} = \frac{w_1^2}{w_1^2+w_2^2}$. (8) Derived in Step 3 of Section 7.1 above. (9) Notice $\text{row}_3 = -\text{row}_1$ and $\text{row}_4 = -\text{row}_2$, and $A = 8\boldsymbol{u}_1\boldsymbol{v}_1^T + 4\boldsymbol{u}_2\boldsymbol{v}_2^T$ with $\boldsymbol{u}_1 = \frac{1}{2}(-1, -1, 1, 1)^T, \boldsymbol{v}_1 = \frac{1}{2}(-1, 1, -1, 1)^T, \boldsymbol{u}_2 = \frac{1}{2}(1, -1, -1, 1)^T, \boldsymbol{v}_2 = \frac{1}{2}(1, 1, -1, -1)^T$ (singular values $\sigma_1 = 8, \sigma_2 = 4$, rank 2).

PROBLEM 8.7.1 – 8.7.17 Computer Graphics in Homogeneous Coordinates

Question 1–6: (1) Homogeneous coordinates of (x, y, z) are $(x, y, z, 1)$ and (cx, cy, cz, c). (2) Affine transformation of $(x, y, z, 1)$. (3) Product TT_1 of translations along $(1, 4, 3)$ and $(0, 2, 5)$. (4) Scale matrix S and blowing up around $(1, 4, 3)$: $\boldsymbol{v}ST$ or $\boldsymbol{v}TS$? (5) 3×3 scaling matrix from 8.5×11 page to 1×1. (6) 4×4 matrix moving $(1, 1, 2)$ to origin and doubling all lengths.

Question 7–17: (7–8) Rotation matrix $Q = (\cos\theta)I + (1 - \cos\theta)\boldsymbol{a}\boldsymbol{a}^T + \sin\theta(\boldsymbol{a} \times \cdot)$ action on $\boldsymbol{a}$ and $\boldsymbol{b} \perp \boldsymbol{a}$. (9–11) 3×3 and 4×4 projection matrices onto $\frac{2}{3}x + \frac{2}{3}y + \frac{1}{3}z = 0$ and $\frac{2}{3}x + \frac{2}{3}y + \frac{1}{3}z = 1$, and projection of $(3, 3, 3)$. (12–13) Projection of a square and of a cube perpendicular to its main diagonal. (14–15) Reflection $M = I - 2\boldsymbol{n}\boldsymbol{n}^T$ and $T_- M T_+$ of $(3, 3, 3)$ across both planes. (16–17) Difference of two points and rescaling the last coordinate

(x, y, z, c).

Solution:

- **1–6:** (1) $(x, y, z, 1)$ and $(2x, 2y, 2z, 2)$. (2) Need $T(\textbf{origin})$; output is $xT(\boldsymbol{i}) + yT(\boldsymbol{j}) + zT(\boldsymbol{k}) + T(\textbf{origin})$. (3) Translation along $(1, 6, 8)$. (4) $S = \text{diag}(c, c, c, 1)$; to blow up around $(1, 4, 3)$, translate $(1, 4, 3)$ to origin, scale by S, and translate back: $\boldsymbol{v}T^{-1}ST$.Between $\boldsymbol{v}ST$ and $\boldsymbol{v}TS$, $\boldsymbol{v}ST$ scales around the origin then shifts by $(1, 4, 3)$, while $\boldsymbol{v}TS$ shifts the center too. (5) $S = \text{diag}(1/8.5, 1/11, 1)$. (6) $T_{-}S = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -1 & -1 & -2 & 1 \end{bmatrix} \text{diag}(2, 2, 2, 1) = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ -2 & -2 & -4 & 1 \end{bmatrix}$.
- **7–11:** (7–8) $\boldsymbol{a}Q = (\cos\theta)\boldsymbol{a} + (1 - \cos\theta)\boldsymbol{a} + \boldsymbol{0} = \boldsymbol{a}$; for $\boldsymbol{b} \perp \boldsymbol{a}$, $\boldsymbol{b}\boldsymbol{a}\boldsymbol{a}^T = \boldsymbol{0}$, leaving $(\cos\theta)\boldsymbol{b} + (\sin\theta)(\boldsymbol{b} \times \boldsymbol{a})$. (9) With unit normal $\boldsymbol{n} = (\frac{2}{3}, \frac{2}{3}, \frac{1}{3})^T$, $P_3 = I - \boldsymbol{n}\boldsymbol{n}^T = \frac{1}{9}\begin{bmatrix} 5 & -4 & -2 \\ -4 & 5 & -2 \\ -2 & -2 & 8 \end{bmatrix}$, and $P_4 = \begin{bmatrix} P_3 & \boldsymbol{0} \\ \boldsymbol{0}^T & 1 \end{bmatrix}$. (10) Choose $(0, 0, 3, 1)$ on the plane $\frac{2}{3}x + \frac{2}{3}y + \frac{1}{3}z = 1$: T_{-} shifts by $(0, 0, -3)$ and T_{+} shifts by $(0, 0, 3)$, giving $T_{-}P_4T_{+} = \begin{bmatrix} P_3 & \boldsymbol{0} \\ \frac{2}{3}\ \frac{2}{3}\ \frac{1}{3} & 1 \end{bmatrix}$. (11) Onto plane $= 0$: $(3, 3, 3)P_3 = (-1, -1, 4)$; onto plane $= 1$: $(-1, -1, 4) + (\frac{2}{3}, \frac{2}{3}, \frac{1}{3}) = (-\frac{1}{3}, -\frac{1}{3}, \frac{13}{3})$.
- **12–17:** (12) A **parallelogram** (or line segment if edge-on). (13) A **regular hexagon**. (14) $M = I - 2\boldsymbol{n}\boldsymbol{n}^T = \frac{1}{9}\begin{bmatrix} 1 & -8 & -4 \\ -8 & 1 & -4 \\ -4 & -4 & 7 \end{bmatrix}$; reflecting $(3, 3, 3)$ across plane $= 0$ gives $(3, 3, 3) - 2(5)(\frac{2}{3}, \frac{2}{3}, \frac{1}{3}) = (-5, -5, -1)$. (15) Across plane $= 1$: $(3, 3, 3) - 2(5 - 1)(\frac{2}{3}, \frac{2}{3}, \frac{1}{3}) = (-\frac{7}{3}, -\frac{7}{3}, \frac{1}{3})$. (16) $\boldsymbol{v} = (x, y, z, 0)$; vectors end in $\boldsymbol{0}$, so we add a **vector** to a point. (17) Rescales the whole space by $\boldsymbol{1/c}$ because (x, y, z, c) is the same point as $(x/c, y/c, z/c, 1)$.

Master Syllabus Formula Sheet (ASD-STE100)

1. Lengths, Dot Products, and Inequalities (Sections 1.2–1.3)

$$v \cdot w = v^T w = \sum_{i=1}^{n} v_i w_i, \quad \|v\| = \sqrt{v \cdot v}, \quad u = \frac{v}{\|v\|},$$
$$\cos\theta = \frac{v \cdot w}{\|v\|\|w\|}, \quad |v \cdot w| \le \|v\|\|w\|, \quad \|v + w\| \le \|v\| + \|w\|$$

2. Four Fundamental Subspaces and Orthogonality (Section 4.1)

$$N(A) = C(A^T)^\perp \text{ in } \mathbb{R}^n \quad (\dim r + (n - r) = n),$$
$$N(A^T) = C(A)^\perp \text{ in } \mathbb{R}^m \quad (\dim r + (m - r) = m), \qquad x = x_r + x_n$$

3. Projections and Least Squares Approximations (Sections 4.2–4.3)

$$\text{Line: } \hat{x} = \frac{a^T b}{a^T a}, \quad p = a\hat{x}, \quad P = \frac{a a^T}{a^T a}$$
$$\text{Subspace: } A^T A\hat{x} = A^T b, \quad p = A\hat{x}, \quad P = A(A^T A)^{-1} A^T \quad (P^2 = P = P^T)$$

4. Orthonormal Columns, Gram-Schmidt, and $A = QR$ (Section 4.4)

$$Q^T Q = I, \quad \hat{x} = Q^T b, \quad P = QQ^T, \quad B = b - \frac{A^T b}{A^T A} A,$$
$$C = c - \frac{A^T c}{A^T A} A - \frac{B^T c}{B^T B} B, \quad A = QR \implies R\hat{x} = Q^T b$$

5. Positive Definite Matrices, Cholesky, Ellipses, and Statistics (Sections 6.5, 8.6, 8.7)

$$x^T A x > 0 \iff \lambda_i > 0 \iff d_i > 0 \iff \det(A_k) > 0 \iff A = R^T R,$$
$$A^T \Sigma^{-1} A\hat{x} = A^T \Sigma^{-1} b, \qquad P_{\text{cov}} = (A^T \Sigma^{-1} A)^{-1}$$